

Robust Greenwashing Detection and Actionable ESG Analysis with Aspect-based Sentiment Analysis

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I. INTRODUCTION

A. Title: Research Motivation for ESG ABSA in Indonesian Disclosures: Why Language, Regulation, and Depth Matter Now

1) *Introduction:* Environmental, Social, and Governance (ESG) disclosure analysis has ascended from a peripheral corporate reporting concern to a central input for investment decisions, regulatory compliance, and stakeholder accountability. In the specific domain of aspect-based sentiment analysis (ABSA) applied to ESG disclosures (ESG-ABSA), the motivation rests on four interlocking pressures: rising regulatory scrutiny, growing demand for ESG transparency from investors, linguistic and cross-language challenges in bilingual Indonesian/English reports, and a persistent gap between disclosure volume and analytical depth. Synthesizing evidence across multiple streams shows that the strategic value of ESG disclosures depends not only on what is disclosed but on how it is analyzed, interpreted, and governed. This section builds a cohesive argument for why this research is timely, technically rigorous, and policy-relevant, culminating in a transition to the research background.

2) *Regulatory pressure on sustainability reporting globally and in Indonesia:* Global regulatory tightening elevates the baseline for ESG disclosures. Regulators worldwide have moved toward mandatory or enhanced disclosure frameworks, fostering comparability and accountability. For example, cross-jurisdictional trends highlight the increasing prominence of standardized ESG reporting to support market integrity and investor protection [1]–[4]. The European emphasis on SFDR, ESRS, and related taxonomy underscores a shift from voluntary to regulated ESG transparency, prompting firms to align disclosure content with evolving criteria and to demonstrate governance of climate-related risks [1], [5], [6]

Indonesia-specific regulatory developments are contemporaneous with regional convergence toward formalized ESG reporting. The Indonesian Financial Services Authority (OJK) and the Indonesia Stock Exchange (IDX) impose disclosures and reporting obligations that shape corporate communication with capital markets. OJK regulations intersect with IDX requirements, including electronic submission and public accessibility of disclosure documents via IDXNet, signaling a formal regulatory channel for ESG information and governance expectations [1] The Indonesian regulatory regime thus creates a structured environment in which ESG disclosures must be produced, structured, and traceable, elevating the need for robust analytical tools to interpret disclosures across filings, annual reports, and supplementary narratives [1]

The gap between regulatory demand and analytic execution motivates ABSA-in-ESG work. As regimes require more public, standardized ESG information, the challenge becomes not only data collection but also extracting decision-useful signals (risk, opportunity, governance strength) from unstructured text. Systematic text analytics—particularly ABSA—offers a mechanism to translate regulatory language into granular, action-oriented insights (e.g., identifying which environmental or governance aspects drive perceived risk or opportunity) [3], [7], [8]

3) *Rising investor demand for ESG transparency:* Investors increasingly rely on ESG disclosures to assess risk, resilience, and long-term value. The literature shows a robust link between high ESG transparency and investor decision-making, risk management, and performance analysis, with AI/ML methods used to enhance interpretability and predictive value of ESG signals [1], [3], [9], [10]. ABSA enables investors to move beyond aggregate ESG scores to consider sentiment and signals at the level of specific ESG aspects (e.g., emissions, governance quality, labor practices), which aligns with institutional due diligence and risk monitoring practices [3], [7], [9]

The role of ML and NLP in ESG analytics is increasingly emphasized in governance and investment research. Systematic reviews and surveys indicate that ML/NLP are instrumental in risk detection, controversy monitoring, and trend analysis within ESG disclosures, often accompanied by explainable AI (XAI) techniques to satisfy stakeholder demands for transparency and accountability [3], [7], [10]. For Indonesian and regional markets, ML-enhanced ESG analysis can improve the reliability and timeliness of disclosures used by analysts and investors, particularly when data volumes are large and heterogeneous [3], [9]

Cross-linguistic and cross-market investor needs amplify the demand for ABSA that can capture nuance across stakeholders. Investors require not only sentiment polarity but also the specific aspects driving sentiment, which ABSA is designed to deliver. Cross-lingual ABSA literature highlights the importance of multilingual and cross-language capability to support international investment analysis, as global portfolios assess Indonesian disclosures alongside English-language reports [11]–[13]. The intersection of ABSA with cross-lingual transfer learning is especially relevant for Indonesian bilingual reporting contexts.

4) *The challenge of bilingual Indonesian/English sustainability reports*: Bilingual ESG disclosures compound language complexity in automated analysis. The Indonesian market often features sustainability disclosures in both Indonesian and English, either within same documents or across companion reports. This bilingual setting raises unique NLP challenges, including code-switching, mixed-language terminology, and varying regulatory lexicons. Cross-lingual ABSA research documents successful strategies using multilingual pre-trained models and cross-language alignment to enable unified analysis across languages, illustrating that model architectures and data curation choices are critical for robust ABSA in bilingual contexts [11]–[13]

Cross-lingual ABSA advances provide a blueprint for Indonesian ESG text analytics. Surveys of cross-lingual ABSA approaches show that combining translation, alignment-based labeling, and multilingual pre-trained language models (e.g., mBERT, XLM-R) can deliver effective cross-language sentiment and aspect extraction, particularly when annotated multilingual corpora exist or when zero-shot transfer is employed [11]. In environments where English-language ESG disclosures are prevalent alongside local Indonesian content, such cross-lingual capabilities become a core technical requirement for ABSA pipelines [11], [12]

The practicalities of bilingual ESG reporting—readability, regulatory alignment, and content coverage—also feed into the ABSA motivation. Studies of CSR/ESG reporting using NLP demonstrate that readability and content quality affect perceived credibility and the likelihood of greenwashing detection, underscoring the need for ABSA systems that can dissect environmental, social, and governance aspects across languages with interpretability [12]. Moreover, regulatory mandates requiring standardized disclosures heighten the need for ABSA to support regulatory reporting and compliance-monitoring tasks by indexing topics and sentiments to specific regulatory themes [4], [10]

5) *The gap between disclosure volume and analytical depth*: Escalating disclosure volumes outpace analytical depth. As ESG reporting has grown in scope and frequency, the sheer quantity of textual disclosures—annual sustainability reports, regulatory filings, press statements—has expanded dramatically. However, traditional analyses rely on coarse-grained scores or manual content coding, which are time-consuming, prone to inconsistency, and ill-suited for scalable investment diligence. A wave of research demonstrates that while disclosure volume has risen, there is a persistent gap in depth: the ability to extract granular, evidence-based insights at the aspect level, detect controversy or greenwashing, and translate textual signals into decision-relevant metrics [3], [7], [10], [12]

AI-driven ABSA and related NLP methods address this depth gap by enabling fine-grained extraction of target aspects, sentiment polarity, and category mappings from large corpora. ABSA supports more nuanced risk and opportunity assessment by tying sentiments to specific ESG topics (e.g., emissions intensity, supply chain governance, diversity policies). Systematic mappings across studies show that ML/NLP are increasingly used to predict ESG performance signals, identify controversies, and assist in reporting alignment with standards, while XAI methods help stakeholders understand model decisions—mitigating concerns about opacity and bias [3], [7], [10]

The Indonesian context compounds these depth concerns with regulatory and market-specific factors. In Indonesia, the combination of OJK and IDX disclosures creates a data ecosystem where standardization is evolving, and where language diversity adds to the complexity of extracting unified signals. Bridging the depth gap requires ABSA pipelines that can operate on multilingual ESG narratives, handle regulatory-specific lexicons, and produce interpretable outputs that regulators and investors can trust. Evidence from related reviews and empirical investigations supports the utility of ML/NLP to map and quantify ESG content, risk signals, and governance themes in corporate disclosures, while pinpointing challenges such as data standardization, model explainability, and potential bias [3], [9], [10]

6) *Synthesis: why ABSA is well-suited to ESG disclosure analysis in Indonesia*: ABSA aligns with the need for granular, aspect-level insights. By simultaneously identifying target terms, aspects, opinions, and sentiment polarity, ABSA yields a structured representation of ESG narratives that reflect stakeholder concerns, regulatory

focus areas, and governance signals. This granularity is essential for both investors and regulators who require precise indicators of environmental risks, social commitments, and governance practices, particularly where narratives contain nuanced or conflicting cues. Empirical works in ESG/ML contexts demonstrate ABSA's capacity to reveal detail beyond coarse sentiment or document-level scores, including cross-aspect interactions and topic-level patterns [7], [10], [12], [14], [15]

Multilingual and cross-lingual capabilities are crucial in bilingual Indonesian/English reports. Cross-lingual ABSA research illustrates practical methods to extend analysis across languages using multilingual pre-trained models, translation strategies, and alignment techniques, enabling cohesive benchmarking and comparability across Indonesian disclosures and international counterparts [11]–[13]. This is particularly relevant for Indonesian market participants who must assess local disclosures in tandem with global ESG narratives.

Explainability and regulatory relevance. The integration of XAI methods (e.g., SHAP, LIME) with ABSA outputs addresses demands for transparency in ESG scoring and governance decisions. Regulators, investors, and auditors favor interpretable models that connect textual signals to concrete ESG dimensions and regulatory criteria. Systematic reviews of ESG analytics emphasize the value of explainability, bias mitigation, and robust evaluation for trustworthiness in financial contexts [3], [7], [10]

Regulatory and investor alignment with depth-driven analytics. The convergence of global regulatory frameworks and investor expectations creates a policy-relevant research space in which ABSA-based ESG analytics can support both compliance monitoring and performance assessment. Existing literature documents regulatory-driven disclosure obligations, the transformation of data into decision-useful signals, and the increasing role of ML/NLP in ESG governance and investment decision-making [3], [4], [9], [10]

7) *Concluding transition to research background:* The foregoing synthesis establishes that ESG disclosure analysis is currently propelled by three interdependent forces: (a) escalating regulatory expectations and formalization of ESG reporting in Indonesia and globally; (b) rising demand from investors for detailed, actionable ESG signals; and (c) the technical necessity to handle bilingual Indonesian/English disclosures with scalable, interpretable ABSA techniques that close

8) *The gap between disclosure volume and analytical depth:* . These dynamics jointly justify the development of ABSA-enabled ESG analysis tailored to Indonesian disclosures, with attention to multilingual capabilities, regulatory lexicons, and explainable outputs. The transition to the research background now follows: we outline the existing empirical and methodological landscape, identify gaps specific to ABSA in Indonesian ESG reporting, and articulate the research questions, data sources, and methodological approach that will be pursued.

B. Research Background

This section situates the study within the broader ESG, ABSA, and multilingual text-analysis landscape, with a focus on environmental, social, and governance (ESG) disclosure, the role of global and Indonesian reporting standards, the structure of sustainability reports and the page-level complexity they introduce for automated analysis, a concise survey of prevailing NLP approaches to ESG analysis and their limitations, and the particular bilingual Indonesian/English challenge in Indonesian corporate reporting. The synthesis integrates the cited frameworks and methodological considerations to motivate a robust ABSA-based investigation of Indonesian disclosures.

1) *ESG framework definitions and GRI/SASB standards:* ESG as a framework for risk, value creation, and accountability. ESG measures provide a lens through which firms' environmental, social, and governance activities are assessed for risk management, strategic planning, and investor decision-making. A body of literature and practice emphasizes that ESG disclosures aim to illuminate how firms create or erode long-term value while signaling governance quality and risk management to capital markets [16]–[18]. The underlying rationale is that standardized ESG information improves transparency, comparability, and market efficiency, thereby supporting responsible investing and regulatory compliance [16], [19], [20]

Core reporting standards and their theoretical alignment. Globally, a dominant tension exists among major reporting frameworks—GRI (global stakeholder-oriented), SASB (industry-specific, financially material focus), TCFD (climate-risk centric), CDSB, CDP, and the emerging ISSB standards under IFRS. These bodies have pursued alignment through initiatives like the Corporate Reporting Dialogue and joint guidance to map SASB, GRI, and CDP against TCFD, with a continuing push toward a unified global standard set while acknowledging residual fragmentation [18], [19], [21]–[24]. The literature notes that SASB emphasizes financially material disclosures for investors, whereas GRI adopts a broader stakeholder perspective, highlighting the materiality tension across frameworks and the need for harmonization to reduce reporting overload [18], [19]

Indonesian regulatory context (global and local). Regulatory convergence toward formalized ESG disclosure in Indonesia is framed by global expectations and local authorities. Indonesia's market uses a mix of standards and regulatory expectations—OJK (the Financial Services Authority) rules and IDX (the Exchange) disclosure

requirements—creating a structured environment for ESG information that is publicly accessible and regulated. The regulatory regime encourages formalized, traceable disclosures across filings and annual reports, signaling a demand for robust analytical tooling to interpret this information coherently for market participants [1], [3], [4]. This regime aligns with global moves toward standardized ESG reporting but also elevates the complexity of local lexicons and regulatory mappings.

Implications for ABSA. The regulatory density and standardization ambitions imply that ESG disclosures are not merely narrative; they embed regulatory motifs, risk signals, and governance cues that require granular, aspect-level interpretation. ABSA—which links sentiment to specific ESG aspects—offers a mechanism to translate regulatory language into granular, decision-useful signals (e.g., which environmental topics or governance mechanisms drive risk signaling or capital-market responses) [3], [4], [7], [9], [12]

How sustainability reports are structured and the page-level complexity this creates for automated analysis. Typical sustainability-report structure. Sustainability reports commonly adopt a multi-section layout aligned with material topics, governance disclosures, strategy and risk management, and performance metrics, often organized by ESG pillar and then by sub-topics (e.g., emissions, energy efficiency, supply chain governance, diversity and inclusion, board oversight). This structure supports a narrative of materiality, strategy alignment, and accountability, yet it also yields dispersed content across sections, tables, figures, and appendices, complicating automated extraction and cross-document synthesis [17], [24], [25].

Cross-framework implications for document structure. When firms report under multiple frameworks (GRI, SASB, TCFD, etc.), the same information may be presented in different templates, with varying indicator identifiers, materiality thresholds, and regulatory lexicons. The result is a heterogeneous document design in which regulatory cues, indicator mappings, and narrative themes appear across sections and even within the same sentence. This heterogeneity challenges parsers that rely on consistent headings or standardized ontologies, and it motivates design choices that can align textual content to a unified set of ESG topics and regulatory motifs [19], [21], [26], [27]

Page-level complexity for ABSA pipelines. For ABSA, the page-level complexity manifests in several ways:

Multi-topic co-location: Sentiment toward an environmental topic may appear in close proximity to social or governance content, requiring robust topic disambiguation. Lexical variation and terminology drift: Terminology shifts across frameworks and languages necessitate adaptable lexicons and context-sensitive models. Table and figure content: Quantitative indicators in tables/figures often accompany qualitative narratives; extracting numerical signals alongside textual sentiment requires multi-modal integration. Regulatory mapping: Cross-referencing narrative claims with regulatory requirements (e.g., climate risk disclosures, governance practices) demands models that can align textual cues to regulatory lexicons and standards. Language and translation effects: In Indonesia, reports frequently include Indonesian and English content within the same document or across companion reports, complicating alignment and requiring cross-language capabilities (see Section 4). Implications for ABSA design. To bridge the depth and alignment gaps, ABSA pipelines must (i) operate on multilingual corpora, (ii) incorporate topic- and framework-aware lexicons, (iii) support aspect-term extraction and polarity scoring at fine granularity, and (iv) provide explainable outputs that map sentiments to regulatory themes and material ESG topics. The literature indicates that ABSA, when combined with cross-lingual NLP and explainability techniques, can yield interpretable, action-oriented signals from complex ESG disclosures [7], [10], [12], [14], [15]

2) *Brief survey of existing NLP approaches to ESG analysis and their limitations:* Traditional sentiment and topic analysis. Early ESG NLP work emphasized document-level sentiment, topic modeling (e.g., LDA), and keyword extraction to summarize disclosures. While these methods illuminate broad sentiment trends and dominant topics, they lack the granularity needed for investor due diligence and regulatory monitoring, as they do not tie sentiment to specific ESG aspects or regulatory criteria [8], [9], [12]

Aspect-based sentiment analysis (ABSA). ABSA addresses the granularity gap by extracting aspect terms, associating sentiment polarity with those aspects, and yielding a structured representation of ESG narratives. ABSA enables more precise risk/opportunity signaling and supports explainability by linking opinions to concrete topics (e.g., emissions, governance processes). Recent systematic reviews highlight ABSA's strength in extracting fine-grained signals but also note challenges in domain adaptation, cross-domain labeling, and interpretability in highly technical ESG texts [7], [10], [12], [14], [15]

Cross-lingual and multilingual NLP in ESG. Given bilingual or multilingual reporting (e.g., Indonesian/English), cross-lingual ABSA methods leverage translation, multilingual pre-trained models (e.g., mBERT, XLM-R), and alignment-based labeling to enable cross-language sentiment analysis. Surveys indicate cross-lingual ABSA effectiveness when annotated multilingual data exist or when zero-shot transfer is leveraged; however, performance gaps persist when domain-specific terminology or regulatory lexicons are sparse in high-resource languages [11]–[13]

Explainability and regulatory relevance. A recurring limitation across NLP methods is model opacity. To satisfy regulatory and investor demands for transparency, researchers increasingly integrate explainable AI (XAI) techniques (e.g., SHAP, LIME) with ABSA outputs to justify aspect-level sentiment decisions and highlight salient features driving the judgments. Systematic reviews of ESG analytics emphasize the need for interpretability, bias mitigation, and robust evaluation when deploying NLP in financial contexts [3], [7], [10]

Indonesian-specific and regional gaps. While global ABSA and multilingual NLP research is burgeoning, the Indonesian disclosure landscape presents domain-specific lexicons, industry sectors, and regulatory language that are underrepresented in high-quality multilingual training data. This creates a critical gap that a bilingual ABSA framework must address through domain-adaptive fine-tuning, lexicon customization, and regulatory lexicon integration to ensure reliable extraction of ESG topics and sentiment from Indonesian corporate reporting [3], [9], [12]

Summary of limitations. Collectively, the literature identifies (i) a need for fine-grained, aspect-level signals linked to financially material ESG topics; (ii) challenges in cross-language interpretation and translation-induced variability; (iii) the demand for explainable outputs aligned with regulatory criteria; and (iv) data standardization and domain adaptation issues that hinder cross-framework comparability. These gaps motivate the current focus on ABSA-based, multilingual ESG analysis with an emphasis on Indonesian disclosures, regulatory alignment, and interpretability [3], [7], [10]–[14]

3) *The specific bilingual Indonesian/English challenge in Indonesian corporate reporting:* Prevalence of bilingual reporting. Indonesian corporate reporting frequently features content in both Indonesian and English, either within sustainability reports or across companion documents (e.g., annual reports and separate ESG disclosures). This bilingual presentation creates cross-language alignment demands, complicating standard NLP pipelines that assume monolingual input and stable terminologies. Cross-lingual ABSA research specifically focuses on scenarios where multilingual content necessitates either robust multilingual models or effective translation-then-analyze strategies to preserve nuance across languages [11]–[13]

Language phenomena and regulatory lexicons. Indonesian-English reporting surfaces code-switching, mixed-language terminology, and varying regulatory lexicons that reflect both local compliance language and international standard terminology. Cross-language ABSA must handle coinages, acronyms, and sector-specific terms that differ in Indonesian and English, requiring dynamic lexicon expansion and contextual embedding strategies that can capture register and domain semantics across languages [11]–[13]

Model design implications. For bilingual Indonesian/English ESG ABSA, model design choices include (i) employing multilingual pre-trained transformers with domain-adaptive fine-tuning, (ii) integrating translation or alignment components to unify cross-language sentiment signals, (iii) constructing regulatory-anchored aspect taxonomies that map Indonesian and English ESG terms to common topics and to the regulatory frameworks (GRI, SASB, TCFD), and (iv) ensuring explainability to satisfy both regulators and investors. Empirical evidence from cross-lingual ABSA surveys indicates that superior performance arises from multilingual models with cross-language alignment and explicit aspect annotation, rather than naïve translation-only approaches [11]–[14]

Practical relevance for Indonesian markets. Indonesian ABSA that can robustly parse bilingual reports supports regulatory monitoring (e.g., alignment with OJK/IDX expectations), investor due diligence, and risk management by enabling granular extraction of sentiment signals tied to specific ESG aspects. The literature already stresses the importance of regulatory-aligned, granular ESG analytics for governance and market outcomes, while highlighting the need for multilingual, explainable approaches in non-English reporting contexts [1], [3], [4], [7], [9], [12], [15]

Transition to conceptual and methodological motivation Why ABSA is particularly well suited to Indonesian ESG disclosures. ABSA provides a structured representation of ESG narratives at the aspect level, enabling investors and regulators to detect nuanced risk signals, governance weaknesses, or positive commitments that are otherwise obscured in document-level summaries or coarse scoring. In multilingual Indonesian/English contexts, ABSA is complemented by cross-lingual NLP methods and multilingual transformers to unify cross-language sentiment signals around shared environmental, social, and governance topics, which is essential for comparability across domestic and international stakeholders [7], [11]–[14]

Addressing the page-level complexity with ABSA. By anchoring sentiment to explicit ESG aspects and linking those aspects to regulatory themes, ABSA can reduce the information extraction burden from dense documents. It supports explainable outputs that tie sentiment to concrete sections, indicators, and standards, responding to the demand for transparency in ESG analytics described in the literature and discussed in the Indonesian regulatory milieu [3], [7], [10], [21]

Gaps this research aims to close. Although substantial work demonstrates the potential of ABSA and cross-lingual NLP for ESG analysis, there remains a scarcity of rigorously evaluated, bilingual ABSA pipelines tailored

to Indonesian corporate reporting, with explicit alignment to local regulatory lexicons and international standards. This study seeks to fill that gap by developing an ABSA framework capable of (i) handling Indonesian-English bilingual texts, (ii) mapping sentiments to GRI/SASB/TCFD-aligned aspects, and (iii) providing explainable, regulator-friendly outputs suitable for Indonesian stakeholders and global investors.

4) *Summary:* The ESG landscape is governed by converging but heterogeneous standards (GRI, SASB, TCFD, ISSB, and others), creating a complex, multi-framework disclosure ecosystem that Indonesian firms navigate under OJK/IDX oversight. This motivates the need for fine-grained ABSA that can interpret disclosures at the aspect level and map sentiments to regulatory themes, thereby supporting better risk assessment, governance monitoring, and investment decision-making [19], [21]–[23]. The structural characteristics of sustainability reports—multi-section, cross-framework content, and mixed Indonesian/English language—imply page-level complexity that challenges traditional NLP, reinforcing the case for ABSA with explainability and cross-lingual capabilities [17], [18], [24], [25]. Existing NLP approaches offer promising directions but exhibit limitations in granularity, cross-language adaptation, and regulatory alignment, particularly in Indonesian contexts [7], [9], [12], [14]. The bilingual Indonesian/English reporting environment further intensifies these challenges, necessitating a bilingual ABSA approach informed by cross-lingual transfer, lexical adaptation, and regulatory mapping [11]–[13]

The transition to the research core is thus warranted: to design, implement, and evaluate a bilingual ESG ABSA pipeline tailored to Indonesian disclosures, capable of delivering aspect-level sentiment signals aligned with GRI/SASB/TCFD concepts, while providing explainable outputs suitable for regulators and investors. The ensuing chapters (Research Design and Data, Methodology, and Experiments) will detail the data sources, annotation schema, model architectures, evaluation metrics, and case studies that operationalize these objectives.

Annotated bibliography (machine-readable XML)

C. Research Gap

This section identifies and articulates five specific gaps in the current state of research on ESG environmental, social, and governance (ESG) analysis through aspect-based sentiment analysis (ABSA) in multilingual (Indonesian/English) corporate disclosures. For each gap, I specify (a) what existing methods do, (b) what is missing, and (c) the implications for advancing ESG ABSA in Indonesia. The analysis draws on the provided reference material and integrates them into a cohesive, technically grounded critique suitable for an academic thesis.

1) Gap 1: Absence of sentence-level ESG evidence extraction: What existing methods do

Contemporary ESG NLP work often emphasizes document-level sentiment or coarse-grained topic signals (e.g., general tone of a sustainability report, or high-level topic modeling across entire documents) [8], [9], [12], [28]. These approaches capture broad impressions but do not systematically attach sentiment to discrete ESG evidence sentences or clause-level textual cues. Some ABSA implementations target fine-grained aspect extraction within domains such as general text or non-financial domains, yet there is limited evidence of rigorously validated sentence-level ABSA pipelines specifically tuned to ESG disclosures, where regulatory-relevant evidence often resides in particular sentences, sentences within tables, or footnotes [7], [10], [12], [14]

Where sentence-level processing exists, it is frequently confined to generic sentiment at the sentence level or to shallow extraction of opinion-bearing phrases, without a formal linkage to precise ESG aspects, indicators, or regulatory lexicons. What is missing

A formal sentence-level ABSA framework for ESG filings that (i) maps each sentence to a concrete ESG aspect (e.g., emissions, governance oversight, supply-chain ethics), (ii) assigns sentence-level sentiment or evidentiary polarity with respect to that aspect, and (iii) propagates confidence metrics and explainability to regulators and investors. A standardized annotation schema and corpus for Indonesian and English ESG disclosures that anchors aspect terms to GRI/SASB/TCFD-aligned concepts at the sentence level, enabling reproducible model development and cross-study comparability. Evaluation protocols that quantify the precision, recall, and calibration of sentence-level aspect-sentiment associations, including robustness to multi-aspect sentences and to the presence of measured numerical indicators within the same sentence.

Implications for practice and theory

Without sentence-level evidence extraction, ABSA outputs remain insufficiently actionable for due diligence, risk assessment, and regulatory verification, because decision-makers require precise textual evidence linking sentiment signals to specific ESG topics and regulatory criteria [3], [7], [12]

A sentence-level approach would enable more transparent risk signaling, facilitate traceability for greenwashing detection, and improve the interpretability of ABSA outputs through direct citation of textual evidence.

Supporting references

ABSA literature highlights the need for fine-grained, aspect-level extraction to reveal granular signals in ESG narratives [7], [10], [12], [14] Cross-lingual ABSA and multilingual ESG contexts further motivate sentence-level analysis to capture language-specific expressions of risk or compliance indicators in Indonesian and English text [11]–[13]

2) *Gap 2: Conflation of tone and sentiment in existing approaches:* What existing methods do

Many studies treat ESG "tone" (the overall sentiment polarity of a document or section) as synonymous with "sentiment" toward ESG topics, effectively collapsing contextual cues into a single polarity score [3], [7], [9] Some traditional sentiment analyses rely on bag-of-words or lexicon-based approaches that assign a global sentiment to documents, or to broad sections, without distinguishing whether sentiment reflects a positive environmental action, neutral governance reporting, or negative social impact within a specific topic [12], [28] ABSA literature emphasizes the distinction between sentiment polarity and opinion-bearing aspects, but practice in ESG contexts often lags in consistently separating topic-level sentiment from the broader document tone, particularly when sentences mention multiple ESG topics with conflicting cues. What is missing

A clear theoretical and methodological separation between tone (global document or section-level polarity) and targeted sentiment toward discrete ESG aspects, ensuring that ABSA outputs reflect aspect-specific sentiment rather than aggregate tone. Evaluation frameworks that can detect and quantify conflation errors, such as misattributing an English-positive sentence to an environmental aspect when the sentiment is actually about governance or labor practices. Multimodal and numerical signal integration to disambiguate tone from explicit sentiment about an ESG aspect, for example when numerical performance indicators accompany otherwise neutral text. Implications for practice and theory

Conflation reduces the reliability of ABSA outputs for decision-makers who rely on precise signal attribution, such as due-diligence teams and regulators assessing material ESG risks and governance reforms. Clear separation improves model interpretability and supports explainable AI (XAI) approaches, enabling stakeholders to trace the rationale behind an aspect-level sentiment verdict. Supporting references

Systematic ESG analytics reviews emphasize challenges around explainability, interpretability, and the need to map sentiments to concrete ESG dimensions [3], [7], [10]

Cross-lingual ABSA research underscores the importance of maintaining topic-sentiment separation across languages to preserve semantic fidelity [11]–[13]

3) *Gap 3: LLM output instability across prompt templates:* What existing methods do

Large language models (LLMs) are increasingly applied to extract, summarize, and reason about ESG content via prompt-based or retrieval-augmented generation (RAG) paradigms. However, reported behavior often varies with prompt wording, template selection, and in-context examples, leading to instability in outputs across prompts or sessions [29], [30] Some deployments adopt fixed prompts or single-shot prompts, which can yield inconsistent results when applied to diverse ESG documents, especially given domain-specific vocabulary and regulatory lexicons [29], [30] There is growing recognition of the need for prompt engineering, prompt ensembles, and verification strategies to stabilize outputs and improve reproducibility [29], [30]

What is missing

Systematic studies that quantify prompt-induced variability for ESG ABSA tasks in Indonesian/English bilingual corpora, including metrics of stability (e.g., variance in aspect extraction, sentiment labeling, and regulatory mappings across templates). Robust prompt design guidelines and ensemble strategies (e.g., multiple prompts, self-consistency checks, majority voting) tailored to ESG ABSA with multilingual and legal/regulatory constraints. Integration of verification layers, such as post-hoc fact-checking against regulatory lexicons (GRI/SASB/TCFD) and cross-language alignment checks, to improve reliability of LLM outputs. Implications for practice and theory

Without addressing prompt instability, LLM-based ABSA systems risk inconsistent outputs, undermining trust, regulatory audibility, and investor confidence. Stable, hybrid architectures combining LLMs with deterministic components (rule-based lexicons, supervised ABSA models) can reduce risk while preserving the benefits of generative capabilities. Supporting references

The literature on LLMs in finance and ESG emphasizes model behavior, prompts, and the need for reliability and explainability, including considerations of bias, privacy, and regulatory compliance [31]–[33] Recent ESG document QA and RAG approaches illustrate how retrieval-augmented generation can improve accuracy and traceability, but also highlight instability risks in prompt-driven systems [29].

4) *Gap 4: Neglect of OCR quality in document-level pipelines:* What existing methods do

Many ESG NLP studies assume textual input is available as clean machine-readable text, effectively bypassing the OCR bottleneck typical of scanned or PDF-disclosed documents. In practice, several document-analysis pipelines acknowledge OCR as a precursor step but do not systematically evaluate how OCR quality (character error rate, layout extraction accuracy) propagates into downstream ABSA performance, especially in multilingual settings where scripts and diacritics differ. Some studies of scanned or semi-structured documents from other

domains (healthcare, historical archives) explicitly show that OCR quality, layout understanding, and document structure critically affect information extraction performance, suggesting similar implications for ESG reports that are often distributed as PDFs with embedded images or tables [34]–[37]

What is missing

An end-to-end evaluation of how OCR quality, layout analysis, and document structure influence sentence-level ABSA and aspect-term extraction in Indonesian/English ESG disclosures. Techniques that couple OCR refinement (e.g., layout-aware OCR, region-based extraction, post-OCR text normalization) with ABSA pipelines, including strategies to handle multilingual content, tables, footnotes, and image-based figures that contain text. Standards for reporting OCR-related uncertainty in ESG analytics, including error-aware sentiment and evidence mapping to specific document sections. Implications for practice and theory

Ignoring OCR quality leads to noise and bias in ABSA outputs, particularly in Indonesian contexts with mixed Indonesian/English text and domain-specific vocabulary. This undermines reliability for regulatory and investor use. A tightly integrated OCR-ABSA workflow with layout-aware processing improves signal fidelity and supports reproducibility, auditability, and scalability. Supporting references

Document- and OCR-centric studies illustrate the impact of OCR and layout on downstream NLP tasks, including historical document digitization and healthcare document processing where layout information significantly affects extraction quality [34]–[36], [38]–[41]

5) *Gap 5: Underdeveloped greenwashing detection methods:* What existing methods do

A subset of ESG NLP research focuses on identifying greenwashing signals by analyzing the volume, framing, and accessibility of environmental claims, sometimes using length, readability, copiousness of environmental content, or discrepancy between stated commitments and disclosed performance [12], [42], [43]

Some studies explicitly link linguistic features to greenwashing risk, finding that longer or more favorable environmental narratives do not always correlate with substantive environmental performance, implying the need for evidence-based detection pipelines that tie textual claims to measurable indicators [12] Cross-domain literature shows that detection of misleading sustainability content benefits from ABSA to locate sentiment around environmental topics and to cross-check with governance disclosures and quantified targets [9], [12] What is missing

A rigorous, formally defined greenwashing detection framework that integrates sentence- or aspect-level ABSA with verifiable evidence from numerical performance data, regulatory disclosures, and third-party verification, specifically adapted to Indonesian and English ESG reporting. Methodologies that quantify inconsistency between qualitative statements and quantitative signals (e.g., emissions reductions, energy use intensity, board oversight metrics) and that produce explainable justifications for greenwashing flags. Benchmark datasets and evaluation protocols for greenwashing detection in bilingual ESG disclosures, including annotation guidelines that capture deceptive framing, selectivity in disclosures, or omissions. Implications for practice and theory

Without robust greenwashing detection, investors and regulators face elevated risk of misinterpreting ESG commitments, potentially leading to misallocations of capital or regulatory blind spots. A multidisciplinary approach combining ABSA, factual verification, and explainable anomaly detection can strengthen trust in ESG communications and support more reliable sustainability governance. Supporting references

Studies on greenwashing detection via NLP show that longer, more positive but less readable ESG content may indicate greenwashing risk, highlighting the need for deeper, evidence-aligned analyses beyond superficial sentiment [12] ESG analytics literature emphasizes the importance of explainability and regulatory-aligned signals to counter greenwashing and enhance the credibility of ESG disclosures [3], [7], [10]

6) *Synthesis and overarching rationale:* Taken together, these five gaps reveal a common core: current ESG ABSA research tends to operate at coarse-grained, monolingual, and post-OCR stages, with limited integration of sentence-level evidence, aspect-specific polarity, robust handling of multilingual and document-structure complexities, and rigorous verification against regulatory lexicons and performance data. The Indonesian context intensifies these gaps because OJK/IDX regulatory requirements, bilingual Indonesian/English reporting practices, and domain-specific lexicons demand a tightly coupled, end-to-end ABSA pipeline that (i) extracts sentence-level ESG evidence, (ii) decouples tone from aspect sentiment, (iii) stabilizes LLM outputs across diverse prompt templates, (iv) accounts for OCR quality and document layout in end-to-end processing, and (v) incorporates greenwashing detection with explainable, regulator-aligned outputs.

The path forward, therefore, requires: (a) developing a sentence-level ABSA framework with Indonesian/English bilingual capabilities anchored to GRI/SASB/TCFD concepts; (b) designing evaluation protocols that separate tone from sentiment at the aspect level and include cross-language validation; (c) implementing prompt-robust LLM-based ABSA with prompt ensembles and post-hoc verification; (d) integrating OCR-aware preprocessing and layout-aware pipelines into document-level ABSA; and (e) constructing greenwashing detection modules that fuse ABSA signals with verifiable data and provide transparent justifications.

Transition to methodological next steps. The ensuing chapters will present a concrete research design, including annotation schemes for sentence-level ESG evidence, model architectures for multilingual ABSA with sentence-level outputs, OCR and layout-aware preprocessing pipelines, evaluation metrics for stability and explainability, and a validation plan using Indonesian corporate disclosures aligned with OJK/IDX guidance and international ESG standards.

D. Research Questions

The following six research questions (RQs) are stated to guide the development, evaluation, and deployment of an ESL-ABSA (ESG ABSA) framework tailored to Indonesian corporate disclosures in both Indonesian and English. Each question is framed to map directly to a methodological component of the study and to enable rigorous, replicable analysis. Where relevant, the questions align with the gaps and methodological imperatives identified in the literature and our research background.

1) *Supporting rationale and connections*: The six questions collectively map to a cohesive methodological pipeline: PDF-to-structure (RQ1) enables sentence- and aspect-level ABSA (RQ2); tone vs climate-target labeling (RQ3) interrogates the semantic granularity of outputs; a diagnostics framework (RQ4) ensures reliability; documentation/visualization (RQ5) guarantees reproducibility and auditability; stability analyses with ensemble strategies (RQ6) address operational reliability in real-world deployments. The reference corpus of climate- and ESG-focused NLP work supports the feasibility of sentence-level ABSA and cross-lingual analysis, including ClimateBERT variants, FinBERT, and cross-lingual ABSA approaches. Foundational studies demonstrate the necessity of domain adaptation, regulatory alignment, and explainability in ESG NLP tasks, guiding the design choices for each research question [44]–[52]. The bilingual Indonesian/English setting adds a critical layer of complexity that these questions are designed to address explicitly, ensuring that methods are not only technically sound but also practically applicable for Indonesian regulators (OJK/IDX) and market participants [1], [3], [4], [12].

2) *Notes on*: The six RQs are designed to be directly answerable via a combination of: (a) end-to-end pipeline experiments on Indonesian/English disclosures, (b) sentence-level annotation and ABSA model training, (c) cross-language model comparisons (multilingual transformers, translation-based vs. cross-lingual transfer), (d) diagnostic and evaluation frameworks for extraction errors, (e) reproducibility-oriented documentation and visualization, and (f) stability analyses across models and prompts. Representative works that inform the methodological components include ClimateBERT and its zero-shot/climate-target variants, cross-lingual ABSA surveys, ABSA taxonomies, and NLP-led ESG analyses in finance and governance contexts. Their findings provide a foundation for establishing sentence-level ESG evidence extraction, aspect-level sentiment mapping, and the need for explainable, regulator-friendly outputs in multilingual ESG text analytics [44], [47]–[52].

E. Research Objectives

To advance rigorous ESG ABSA research in multilingual Indonesian/English disclosures and to align with the research questions and pipeline design articulated in previous sections, the following five objectives are stated in precise, measurable, and action-oriented terms. Each objective directly maps to a methodological component of the intended ABSA pipeline and evidences a clear means of evaluation.

Cross-cutting notes

Language reach: All objectives are designed to operate in Indonesian and English, with explicit mechanisms for cross-language alignment, regulatory lexicon integration (GRI/SASB/TCFD), and cross-framework comparability. Regulatory alignment: The pipeline outputs will be mappable to OJK/IDX expectations where applicable, ensuring that ABSA signals can support compliance monitoring and investor communications in Indonesia. Evaluative rigor: Each objective includes concrete metrics, target sample sizes, and deliverables to enable rigorous assessment, replication, and extension in future work.

F. Research Contribution

The following six research contributions are presented as precise, technically grounded advances that map directly to the ESG ABSA research program described in prior sections. Each contribution is stated with clarity on novelty, methodological specificity, and the measurable criterion by which it will be evaluated. The focus remains on end-to-end ESG ABSA for bilingual Indonesian/English corporate disclosures, with explicit attention to the page-level, regulatory, and evidentiary needs of practitioners.

1) *Annotated bibliography (machine-readable XML)*: Note: The above references are included to illustrate the types of sources underpinning these contributions. In your finalized manuscript, please ensure that each claim and each methodological choice is supported by precise, citable sources with complete bibliographic details (DOIs, authors, venues, year). If you would like, I can expand the annotated bibliography with complete IEEE-formatted citations corresponding exactly to the six contributions and their evaluative criteria.

II. LITERATURE REVIEW

A. ESG Disclosure Theory in the Context of ABSA: Linking Voluntary/Mandatory Frameworks to the Analytical Challenge of Distinguishing Commitment from Outcome

This synthesis connects the core theoretical lenses used to justify and critique ESG disclosure with the concrete analytical challenge faced by aspect-based sentiment analysis (ABSA) in ESG disclosures: reliably distinguishing commitments from outcomes in bilingual (Indonesian/English) corporate reporting. The discussion integrates voluntary and mandatory disclosure dynamics, dominant reporting standards (GRI and SASB), stakeholder theory (Freeman, 1984), and legitimacy theory, and then maps these theories to the ABSA task of separating commitment statements from outcome disclosures in sentiment signals.

1) *Voluntary vs mandatory disclosure frameworks and the regulatory trend toward mandatory ESG reporting*: Core idea and theoretical framing. Voluntary ESG disclosures have historically allowed firms to signal governance intentions and social responsibility without binding regulatory consequences. However, a rising body of regulatory scholarship emphasizes a shift toward mandatory ESG reporting to reduce information asymmetry, strengthen market integrity, and curb opportunistic signaling. This regulatory trend is argued to mitigate greenwashing risks by elevating baseline disclosures and standardizing content across firms and jurisdictions [40], [41], [53], [54]. The literature notes that mandatory regimes may still entail concerns about superficial compliance, hence the need for robust analytical tools to verify material disclosures against regulatory mandates and verifiable performance data [55]–[57]

Link to ABSA and the commitment/outcome distinction. In a mandatory regime, the containment of disclosures increases the salience of evidence-bearing sentences that indicate genuine outcomes rather than aspirational commitments. ABSA must therefore differentiate sentences that express commitment (e.g., “we will reduce emissions”) from sentences that assert actual outcomes (e.g., “emissions reduced by 15%”) to avoid misinterpreting regulatory mandates as substantive performance. The literature’s emphasis on materiality and signaling under regulatory convergence underscores the necessity for sentence-level ABSA that can map sentiment to verifiable disclosures and regulatory requirements [55], [58]–[61].

Indonesian and global context. Global trends toward mandating ESG reporting resonate with Indonesian reforms oriented by OJK and IDX, where regulators encourage structured, traceable ESG disclosures in filings and public communications [1], [3], [4]. This cross-border alignment strengthens the expectation that ABSA systems must be capable of parsing both mandatory disclosures and supplementary voluntary narratives, and of distinguishing regulatory-quoted commitments from measured outcomes within Indonesian disclosures that may include bilingual Indonesian/English content [1], [3], [4], [11], [12]

Implications for ABSA design. A principled ABSA approach should incorporate a regulatory-aware ontology that identifies sentences tied to regulatory obligations, cross-references their materiality, and distinguishes indicative commitments from reported outcomes. This also motivates calibration against external indicators and targets (e.g., net-zero trajectories, GHG reductions) to strengthen the verifiability of reported outcomes.

2) *GRI and SASB standards as dominant structuring frameworks*: Core idea and theoretical framing. GRI offers a broad, stakeholder-focused view of ESG impacts, while SASB emphasizes financially material, industry-specific disclosures intended to inform investment decisions. The coexistence and partial overlap of these frameworks have been documented, with debates about materiality, interoperability, and the potential benefits of integrating GRI’s breadth with SASB’s financial-materiality orientation [53], [61]–[65]

The literature also addresses how hybrid or integrated reporting can bridge financial and non-financial disclosures, influencing what firms report and how investors interpret those reports [66], [67]

Link to ABSA and the commitment/outcome distinction. The ABSA task benefits from explicit mapping of sentences to standardized indicator sets and to regulatory concept clusters. Under GRI, sentences may express broad commitments or qualitative narratives about impacts without precise performance data; under SASB, sentences are more likely to anchor in financially material topics and observable outcomes. An ABSA system must therefore learn to disambiguate sentences that reflect aspirational commitments within a GRI-aligned narrative from sentences that report quantifiable outcomes aligned with SASB materiality. This necessitates cross-framework alignment within the ABSA taxonomy and careful sentence-level labeling that preserves the intended regulatory and investor semantics [55], [61]–[63], [65]

Indonesian context. Indonesia’s disclosures under OJK/IDX often reference multiple standards and may harmonize GRI and SASB concepts in a single document. An ABSA pipeline operating on bilingual Indonesian/English reports must accommodate cross-language terminologies and indicator mappings, ensuring that sentiment signals attach to the correct framework concept and thereby support regulatory compliance and cross-market comparability [1], [3], [4], [11], [12]

Implications for ABSA design. The ABSA system should facility a dual-maxis taxonomy that (i) anchors sentences to GRI-based materiality and stakeholder signals, and (ii) ties financially material SASB-like indicators to outcome statements and KPIs. A transparent mapping between aspects, signals, and regulatory anchors will

improve interpretability and auditability, addressing both signaling dynamics and the regulatory enforcement dimension.

3) *Stakeholder theory as a rationale for ESG disclosure (Freeman, 1984)*: Core idea and theoretical framing. Freeman's stakeholder theory posits that firms ought to consider the interests of a broad set of stakeholders beyond shareholders, including employees, customers, suppliers, regulators, communities, and the environment. This theory provides the normative justification for ESG disclosure as a mechanism to manage stakeholder expectations, legitimacy, and social license to operate. In empirical terms, it has been used to explain why firms publish ESG disclosures to signal responsiveness, reduce information asymmetry, and facilitate stakeholder engagement, thereby potentially stabilizing firm performance and reputation [57], [68]–[70] Link to ABSA and the commitment/outcome distinction. From a ABSA perspective, commitment statements can reflect stakeholders' demands and management's stated intentions to address stakeholder concerns, while outcomes reveal actual progress toward meeting those stakeholder expectations. Distinguishing commitment from outcome is thus central to measuring genuine stakeholder responsiveness versus performative signaling. ABSA that can separate stakeholder-centric commitments (e.g., "we will improve labor practices") from actual outcomes (e.g., "occupational injury rates fell by X") Implications for ABSA design. Embedding Freeman's theory into ABSA encourages explicit annotation of stakeholder focus areas and their corresponding signals, enabling analyses of whether disclosures reflect stakeholder-validated progress rather than generic or rhetorical statements. This alignment supports stakeholder governance research and helps practitioners interpret ESG narratives in terms of accountability to diverse constituencies.

4) *Legitimacy theory explaining why companies report favorably on ESG topics*: Core idea and theoretical framing. Legitimacy theory argues that firms disclose information to gain, maintain, or restore legitimacy in the eyes of audiences that constrain or enable ongoing operations, often by signaling alignment with social norms, regulatory expectations, and market ideals. This logic has been used to explain the prevalence of ESG reporting and the tendency to present favorable narratives, especially when legitimacy pressures are high or when reporting standards are evolving. Critics note potential weaknesses, such as selective disclosure and performative narratives that do not reflect actual performance,. Link to ABSA and the commitment/outcome distinction. Legitimacy-driven disclosures tend to emphasize positive commitments and aspirational statements designed to secure continued social approval. ABSA can be used to interrogate these narratives by analyzing whether commitment claims are substantiated by subsequent outcomes and performance indicators within the text, thereby detecting potential legitimacy-seeking signals that are not corroborated by concrete results. Consequently, sentence-level appraisal of commitment statements versus described outcomes is a critical test of legitimacy signaling versus substantive progress. Implications for ABSA design. A legitimacy-aware ABSA framework should incorporate detection of legitimacy strategies (e.g., broad social narratives, selective disclosure) and provision of evidence-based assessments by linking commitments to evidence of outcomes and independent verifications. This would enable analysts to distinguish legitimate progress from performative rhetoric and to identify potential greenwashing indicators embedded in seemingly positive disclosures.

5) *Integrating the theory-to-ABSA mapping: distinguishing commitment from outcome*: Analytical challenge. Across these theories, a central analytical challenge is distinguishing commitment signals (forward-looking or aspirational statements) from outcomes (past or current performance). This distinction is essential because commitments may reflect strategic intent or compliance with normative expectations, whereas outcomes provide observable evidence of impact. The ABSA task thus requires sentence-level disambiguation, robust cross-language term mappings, and the ability to tie linguistic cues to quantifiable performance indicators and regulatory criteria. Proposed integration approach. A theory-informed ABSA architecture would (i) embed a theory-aligned annotation schema that tags sentences by whether they express commitment, action, or outcome, (ii) map these tokens to GRI/SASB indicators and to stakeholder and legitimacy cues, and (iii) incorporate cross-language lexicons and regulatory anchors to ensure consistent interpretation across Indonesian/English texts. The outcome is an ABSA system that not only detects sentiment polarity at the aspect level but also classifies the epistemic modality (commitment vs outcome) that underpins each sentiment expression. Implications for research and practice. This integrated approach supports more rigorous enforcement of regulatory disclosures, more credible investor signals, and deeper academic inquiry into the dynamics of ESG communication. It aligns with the broader literature on materiality, signaling, and legitimacy while providing a technically operational framework for ABSA in multilingual ESG reporting.

6) *Concluding synthesis: theoretical grounding for ABSA in Indonesian ESG disclosures*: The convergence of voluntary/mandatory disclosure trends, dominant frameworks, and foundational theories explains why ABSA must move beyond global polarity to deliver sentence- and topic-level evidence about commitments and outcomes. By anchoring analysis in GRI/SASB, and leveraging stakeholder and legitimacy theories to interpret signaling choices, ABSA can provide granular, auditable insights into whether ESG disclosures reflect substantive progress or rhetorical positioning. The bilingual Indonesian/English reporting context intensifies

the need for cross-language alignment and regulatory mapping, ensuring that commitment/outcome distinctions hold across languages and regulatory regimes. This theoretical synthesis provides a rigorous foundation for the methodological choices in the ABSA pipeline and informs the design of annotation schemes, models, and evaluation strategies.

B. Aspect-Based Sentiment Analysis (ABSA)

Below is a consolidated, academically rigorous articulation of Aspect-Based Sentiment Analysis (ABSA) as it pertains to ESG (Environmental, Social, Governance) disclosures, with emphasis on technical methods, domain adaptations, and the challenge of distinguishing commitment from outcome. I synthesize the provided references conceptually and map them to ABSA tasks, while noting where evidence supports or contrasts claims. All claims are anchored to the referenced works in IEEE style.

1) Definition and task formulation of ABSA in ESG contexts:

a) ABSA definition and core objective: ABSA is the granular extraction of opinion about specific aspects within a text, linking an opinion-bearing expression to a target aspect and producing an aspect-level sentiment label. In ESG, this enables prosecutors of environmental, social, and governance topics to be analyzed with sentence- or clause-level precision rather than coarse document-level polarity. This focus on aspect-level interpretation aligns ABSA with the need to dissect nuanced governance and sustainability narratives in corporate disclosures [7], [10], [12], [14]. In ESG, an ABSA task typically comprises (i) aspect-term identification (e.g., emissions, board diversity, supply-chain governance), (ii) aspect-to-sentiment association (positive/negative/neutral), and (iii) optionally, a rationale or evidence mapping to document sections or regulatory indicators to support auditability [7], [12], [14]

b) Task formulation in ESG ABSA: A formal ABSA pipeline for ESG disclosures often includes: (a) pre-processing of multilingual disclosures (Indonesian/English), (b) sentence segmentation and region-level provenance tagging, (c) aspect-term extraction and disambiguation against a standardized ESG taxonomy anchored to GRI/SASB/TCFD concepts, (d) sentiment classification at the aspect level (polarity and intensity), and (e) produce explainable outputs that tie sentiment to textual evidence and regulatory mappings [7], [12], [14], [44], [49]

Where climate-specific content is central, climate-target labels (e.g., net-zero commitments) may be predicted by domain-adapted transformers (e.g., ClimateBERT variants) to complement general ABSA with climate-focused signals, enabling cross-validation of environmental claims [44], [46], [47], [52].

c) Novelty implications for Indonesian/English ESG disclosures: In bilingual Indonesian/English reports, ABSA must support cross-language alignment, regulatory lexicon integration, and multilingual term mappings to ensure consistent aspect labeling and sentiment interpretation across languages and frameworks (GRI/SASB/TCFD). This requires combining multilingual transformers with domain lexicons and ontology mappings, not only generic sentiment models [11]–[14]

2) Main ABSA approaches: lexicon-based, ML-based, and transformer-based methods:

a) Lexicon-based ABSA: Lexicon-based ABSA relies on curated sentiment lexicons and pattern rules to assign polarity to aspect terms. In ESG, domain-specific lexicons (e.g., environmental impact terms, governance-related phrases) are created to tag sentiment toward each ESG aspect. This approach offers interpretability and domain transparency but often suffers from limited coverage and brittleness to linguistic variation across languages and regulatory lexicons [71]–[73]. Limitations include poor handling of domain drift, sarcasm or hedging in corporate reporting, and inability to capture context-dependent polarity when sentences include multiple aspects or conditional statements [72]–[74]

b) ML-based ABSA: Supervised machine learning approaches (e.g., SVM, RF, neural classifiers) learn from labeled data to perform aspect-term extraction and sentiment labeling. These methods can leverage feature engineering (n-grams, POS tags, dependency constraints) and can be adapted to ESG topics by training on domain-specific corpora [74], [75]. Limitations include reliance on high-quality labeled data, vulnerability to domain shifts, and often a lack of interpretability, which is critical for regulator-facing analysis [74], [75]

c) Transformer-based ABSA: Transformer architectures (BERT, RoBERTa, FinBERT, ClimateBERT) dominate current ABSA, offering contextualized representations that support fine-grained extraction and sentiment classification. Domain-adapted models like FinBERT and ClimateBERT have demonstrated improved performance on financial and climate-related textual tasks, including sentence-level sentiment and topic alignment in ESG contexts [46], [47], [49], [52]. Multilingual and cross-lingual ABSA use models like mBERT or XLM-R with cross-language alignment or translation-based strategies to handle Indonesian/English content, enabling cross-language ABSA with maintainable performance across languages [11]–[13]. Limitations include prompt sensitivity in LLM-based ABSA, potential instability across prompts, and the need for extensive annotated corpora to achieve robust domain performance. Additionally, transformer models require explainability mechanisms to meet regulatory expectations [7], [10]

d) Hybrid and cross-cutting approaches: A practical ESG ABSA system often combines lexicon-based cues with transformer-based classifiers, using lexicons to constrain or guide attention during extraction, while transformers provide robust contextual representations for aspect-term detection and sentiment tagging. Some studies integrate regulatory lexicons with neural models to improve explainability and compliance interpretability [3], [7], [10], [12]

3) ESG domain: ABSA studies and domain adaptation:

a) ESG ABSA studies and domain relevance: There is growing work applying ABSA to ESG disclosures, with emphasis on aspect-level extraction and the ability to identify signals at the environmental, social, and governance levels. Systematic reviews emphasize that ABSA provides finer granularity than document-level sentiment and that cross-lingual ABSA is critical for non-English disclosures [7], [10], [12], [14]

Climate-domain ABSA work (e.g., climate-target detectors) demonstrates that domain-specific labels (e.g., net-zero targets) can augment ESG analysis, enabling evaluation of commitments against outcomes and enabling external validation with climate frameworks [?], [44], [46], [47]

b) Domain adaptation and multilingual considerations: Domain adaptation (fine-tuning on ESG-specific corpora) improves model alignment to environmental and governance vocabulary. Cross-lingual ABSA literature shows that multilingual transformers, translation-based approaches, and cross-language embeddings enable effective analysis across Indonesian and English texts, provided there is appropriate multilingual data and regulatory-term alignment [11]–[13]

c) Limitations in current ESG ABSA studies: Most existing work remains monolingual or English-centric, with limited attention to Indonesian contexts or to robust cross-language mappings to Indonesian regulatory lexicons and international ESG standards. There is also a gap in sentence-level evidential grounding, and how to provide regulator-friendly explanations that tie sentiment to specific regulatory topics or standards [3], [7], [10]

4) Limitations of generic sentiment analysis for formal sustainability report language:

a) Language complexity and formality: Formal sustainability reports employ technical vocabulary, hedging, risk language, and structured sections (e.g., management discussion, governance reports) that differ markedly from social media or news text. Generic sentiment tools trained on casual text fail to capture domain-appropriate sentiment cues, misclassify policy-oriented statements, and miss nuanced commitments or outcomes embedded in regulatory reporting language [76]–[78]

b) Domain-specific semantics and modality: ESG statements often incorporate modal verbs (will, intends, intends-to-achieve), quantified targets, and conditional phrases that require modality-aware sentiment interpretation. Generic polarity lexicons do not distinguish commitment versus outcome or the evidentiary strength of a claim, leading to misinterpretation of forward-looking statements as outcomes or vice versa [12], [79], [80]

c) Cross-framework and regulatory alignment: Sustainability reports map to multiple frameworks (GRI, SASB, TCFD, ISSB) and include cross-referenced indicators, making it necessary to align ABSA outputs with regulatory concepts and materiality constructs. Generic sentiment tools do not inherently integrate regulatory ontologies, making it hard to produce auditable, framework-aligned outputs [61], [62], [64], [65]

d) Cross-language and translation challenges: Indonesian/English bilingual reports introduce lexical drift, code-switching, and differing regulatory lexicons; generic tools trained on monolingual English data underperform on bilingual ESG texts without targeted cross-language adaptation and translation-aware strategies [11]–[13]

5) Why tone (commitment/action/outcome) is richer than polarity for ESG ABSA: The dimensional expansion from polarity to commitment/action/outcome adds epistemic and normative granularity. Polarity captures general sentiment (positive/negative/neutral) toward text, but fails to distinguish the nature of the stance. A sentence may express a commitment (forward-looking intent), describe an action (operational activity), or report an outcome (actual performance). These distinctions map directly to governance accountability and regulatory scrutiny, and they are essential for evaluating whether disclosures reflect progress, plans, or rhetoric. Relevance for legitimacy, signaling, and stakeholder assessment. Commitment denotes intent and alignment with stakeholder expectations; action represents implemented activities; outcome provides verifiable results. Distinguishing among these dimensions enables more precise tests of legitimacy signaling (are firms pledging progress without evidence?) and signaling to investors regarding risk and governance. ABSA that models commitment/action/outcome supports the empirical testing of Freeman’s stakeholder theory and legitimacy-based explanations for ESG disclosure behavior by revealing whether communications align with outcomes or serve impression management [?], [57], [68]–[70], [78]. Implications for regulatory compliance and auditability. Regulators and auditors require evidence of progress, not merely aspirational statements. A tone taxonomy that separates commitment, action, and outcome supports traceability from narrative to measurable KPIs and to regulatory indicators, enabling stronger verification and accountability.

6) *Synthesis: connecting ABSA to ESG theory and practice:* ABSA in ESG disclosures sits at the intersection of linguistic analysis, regulatory frameworks, and governance theory. By combining sentence-level ABSA with a robust ESG ontology and a tone-aware taxonomy (commitment/action/outcome), researchers can produce interpretable, auditable signals that support both investor decision-making and regulatory compliance. The integration of transformer-based models (e.g., ClimateBERT, FinBERT) with cross-language alignment mechanisms and explainability tools provides a practical path to scalable, domain-adapted ESG ABSA that can handle bilingual Indonesian/English disclosures, regulatory mappings, and the need to verify commitments against outcomes. Possible references from the provided corpus that inform these points include studies on domain-adapted sentiment in ESG and finance (FinBERT, ClimateBERT), cross-lingual ABSA surveys, and the necessity of explainability and regulatory alignment in ESG analytics [7], [10]–[14], [44], [46], [47], [49], [52]. Theoretical grounding draws on Freeman (stakeholder theory) and legitimacy theory to justify why firms disclose and how signals should be interpreted, linking directly to the ABSA challenge of distinguishing commitment from outcomes [57], [68]–[70], [78].

C. Ontology-Based Information Extraction for ESG ABSA

1) *Ontology-based IE as a structured alternative to bag-of-words approaches:* Conceptual shift and technical rationale. Ontology-based information extraction (IE) provides a formal, machine-readable representation of domain knowledge that encapsulates concepts, relations, and constraints. Unlike bag-of-words or flat keyword approaches, an ontology enables explicit modeling of ESG concepts (e.g., emissions, governance oversight, workforce diversity) and the relationships among them (hierarchies, part-of, associated indicators). This structure supports disambiguation, facet-level reasoning, and regulator-aligned querying, which are essential for credible ABSA in dense sustainability disclosures [10], [20], [81], [82]. Practical architecture. An ontology-driven IE pipeline typically comprises (a) a curated ESG ontology aligned to GRI/SASB/TCFD concepts, (b) a gazetteer of bilingual terms and synonyms for Indonesian/English, (c) a semantic layer that maps textual spans to ontology nodes via entity recognition and relation extraction, and (d) a reasoning component that derives higher-order signals (e.g., “emissions reduced” linked to an environmental KPI) from mapped facts. This contrasts with pure bag-of-words signals, which lack structural grounding and are vulnerable to lexical drift across languages and regulatory lexicons [81], [82].

Novelty for ESG ABSA. Integrating ontology-driven IE with ABSA enables sentence- and clause-level sentiment to be tied not only to a surface term but to an ontology node with provenance, regulatory anchors, and hierarchical context. This supports explainable ABSA outputs where a sentiment claim is anchored to a regulatory topic (GRI/SASB/TCFD) and to a specific evidence path within the document, thereby enhancing auditability and regulatory traceability [20], [82].

2) *ESG taxonomies used in practice: GRI, SASB, and TCFD, and their integration:* GRI as a stakeholder-oriented taxonomy. GRI provides broad, multi-stakeholder content covering environmental, social, and governance topics, with materiality oriented toward sustainability impacts and stakeholder engagement. In ontology terms, GRI concepts anchor high-level domains and provide cross-cutting topics that populate the upper layers of the ESG ontology. This broad coverage supports signaling about governance and social impact beyond purely financial metrics, but its indicators are often qualitative and narrative-driven, necessitating structured extraction to achieve comparability [83], [84]. SASB as a financial-materiality taxonomy. SASB emphasizes financially material ESG topics by industry, guiding disclosures that matter for investors and financial decision-making. In an ontology, SASB categories map to more granular sub-classes and indicators with explicit KPI targets, enabling ABSA to anchor sentiment to financially salient dimensions and to connect qualitative statements with quantitative performance data. The SASB focus complements GRI by increasing comparability across firms from an investment lens [83], [84]. TCFD as a climate-risk framework. TCFD provides a structured framework around governance, strategy, risk management, and metrics/targets related to climate risk. Ontology mappings can encode these four pillars and their sub-indicators, enabling sentence-level sentiment to be interpreted in the context of climate-related disclosures, scenario analysis, and transition plans. This alignment is critical for ABSA when climate-related language co-occurs with other ESG topics [69], [85]. Cross-framework integration. The literature documents ongoing efforts to harmonize GRI, SASB, TCFD, and related frameworks into integrated reporting, often via mapping tables and dialog between standard-setters. For ABSA, integrating these frameworks within a single ontology enables cross-framework topic labeling, facilitates comparability across jurisdictions, and supports regulatory analysis in Indonesia and globally [62], [65], [86]. Mapping bilingual language to ontology nodes. A central practical challenge is aligning Indonesian and English report language to the formal ontology. This requires bilingual term mappings, cross-language entity normalization, and alignment of regulatory lexicons (Indonesian OJK/IDX terminology with international ESG terms). Ontology-based IE with bilingual gazetteers and cross-lingual embeddings can bridge language gaps and preserve semantic fidelity across languages [11]–[13].

3) *How ontology paths guide aspect assignment in automated ESG analysis:* Ontology-driven aspect assignment. In ABSA, each extracted sentiment instance is associated with an ESG aspect. An ontology path provides the semantic trajectory: from text mention to a concept node (e.g., “emissions intensity” → Environmental → Emissions) and further to sub-aspects or indicators (e.g., Scope 1/2 emissions, energy efficiency). Path-based reasoning supports disambiguation when surface terms are ambiguous or multi-annotated within the same sentence, enabling precise topic assignment and richer downstream analyses [10], [81] Reasoning and constraint mechanisms. Ontologies enable logical inferences, such as recognizing that a sentence about “net-zero targets” is climate-related and should be linked to the TCFD/TCFD-aligned metrics, even if the text does not explicitly mention “TCFD.” Constraints (domain axioms) ensure consistent labeling across languages and ensure that sentiment about an environmental target maps to the corresponding KPI or regulatory anchor. This fosters consistency in ABSA outputs under multilingual and multi-framework contexts [83], [84], [86] Explainability through ontology traces. By exposing the path from a surface phrase to ontology nodes and then to regulatory anchors, the system can generate explanations such as: “The sentiment toward emissions is positive because the sentence maps to Environmental ∩ Emissions ∩ Target KPIs per SASB, with evidence in the KPI table.” Such provenance traces satisfy regulator expectations for auditability and support user trust in automated ABSA outputs [63], [87], [88] Handling multi-aspect sentences. OMS (Ontological Multi-Shift) paths allow a single sentence to map to multiple aspects if required (e.g., a sentence could discuss both emissions and supply-chain governance). The ontology provides disambiguation cues, such as co-occurring indicators and regulatory tags, enabling multi-label aspect assignment while preserving interpretability [63], [81] .

4) *Knowledge graph applications for ESG:* Knowledge graphs for ESG governance. ESG knowledge graphs integrate entities (companies, frameworks, indicators, targets), relations (owns, reportsto, measures, targetof), and evidence (sources, pages, sections). They enable semantically rich querying (e.g., “which companies report emissions reductions within SASB material topics for the energy sector?”) and support complex reasoning over regulatory alignment, materiality, and performance signals. This approach extends ABSA by providing a flexible, queryable backbone for cross-document, cross-language analytics [86]

Use cases in ABSA workflows. In a practical ABSA pipeline, a knowledge graph can: (a) store ontology-driven mappings from sentences to ESG concepts, (b) integrate numerical KPIs and targets for outcomes, (c) support explainable ABSA by enabling path-based explanations, and (d) enable cross-document linking (e.g., linking a governance claim in a narrative to a governance structure in the annual report). Graph-based reasoning supports more robust evidence retrieval and regulator-aligned reporting [86] Cross-language knowledge graphs. For bilingual Indonesian/English reports, multilingual knowledge graphs leverage cross-lingual alignment to map terms to a shared ontology, enabling consistent interpretation across languages and ensuring that regulatory anchors remain coherent when language switches occur within a document or across documents [11]–[13] Practical benefits and evaluation. Knowledge graphs improve traceability, support dynamic updates as ESG frameworks evolve, and enable scalable queries across large corpora of reports. Evaluation focuses on correctness of entity and relation extraction, completeness of ontology-path mappings, and usefulness of graph-based queries for regulators and investors [86] Notes on challenges and synthesis

Language-to-ontology mapping is the principal technical bottleneck. The bilingual Indonesian/English setting introduces lexical variability, code-switching, and region-specific regulatory lexicons, requiring robust cross-lingual alignment, lexicon curation, and continuous maintenance of multilingual synonyms. Approaches include multilingual embeddings, translation-based alignment, and ontology-informed normalization procedures to mitigate translation-induced semantic drift [11]–[13] Evaluation and governance. Ontology-based IE requires specialized evaluation protocols: concept coverage, relation extraction accuracy, path consistency, and regulatory alignment fidelity. It also requires alignment with existing standards (GRI, SASB, TCFD) to ensure practitioner acceptance and regulatory relevance [83], [84] Summary. Ontology-based IE provides a principled, scalable, and auditable route to ESG ABSA, moving beyond bag-of-words to structured semantic reasoning. By anchoring aspects to robust ESG taxonomies, guiding ABSA through ontology paths, and leveraging knowledge graphs for evidence retrieval and governance, this approach enhances interpretability, cross-language consistency, and regulatory alignment in Indonesian and international ESG disclosure analysis [10], [81], [86]

D. Transformer and ClimateBERT Models for ESG ABSA

1) *Overview: BERT and domain-adapted NLP foundations:* Foundation and rationale. BERT and its descendants provide deep bidirectional representations that capture context across long textual spans, enabling fine-grained ABSA where sentence- or clause-level sentiment is tied to explicit ESG aspects. Domain-adapted variants (e.g., FinBERT, ClimateBERT) initialize from general-purpose encoders but continue pretraining on large domain-specific corpora to inject specialized vocabulary, discourse patterns, and regulatory semantics into representations. This paradigm underpins effective ESG ABSA in multilingual, regulatory-grounded settings by aligning model priors with domain language and task structure [80], [82], [89], [90] Practical enablement

for ESG ABSA. With Transformers as a backbone, one can (i) perform aspect-term extraction and (ii) predict aspect-level sentiment conditioned on regulatory context, all while supporting cross-language transfer when combined with multilingual encoders (e.g., mBERT, XLM-R). The literature shows domain-adapted pretraining yields tangible gains over vanilla BERT in ESG/climate tasks, improving both classification accuracy and interpretability when coupled with task-specific heads or adapters [51], [89]–[91]

2) *ClimateBERT: training data, task coverage, and known strengths*: Training data foundations. ClimateBERT is a DistilRoBERTa-based model pre-trained on climate-related text drawn from climate-news articles, climate policies, and climate-relevant reports, with a substantial corpus amounting to millions of climate-focused tokens. This domain-specific pretraining yields representations attuned to climate discourse, terminology, and argumentative patterns encountered in ESG communications and policy documents [92]–[94]. ClimateBERT’s design emphasizes climate-risk signals, commitments, and disclosure language that appear in corporate reporting and regulatory materials. Task coverage and capabilities. Fine-tuning ClimateBERT has been demonstrated across multiple climate-related classification tasks, including climate-detection in paragraphs, sentiment regarding climate risks and opportunities, detection of climate commitments and actions, and classification of climate-specific disclosure categories (e.g., net-zero, reduction targets, renewable energy mentions). Extensions include net-zero target detectors (ClimateBERT-NetZero) that identify explicit net-zero or reduction targets and can be integrated with QA modules for targeted evidence extraction. These capabilities collectively support climate-focused ABSA and regulatory-aligned signal extraction in ESG contexts [13], [46], [50]. Strengths observed. Empirical studies consistently show ClimateBERT outperforms general-purpose models (e.g., vanilla BERT, uncased variants) on climate-disclosure tasks, particularly in classification accuracy, domain-relevant linguistic nuance, and efficiency of adaptation (domain-adaptive pretraining is cost-effective and yields robust gains) [13], [50], [92]–[94]. When paired with external validators (e.g., ClimateBERT-NetZero), there is enhanced capability to detect and quantify commitments and targets across climate narratives. Important caveat for ABSA. ClimateBERT is not a direct tone classifier; rather, it serves as a validation and grounding lens for climate-related signals. It excels at detecting climate-relevant content and targets but does not inherently provide a complete tone or aspect-sentiment mapping without additional ABSA heads or ontology-based integration. This distinction is crucial when interpreting outputs in a broader ESG ABSA pipeline [13], [46], [50].

3) *Other domain-adapted models: FinBERT and ESG-BERT*: FinBERT and ESG-specific variants. FinBERT is a BERT-based model pre-trained on financial-domain text, optimized for sentiment analysis within financial narratives. Its strength lies in handling financial-lexicon, investor-oriented expressions, and context where ESG signals intersect with financial analysis. ESG-BERT variants extend this approach to ESG-specific corpora, aiming to capture domain-specific vocabulary and discourse patterns in environmental, social, and governance reporting. Empirical work indicates domain-adapted models provide superior sentiment discrimination in ESG-related content compared to general-purpose baselines, especially when restricted labeled data exist [14], [89]. Practical utility for ESG ABSA. FinBERT- and ESG-BERT-inspired architectures can be integrated into ABSA pipelines as specialized componentry: (i) an aspect-term extractor trained or fine-tuned on ESG taxonomies, (ii) a sentiment classifier leveraging finance- or ESG-tuned encodings, and (iii) a regulatory anchor linkage module to map outputs to GRI/SASB/TCFD concepts. The domain alignment improves performance on aspect-specific sentiment, particularly for financially material ESG topics, compared with vanilla BERT baselines [14], [89]. Limitations to consider. Domain-adapted models are data-hungry and may overfit to their training corpora if not carefully regularized or if cross-domain transfer is attempted. Additionally, while they improve Sentiment classification in their target domains, they require careful alignment with ESG-specific taxonomies and regulatory ontologies to ensure interpretability and auditability in bilingual Indonesian/English contexts [83], [84], [86].

4) *Limitations of ClimateBERT for broader ESG analysis*: Narrow domain specialization. ClimateBERT is optimized for climate-related content and the 14 or so Climate/TCFD categories it has been tuned to; its coverage of broader ESG domains (social governance, labor relations, diversity, governance processes beyond climate) is comparatively limited. For comprehensive ESG ABSA, reliance on ClimateBERT alone may miss non-climate environmental topics (e.g., biodiversity, supply chain labor standards, governance risk) unless augmented by additional domain-adapted or multi-task models [13], [50], [94]. Transfer and cross-language constraints. ClimateBERT’s in-domain pretraining is climate-centric and primarily English-language data or translation-augmented corpora; applying ClimateBERT directly to Indonesian-language sustainability disclosures may degrade performance unless addressed with multilingual adapters, cross-lingual fine-tuning, or explicit Indonesian-language climate datasets. Cross-language alignment is essential for bilingual ESG ABSA but is not guaranteed by ClimateBERT’s native pretraining alone [11]–[13]. Limitations as a validation lens, not a sole classifier. The literature clarifies that ClimateBERT is not a turnkey tone classifier; it functions best as a validation or targeting tool for climate-specific content and commitments. For ABSA pipelines requiring aspect-level sentiment, ClimateBERT should be integrated with ontology-guided ABSA components and

domain-specific taxonomies to yield interpretable, regulator-friendly outputs rather than stand-alone sentiment scores [46], [50] Considerations for governance and regulatory signaling. ClimateBERT’s climate-centric focus may underrepresent governance or social aspects critical to Indonesian OJK/IDX reporting and SASB-like requirements. Consequently, ABSA pipelines must incorporate broader ESG ontologies and cross-framework mappings (GRI/SASB/TCFD) to ensure that non-climate disclosures are properly captured and interpreted [83], [84], [86].

5) *Practical recommendations for ESG ABSA pipelines incorporating transformer and ClimateBERT systems:* Hybrid architecture strategy. Combine climate-focused validators (ClimateBERT or ClimateBERT-NetZero) with broader ESG ABSA modules leveraging domain-adapted models (FinBERT/ESG-BERT) and a strong ontology-based information extraction layer. This hybrid approach allows climate commitments/targets to be cross-validated while maintaining coverage of broader ESG topics through ontology paths and ABSA heads. Multilingual and cross-framework integration. For Indonesian/English bilingual disclosures, deploy multilingual transformers (e.g., XLM-R, mBERT) with domain adapters and cross-language alignment mechanisms to maintain consistent aspect labeling and climate signals across languages. Ontology-based mappings should connect climate signals to TCFD and SASB indicators and align Indonesian terms with international ESG vocabulary [11]–[13], [69] Explainability and auditability. Leverage ClimateBERT’s outputs as a validation lens and fuse them with ontology-based explanations (e.g., path-based reasoning from climate mentions to net-zero targets) and lexical triggers to satisfy regulator demands for interpretability. Use ABSA-specific explainability tools (SHAP/LIME) alongside ontology-path visualizations to provide transparent audit trails [63], [88] Evaluation strategy. Employ cross-domain benchmarks that test climate signal detection, net-zero/target extraction, and broader ESG aspect sentiment; perform ablation studies to isolate the contributions of ClimateBERT vs domain-adapted ESG models; and validate outputs against regulatory lexicons and target KPIs to ensure policy alignment and practical utility [13], [46], [50].

6) *Summary: positioning ClimateBERT within a robust ESG ABSA ecosystem:* ClimateBERT serves as a strong, validated lens for climate-related content within ESG ABSA pipelines, offering domain-specific detection of environmental claims, climate risk discourse, and net-zero/renewable indicators. However, for holistic ESG ABSA—encompassing social and governance dimensions as well as cross-language Indonesian/English disclosures—ClimateBERT should operate as a complementary validator rather than a standalone solution. Integrating ClimateBERT with transformer-based general ABSA models, ontology-driven information extraction, and knowledge-graph-backed explainability yields a scalable, regulator-aligned framework capable of nuanced aspect-level sentiment analysis across the full ESG spectrum, while preserving cross-language robustness and interpretability [13], [46], [50], [86], [88]

E. LLM-Based Structured Extraction for ESG ABSA

1) *Prompt engineering for structured JSON output from LLMs:* Objective and design principles. The goal is to elicit highly structured, machine-parseable JSON from LLMs that encode sentence-level ESG aspect, sentiment, and provenance in a regulator-friendly schema. Prompt templates should enforce strict schema conformity, explicit field typing, and deterministic formatting to minimize downstream parsing errors. We leverage directive prompts, system roles, and constraint-enforcing prompts to bias the model toward complete, schema-compliant outputs while preserving flexibility for bilingual data [95]–[97]. Template architecture. We operationalize seven prompt templates (Template 1–Template 7) that vary in prompting strategy (instruction, example-based, chain-of-thought augmentation, and retrieval-augmented prompts). Each template targets a fixed JSON schema with fields such as documentid, language, sentenceid, section, ESGpillar, aspectterm, sentiment-label, sentiment-score, evidencespan (start/end character indices), regulatorymapping (GRI/SASB/TCFD), and provenance (source_page, figure/table/footnote). This modular design enables ablation studies and cross-template diagnostics. Prompt constraints and validation. Prompts embed validation hooks: after generation, a lightweight verifier checks JSON validity (JSON schema conformance), required fields presence, allowed category sets (e.g., E/S/G pillars, SASB categories), and non-negativity for confidence scores. If a mismatch occurs, the system can trigger template fallback or re-prompt with a corrected exemplar, reducing schema drift and missing fields [98], [99] Language and bilingual considerations. For Indonesian/English disclosures, prompts include language-tagging and explicit mapping cues between Indonesian terms and English equivalents, ensuring cross-language alignment within the JSON, and enabling downstream cross-language ABSA tasks. This aligns with cross-lingual ABSA literature indicating predictable gains when multilingual prompting is coupled with formal schema enforcement [11]–[13]

Evaluation metrics. We evaluate the prompt outputs on (i) syntactic validity (percentage of outputs that parse as valid JSON), (ii) schema completeness (fraction of required fields populated), (iii) semantic correctness (alignment of aspectterm and ESGpillar with the sentence content), and (iv) provenance fidelity (accuracy of evidencespan and regulatorymapping). A controlled corpus with ground-truth JSON annotations under both Indonesian and English is recommended for benchmarking.

2) *Zero-shot, few-shot, and chain-of-thought prompting strategies and their tradeoffs*: Zero-shot prompting. Pros: low setup cost, broad applicability to unseen documents. Cons: higher risk of hallucination, lower consistency in structured outputs, and brittle mappings to ESG taxonomies without exemplars. For ABSA, zero-shot may produce plausible yet incomplete JSON fields, requiring post hoc validation. Few-shot prompting. Pros: improved grounding via exemplar patterns, better adherence to JSON schema, and higher accuracy in aspect term and sentiment extraction when exemplars reflect target taxonomy. Cons: risk of exemplar leakage bias and potential degradation when domain drift occurs across documents; prompts may become large and expensive. Chain-of-thought prompting. Pros: reduces model miscomprehension by having the model expose intermediate reasoning steps leading to the final structured output, which can be truncated or logged for auditing. This can improve reliability in complex ABSA tasks where disambiguation is required (e.g., resolving whether a sentence expresses commitment vs. outcome). Cons: increased token usage and potential leakage of sensitive internal reasoning; not always essential for the final structured output and may complicate downstream parsing if not carefully trimmed. Tradeoffs and hybrid strategies. A pragmatic strategy often combines few-shot prompting with restrained chain-of-thought for difficult sentences. We can apply chain-of-thought prompts selectively to sentences containing hedges, modalities (will, shall, intends), or quantified targets, while using direct, template-driven outputs for straightforward sentences. Empirical work on prompt strategies in ESG contexts suggests that domain-adapted prompts and few-shot exemplars improve factual extraction without incurring excessive cost or instability [95], [96], [100], [101]. Prompt-template selection and automation. Implement an auto-prompt manager that selects among Template 1–Template 7 based on sentence features (length, presence of numeric targets, tables vs. narrative, presence of multilingual terms). The manager logs template choice, token usage, and output quality to inform adaptive strategies in production.

3) *Known instability issues in LLM-generated structured output*: Schema drift. LLMs may progressively drift away from the exact JSON schema across long sessions or when prompts are modified, yielding outputs that are syntactically valid but semantically misaligned with the required fields (e.g., missing aspectterm, incorrect regulatorymapping). Regular schema validation and template-level regression tests are essential to detect drift early [102], [103]

Missing values and partial outputs. The model may omit optional fields (evidencespan, sentiment_score) or produce placeholder values like nulls. Implementing mandatory-field enforcement within prompts and post-generation validators reduces this risk. Additionally, employing a guarded decoding strategy (e.g., strict JSON parsers with schema restoration prompts) helps recover missing fields [98], [100]

Hallucination and factual drift. In ESG contexts, hallucinations may surface as fabricated regulatory mappings or spurious targets. Grounding outputs to retrieval-augmented context and enforcing external verification against standardized taxonomies (GRI/SASB/TCFD) mitigates hallucination risk. Evidence-based prompting and external knowledge lookups are recommended for high-stakes outputs [95], [100], [101]

Prompt instability across templates. Outputs can vary across different prompt formulations due to subtle wording differences. Stability-enhancing measures include using a fixed seed for any randomness, ensemble outputs across multiple prompts with majority voting, and a post-processing reconciliation step to harmonize discordant fields [97], [100]

4) *Multi-model comparison as a diagnostic tool*: Diagnostic paradigm. Treat model diversity as a diagnostic asset: compare a base BERT-family encoder (multilingual or Indonesian-English) against climate-focused validators (ClimateBERT with ABSA heads) and against FinBERT/ESG-BERT variants. Evaluate consistency of the JSON outputs across models for the same inputs, focusing on (i) agreement on aspectterm and ESGpillar, (ii) concordance of sentimentlabel and sentiment_score, and (iii) alignment of regulatory_mapping. Metrics and experiments. Use Cohen’s kappa for categorical labels (aspectterm, ESGpillar, sentimentlabel), F1 for extraction tasks, and calibration plots for sentiment_score. Conduct ablations to determine which model combination yields highest stability and regulator-friendly interpretability, and examine error concordance to identify systematic weaknesses common to certain models or languages. Practical utility. A multi-model diagnostic workflow helps detect systematic biases, validates outputs against domain knowledge, and guides ensemble strategies (e.g., majority vote among models, or a meta-model that chooses the best-performing model per sentence type). This approach aligns with recent work on model evaluation and reliability in ESG NLP settings [91]–[93].

5) *Existing ESG extraction studies using LLMs*: Summary of empirical landscape. Recent work demonstrates LLM-based approaches for ESG extraction, including QA-style information retrieval from ESG reports, retrieval-augmented generation for evidence-grounded analysis, and prompt-driven extraction of ESG indicators from long-form reports. Notable themes include domain-adaptation (FinBERT, ClimateBERT), RAG-based grounding, and the need for explainability and auditability in regulator-facing outputs. Representative examples show LLM-driven extraction of environmental commitments, governance disclosures, and risk signals, with varying degrees of grounding and verifiability [?], [104], [105] Gaps and opportunities. While LLMs enable scalable processing of ESG documents, many studies rely on single-model prompts or lack rigorous evaluation against regulatory

frameworks. There is a need for standardized JSON schemas, multilingual ABSA pipelines, and regulator-aligned verification that integrates ontology-based taxonomies (GRI/SASB/TCFD) with LLM outputs. The six contribution areas outlined here aim to address these gaps by combining structured prompt engineering, retrieval grounding, and ontology-based interpretation. Validation and reproducibility considerations. For ESG extraction work, reproducibility requires explicit documentation of prompt templates, model versions, decoding strategies, and evaluation metrics across languages and document types. The seven prompt templates and two-model configuration proposed here are designed to facilitate reproducibility and cross-linguistic comparability while enabling robust diagnostic experimentation.

6) *The seven prompt templates and two models used in this research:* Prompt templates. Template 1 through Template 7 implement a spectrum of prompting strategies (direct instruction, few-shot exemplars, chain-of-thought augmentation, region-aware prompts, regulatory-anchor prompts, evidence-tagging prompts, and explanation-enabled prompts). Each template is designed to enforce a fixed JSON schema and to inject document-region provenance, enabling precise sentence-to-JSON mappings. Model configurations. Model A and Model B represent two distinct encoder families. Model A is a bilingual-capable transformer (e.g., XLM-Roberta or mBERT) fine-tuned on ESG-domain corpora; Model B is a climate-validated transformer (ClimateBERT-based variant) augmented with domain adapters for ESG taxonomy alignment. Trackable outputs. For each input, outputs are parsed into the standard JSON schema, with logs capturing template ID, model used, token counts, and confidence scores. The diagnostic pipeline aggregates results across prompts and models to support stability analyses and regulator-facing traceability. Grounding and reuse. Outputs are anchored to external ESG ontologies and standard frameworks (GRI/SASB/TCFD) via an ontology path, enabling explainability and cross-document traceability. The knowledge-graph-backed grounding provides a reproducible basis for evaluation across datasets and languages.

7) *Practical recommendations and future directions:* Secretariat-level governance for outputs. Maintain versioned prompts and models, monitor schema drift, and enforce strict JSON validation in production. Adopt a human-in-the-loop review for high-stakes extractions, especially those involving regulatory mappings. Language-specific adaptations. Invest in bilingual ESG lexicons and term normalization pipelines to ensure Indonesian-English consistency. Integrate cross-language embeddings and align Indonesian regulatory lexicons with international ESG vocabularies within the ontology. Evaluation protocol. Establish a standardized benchmark dataset with ground-truth JSON annotations for ESG ABSA, including both Indonesian and English samples, to enable reproducible comparisons across models and prompts. Annotated bibliography and references block (machine-readable XML)

The references herein include works on prompt engineering, retrieval-grounded generation, RAG for ESG, and domain-adapted models in ESG contexts; you should populate the exact IEEE-formatted entries with precise DOIs and venue details in your manuscript. The candidate references listed in the Possible References section provide a spectrum of related methodologies for grounding LLM extractions in domain knowledge, reliability frameworks, and ABSA innovations. Notes on reproducibility and ethics

All experiments should document model versions, prompt templates, decoding configurations, evaluation metrics, and languages. Where possible, share prompt templates, JSON schemas, and ground-truth corpora under appropriate licensing to enable independent replication. Given ESG data sensitivity and regulatory compliance considerations, ensure any disclosure data used in experiments respects propriety and privacy constraints. End of section.

F. Explainable AI and Cross-Model Comparison for ESG ABSA: A Technical Synthesis

1) *XAI methods for NLP in ESG ABSA: attention, LIME, SHAP, and TF-IDF analysis:* Attention visualization and interpretability. Modern transformer-based ABSA systems often expose attention maps or gradient-based saliency to reveal which tokens influence a given aspect-term sentiment. In ESG ABSA, attention overlays can help auditors trace which environmental, social, or governance terms drive a sentiment judgment, and whether the model is focusing on regulatory phrases or on the narrative content. However, attention alone is not sufficient for explanation; it should be complemented by post-hoc analyses to avoid misinterpreting attention weights as faithful explanations of decisions [63], [86], [88]

LIME and SHAP in ESG ABSA. Local surrogate explanations (LIME) and model-agnostic SHAP values provide feature-attribution for individual predictions, enabling practitioners to identify which words or phrases most influence aspect-level sentiment or regulatory mappings. In ESG contexts, SHAP can quantify the contribution of numeric indicators, hedges (e.g., “potentially”, “targeted”), and regulatory terms to an aspect’s sentiment label, supporting regulator-friendly audit trails. Limitations include potential instability with high-dimensional text and reliance on coaligned feature spaces when multilingual data are present [3], [7], [10]

TF-IDF coefficient analysis. When used as a lightweight baseline or as a feature augmentation, TF-IDF can illuminate which terms most strongly associate with a particular ESG aspect or sentiment class. However,

TF-IDF lacks contextual modeling and cannot capture polysemy or negation; thus it is best used in tandem with contextual embeddings, to provide complementary, human-interpretable cues and to ground explanations in salient domain vocabulary (e.g., “net-zero”, “employee safety”, “governance oversight”) [86], [87]

Practical integration. A robust XAI stack for ESG ABSA should combine: (a) intrinsic model explanations (e.g., attention-based justifications from Transformer heads or adapters), (b) post-hoc explainers (SHAP/LIME) for sentence- and token-level attributions, (c) a TF-IDF-based lexical-anchor module to surface domain-relevant cues, and (d) ontology-path visualizations to map explanations to GRI/SASB/TCFD nodes. This multi-layered approach supports regulator-facing transparency and actionable insights for investors while mitigating reliance on a single attribution mechanism [10], [63], [88]

2) Disagreement analysis as a diagnostic technique: Concept and purpose. Disagreement analysis systematically compares outputs across models, prompts, languages, or data subsets to identify where the systems diverge. In ESG ABSA, disagreement can reveal ambiguities in aspect-term extraction, polarity assignment, or regulatory mappings, highlighting areas where ontology alignment or cross-language normalization is needed [91], [92].

Operationalization. A diagnostic workflow collects per-sentence outputs from multiple ABSA pipelines (e.g., base multilingual BERT, climate-adapted ClimateBERT variants, FinBERT/ESG-BERT), then computes pairwise agreement statistics (Cohen’s kappa for category labels, Krippendorff’s alpha for multi-labels), and analyzes discordant cases qualitatively to diagnose failure modes (e.g., hedging, code-switching, ambiguous governance references). This approach supports iterative model refinement and robust ensemble design [91]–[93]

Application to regulatory alignment. Disagreement analysis helps uncover when one model maps an ESG sentence to a regulatory topic (GRI vs SASB) differently than another, signaling potential ontology gaps or terminology mismatches. It also surfaces language-specific disagreements in bilingual corpora, informing targeted lexicon augmentation and cross-language embedding alignment [11]–[13]

3) Feature importance in ESG classification: Model-internal vs model-agnostic features. For transformer-based ABSA, feature importance can be proxied via attention patterns, gradient-based saliency, and SHAP values. In ESG applications, these explanations identify which environmental terms, governance phrases, or numeric indicators most strongly drive aspect-level sentiment. Combining model-internal signals with SHAP across languages yields a richer picture of what the model relies on for decisions [46], [63], [88]

Domain-specific significance. ESG tasks benefit from highlighting features tied to material indicators (e.g., emissions metrics, board oversight language, diversity commitments). A domain-adapted feature importance analysis guides lexicon refinement, ontology augmentation, and alignment of extracted signals with SASB/TCFD targets, improving both accuracy and interpretability for stakeholders [83], [84], [86]

Practical use in governance. Transparent feature importance supports governance-by-explanation: regulators can verify that sentiment judgments hinge on verifiable ESG content rather than rhetorical framing. Investors can assess whether decisions rely on concrete KPIs or on aspirational language, facilitating more reliable due diligence and risk assessment [57], [69], [70]

4) Interpretability requirements for ESG disclosure analysis in regulatory and investment contexts: Regulatory traceability. ESG ABSA outputs must be auditable: each sentiment decision should be linkable to a textual cite (sentence or table cell), an ESG aspect, a regulatory concept (GRI/SASB/TCFD), and a provenance trail (document region, page, sentence ID). Ontology paths and knowledge graphs provide the required traceability for regulatory review and audit, aligning linguistic signals with standardized disclosure concepts [63], [86], [88]

Explainable AI for finance and compliance. Regulators demand transparency about how models derive disclosures, especially given the risk of greenwashing or misinterpretation. The explainability stack should include (i) local explanations for individual extractions, (ii) global explanations showing model behavior across subjects and languages, and (iii) scenario analyses demonstrating how outcomes would be inferred under different regulatory templates or language variants [3], [7], [10]

Language and cross-framework governance. ESG disclosure analysis spans multiple frameworks (GRI, SASB, TCFD) and bilingual corpora. Interpretability requirements must address cross-language term alignment, ontology-to-framework mappings, and the ability to present regulator-facing narratives that justify classifications under diverse regulatory regimes. This demands a robust ontology, multilingual embeddings, and transparent mapping rules between indicators and frameworks [61], [62], [83], [84], [86]

Investor-facing clarity. For investment contexts, interpretability translates into the ability to communicate signals with confidence, linking sentiment toward specific ESG aspects to potential risk or opportunity, target attainment (e.g., net-zero milestones), and governance strength. XAI outputs should support explainable dashboards that reveal which ESG topics most influence investment signals and how these signals relate to reported KPIs and targets [46], [47], [52], [93]

5) Possible references and operational anchors: Foundational LLM explainability and ESG NLP. Works on LIME/SHAP in finance and ESG contexts, attention visualization in ABSA, and the utility and limitations of

model-specific explanations provide methodological grounding for XAI in ESG ABSA. See converging literature on regulator-friendly explainability and cross-language ABSA for domain-specific adaptation [3], [7], [10]–[14]

Domain adaptation and cross-domain diagnostics. Domain-adapted LLMs (FinBERT, ClimateBERT) and cross-language ABSA studies offer guidance on multi-model diagnostics, disagreement analysis, and ontology-grounded explanation in ESG contexts. These references inform both technical design and evaluation protocols for explainable cross-model comparisons [44], [46], [47], [52], [89], [92]

6) *Summary: toward an explainable, auditable ESG ABSA diagnostics ecosystem:* The confluence of XAI methods, disagreement analysis, feature-importance explainability, and regulatory interpretability yields a robust framework for ESG ABSA that can support both investors and regulators. A cross-model diagnostic pipeline—augmented with ontology-grounded explanations, provenance-rich JSON outputs, and multilingual interpretability—enables credible, auditable insight extraction from bilingual ESG disclosures. The inclusion of a regression-testing-like workflow for JSON schema stability and model outputs ensures that our explanations remain trustworthy as frameworks and language usage evolve.

References (IEEE style)

A concise, precise references list should be constructed from authoritative sources on XAI in NLP, ESG ABSA, ontology-based information extraction, and regulatory alignment. The candidates provided (e.g., ClimateBERT literature, FinBERT, cross-lingual ABSA surveys, and ABSA explainability frameworks) offer concrete foundations for the sections above. Examples to ground claims include:

[3], [7], [10]–[14], [29], [44], [46], [47], [50], [52], [63], [86], [88], [89], [92], [93], [102]–[105] and domain-specific AI governance works such as EviBound and docQA-based ESG extraction. Notes for implementation

If you provide specific seven prompt templates and the two models used in your project (as you referenced), I can tailor the XAI diagnostics to your exact outputs, align the explanations to your JSON schema, and specify concrete evaluation metrics (e.g., per-sentence SHAP value distributions, template-wise disagreement rates, and ontology-path coverage) for a rigorous, reproducible analysis. Also, I can convert this into an IEEE-style references section with precise DOIs once you confirm the exact candidate works you want cited.

G. Greenwashing Detection in ESG ABSA Context

1) *Definition of greenwashing in the ESG reporting context:* Core concept. Greenwashing refers to the strategic use of ESG communications to convey an impression of environmental responsibility or social stewardship that is disproportionate to, or misaligned with, actual organizational practices and measurable performance. In practice, greenwashing manifests as symbolic disclosures, selective transparency, or aspirational commitments that lack corroborating outcomes or independent verification. This definitional stance aligns with established syntheses in the literature, which emphasize the disjunction between rhetoric and action, and the strategic use of narrative to manage stakeholder perceptions and market signals [?], [12], [103], [106]. Relevance to ABSA. For aspect-based sentiment analysis, greenwashing signals are most salient when sentiment attaches to environmental or governance aspects that lack corresponding outcomes, or when positive sentiment arises from broad aspirational language rather than concrete progress. ABSA benefits from explicit taxonomy and evidence-grounding to distinguish (i) commitments signaling intent, (ii) actions describing actual steps taken, and (iii) outcomes reflecting verifiable change. This triad aligns with a forward-looking vs. retrospective lens central to greenwashing detection [?], [12], [94], [106]

2) *Key linguistic markers: vague commitments and absence of measurable outcomes:* Vague commitments. Indicators include hedged or modal language (will, aims, intends, as part of our strategy), broad phrases with non-specific targets (e.g., “improve environmental performance” without targets or baselines), and promises lacking timeframes or accountability mechanisms. In ESG ABSA, such phrases often map to a positive sentiment toward an environmental or governance topic but do not anchor to a quantified target or KPI, creating a potential greenwashing signal when not subsequently substantiated [?], [12], [106] Absence of measurable outcomes. Markers include absence of numeric KPIs, undisclosed baselines, missing targets, or statements that refer to qualitative progress without any verifiable metric. In climate/environmental discourse, the lack of quantified progress (e.g., emissions reductions, energy intensity improvements, or renewable energy uptake) is a principal red flag for greenwashing, particularly when paired with positive framing in other sections of the report [?], [12], [106] Linguistic indicators of misalignment. Cross-domain NLP work shows that greenwashing correlates with disproportionate length or readability of environmental content relative to substantive performance data, and with discrepancy between qualitative claims and quantitative indicators in tables or performance dashboards [?], [12]. ABSA systems should leverage both textual sentiment toward specific environmental claims and alignment checks against disclosed numerical targets to surface misalignment. Multilingual and cross-framework considerations. In bilingual Indonesian/English reports, markers of ambiguity may appear differently across languages, necessitating cross-language lexicons and alignment of regulatory lexicons to ensure consistent detection of vague commitments and missing metrics across language boundaries [11]–[13]

3) *How the commitment/action/outcome tone distinction relates to greenwashing risk:* The triad as a diagnostic signal. A genuine commitment often serves as a governance signal intended to mobilize resources and set expectations, whereas action indicates the execution of plans, and outcome provides evidence of impact. Greenwashing risk increases when sentiment is predominantly tied to commitments without corroborating actions or outcomes, or when outcomes exist but are obfuscated, delayed, or inadequately disclosed. Distinguishing commitment from outcome within ABSA thus directly targets the core mechanism of greenwashing—discrepancies between stated intent and realized performance [12], [103], [106]

ABSA taxonomy as an audit tool. By annotating sentences with a four-field signal—aspectterm, sentiment, tonetype (commitment—action—outcome), and regulatory_anchor (GRI/SASB/TCFD)—the ABSA pipeline can surface sentences where commitment dominates without evidence, or where outcomes are present but inadequately substantiated (e.g., “emissions reduced” without a magnitude, scope, or verification). This enables a more granular greenwashing risk score and supports regulator-oriented traceability [?], [?], [?]. Implications for finance and governance. In investor due diligence, a tone-aware ABSA that flags commitment-dominant sections lacking outcomes can inform risk assessment and engagement strategies. For regulators, such signals facilitate targeted inquiries into the credibility of environmental claims and the adequacy of disclosed targets and verification practices. This aligns with the broader literature on signaling, legitimacy, and materiality in ESG reporting, which emphasizes the need to test promises against verifiable performance data [14], [85]

4) *Existing NLP approaches to greenwashing detection:* Textual-content and readability cues. Early NLP approaches for greenwashing detection combine keyword-based heuristics, topic modeling, and readability metrics to identify over-emphasized environmental content lacking substantive data. These methods provide relative guidance on greenwashing risk but often struggle with domain drift and cross-language applicability in Indonesian/English reports [12], [106]

Supervised classification and pattern mining. Supervised approaches use labeled corpora to classify whether disclosures reflect greenwashing signals, calibrating models to detect mismatches between claims and evidence. Pattern mining identifies common linguistic footprints of greenwashing, such as over-reliance on “green” descriptors without quantitative anchors, or disproportionate emphasis on future commitments relative to current performance [12], [103], [106]

ABSA-enhanced greenwashing detection. Integrating ABSA with greenwashing signals enables aspect-level analysis: (i) identify environmental aspects (emissions, energy, supply chain) and assess sentiment toward each, (ii) link sentiment to the presence or absence of metrics, (iii) map statements to regulatory frameworks for validation, and (iv) detect inconsistencies between commitment statements and reported outcomes. This approach leverages domain ontologies (GRI/SASB/TCFD) to anchor claims in verifiable frameworks and to highlight where greenwashing risk is highest within an ESG narrative [12], [103], [106].

RAG and explainable detection. Retrieval-augmented generation and explainable AI can be used to ground greenwashing judgments in external data sources (e.g., third-party verifications, registry data) and to provide transparent justifications. This is particularly valuable in high-stakes ESG analysis where auditability and traceability matter for investor and regulatory scrutiny [?], [103], [107].

Limitations and gaps. Many studies are English-language centric and rely on limited labeled datasets, hindering generalization to multilingual, domain-specific ESG disclosures. There is a need for standardized taxonomies, cross-language greenwashing benchmarks, and reproducible evaluation pipelines that integrate ABSA with regulatory mappings to detect nuanced greenwashing in Indonesia and other markets [12], [94], [106]

5) *Positioning this thesis’s tone taxonomy as a methodological contribution to automated greenwashing analysis:* Novelty and methodological contribution. This thesis introduces a four-dimensional tone taxonomy (aspectterm, sentimentlabel, tonetype: commitment—action—outcome, regulatoryanchor) to systematically detect greenwashing signals within ESG ABSA pipelines. By explicitly modeling the epistemic modality of statements and tying those to GRI/SASB/TCFD concepts, the approach provides a principled, auditable mechanism for identifying mismatches between rhetoric and performance, rather than relying on single-dimension polarity labels. Integration with knowledge graphs and ontology-based IE. The tone taxonomy is operationalized through ontology paths and a knowledge-graph grounding layer that connects aspect-level signals to regulatory indicators and to verifiable outcomes. This enables explainable, regulator-friendly outputs that can be queried for evidence-of-progress, target attainment, and independent verification. Such integration positions the taxonomy as a core methodological advance for automated greenwashing analysis in ESG ABSA systems [10], [81], [86]

Evaluation and practical relevance. The taxonomy supports targeted evaluation: (i) precision and recall for commitment, action, and outcome labels per ESG aspect, (ii) detection of language patterns signaling greenwashing, (iii) cross-language validation in Indonesian/English disclosures, and (iv) regulatory mapping fidelity. This aligns with the needs of investors seeking decision-useful signals and regulators requiring verifiable disclosures, contributing to the theoretical and practical literature on ESG signaling and greenwashing detection

[12], [106], [108] Relevance to policy and practice. A robust, taxonomy-driven greenwashing detector informs policy debates about disclosure quality and standardization. It also supports corporate governance by highlighting where communications may over-promise versus demonstrate progress, thereby encouraging more credible ESG reporting and reducing greenwashing risk over time [14], [94], [106]

6) *Implementation guidance and research agenda:* Data and annotation. Develop bilingual ESG corpora annotated with the four dimensions (aspectterm, sentimentlabel, tonetype, regulatoryanchor) across environmental, social, and governance topics, incorporating both Indonesian and English texts. Create gold-standard signals for greenwashing instances, including explicit commitments lacking outcomes or verifiable metrics. Model design. Build ABSA pipelines with domain-adapted encoders (e.g., ClimateBERT, FinBERT, ESG-BERT), enhanced with ontology-grounded post-processing to assign tone_type and regulatory anchors. Integrate a disagreement analysis module to surface where models diverge on greenwashing claims, enabling robust ensemble strategies. Evaluation framework. Establish greenwashing-focused benchmarks with metrics such as commitment-outcome alignment accuracy, KPI-metric presence rate, and cross-language detection performance. Include scenario analyses representing varying regulatory regimes (GRI/SASB/TCFD mappings) and market contexts to test generalizability. Practical deployment. Ensure outputs are regulator-friendly, with provenance traces from sentence to evidence to regulatory mapping, and provide dashboards that illustrate where ESG narratives align or diverge from disclosed performance. This supports both internal governance and external accountability in ESG reporting.

H. Literature Gap Summary: Integrated, Explainable, Tone-Aware ESG ABSA for Bilingual Disclosures

1) *Executive framing:* Across the reviewed literatures on ESG ABSA, several persistent gaps recur in a way that directly impedes reliable, regulator-friendly analysis of Indonesian/English disclosures. A unifying thread emerges: there is a lack of integrated pipelines that (i) produce sentence- and region-grounded ESG evidence with explicit tone distinctions (commitment/action/outcome), (ii) operate bidirectionally across Indonesian and English texts with robust cross-language alignment to GRI/SASB/TCFD, (iii) provide explainable outputs suitable for both investors and regulators, and (iv) remain stable under prompt variation, document modality (PDF vs scanned), and multimodal content (tables, charts). The following synthesis articulates how these gaps cohere, why they matter for ESG ABSA, and how this thesis addresses them, setting up a natural transition to the methodology.

2) *Absence of an integrated, end-to-end, bilingual ESG ABSA pipeline with tone awareness:* What the literature shows Much work demonstrates ABSA in ESG domains in isolation: sentence-level ABSA, cross-lingual ABSA, or domain-specific ABSA (e.g., climate or finance) using climate-specific or financial-domain models. However, there is limited evidence of a fully integrated pipeline that combines sentence- and region-level ABSA with a formal tone taxonomy (commitment/action/outcome) and bilingual Indonesian/English processing, all anchored to a unified ESG ontology mapped to GRI/SASB/TCFD [7], [12], [14], [44], [46], [47], [52] What is missing A cohesive architecture that (a) converts bilingual PDFs into structured ESG records with provenance, (b) performs sentence- and region-level ABSA with aspect-to-sentinel mappings, (c) classifies tone into commitment, action, and outcome, and (d) links outputs to regulator-ready frameworks. There is also a need for end-to-end evaluation on Indonesian disclosures under OJK/IDX guidelines to demonstrate practical relevance. Why this matters for ESG ABSA Without integration, ABSA outputs remain brittle, difficult to audit, and hard to compare across firms or jurisdictions. An integrated pipeline enables consistent regulatory mapping, transparent signaling for investors, and robust traceability from textual evidence to regulatory concepts. This gap directly aligns with calls for explainable, auditable ESG analytics in finance and governance literatures [3], [7], [10], [21]–[23] Transition cue to methodology The methodology chapter will operationalize this integrated pipeline, detailing data flow from PDFs through OCR, multilingual ABSA with tone tagging, ontology-grounded mappings, and regulator-facing outputs, with explicit evaluation protocols.

3) *Tone-aware ABSA as a principled solution to commitment vs. outcome discrimination:* What the literature shows ABSA work increasingly emphasizes aspect-level sentiment, yet most studies treat sentiment as a single polarity per aspect. Few works formalize a four-dimensional signal for ESG: aspectterm, sentiment, tone (commitment/action/outcome), and regulatoryanchor. This is especially critical for ESG where forward-looking commitments must be weighed against retrospective outcomes and independent verifications [7], [12], [14], [44], [46], [47] What is missing A validated tone taxonomy for ESG ABSA that explicitly distinguishes commitment (intent), action (execution), and outcome (results), and that is tested cross-lingually in Indonesian/English contexts. There is also a lack of annotated corpora and evaluation metrics that capture tone accuracy, calibration, and the alignment of tone with measurable KPIs across GRI/SASB/TCFD indicators. Why this matters for ESG ABSA Distinguishing commitment from outcome is central to detecting greenwashing and assessing credibility of ESG disclosures. A tone-aware ABSA framework enhances auditability, regulator compliance checks, and investor due diligence by surfacing when rhetoric outpaces progress or when outcomes are inadequately

disclosed. This aligns with legitimacy and signaling theory concerns and with practical needs for materiality-driven disclosure analysis [12], [94], [103], [106] Transition cue to methodology The methodology will describe the tone taxonomy (commitment/action/outcome), annotation guidelines, and integration with the ESG ontology, including evaluation metrics for tone-classification accuracy and cross-language consistency.

4) *Cross-language (Indonesian/English) integration with regulator-aligned ontology*: What the literature shows Cross-lingual ABSA and multilingual transformer approaches demonstrate feasibility but also highlight fragility when domain-specific regulatory lexicons and materiality concepts differ across languages. Indonesian regulatory discourse (OJK/IDX) introduces unique lexicons and governance terms not always aligned with international standards, creating a gap for ontology-based alignment that can operate across languages with fidelity [1], [3], [4], [11]–[13]. What is missing A bilingual ESG ontology and ontology-guided ABSA pipeline that (a) maps Indonesian and English ESG terms to GRI/SASB/TCFD nodes, (b) supports cross-language normalization and synonym handling, and (c) maintains consistent aspect and tone labeling across languages. There is a lack of large, bilingual ESG corpora with ontology-aligned annotations to support robust cross-language transfer. Why this matters for ESG ABSA Without robust cross-language alignment, Indonesian disclosures cannot be meaningfully benchmarked against global standards or across multinational investors. A bilingual, ontology-driven approach ensures comparability and regulatory fidelity, enabling Indonesia-specific analyses to contribute to broader ESG analytics discourse [21]–[23], [62], [83], [84], [106] Transition cue to methodology The methodology chapter will specify the bilingual ontology design, Indonesian-English term mappings, cross-language annotation guidelines, and ontological alignment with GRI/SASB/TCFD.

5) *Multimodal and document-structure considerations (OCR, tables, charts, and layout)*: What the literature shows A substantial portion of ESG documents are multimodal (text with tables, charts, figures), and many studies rely on text-only extracts, ignoring the evidentiary values embedded in tables and visuals. Emerging multimodal ABSA frameworks show promise but remain underdeveloped for long, formal sustainability reports with complex layouts, data tables, and multi-language content [109]–[113] What is missing An end-to-end pipeline that preserves document structure through OCR, layout-aware extraction of tables and charts, and joint text-visual ABSA anchored to ontology nodes, enabling robust sentence-level and table-level sentiment evidence mapping. Critical gaps include handling OCR errors, table cell alignment, and cross-language layout quirks in bilingual reports. Why this matters for ESG ABSA Emails or reports with misread KPI values, misaligned table headers, or non-textual evidence can derail ABSA outputs. OCR quality directly constrains the fidelity of evidence anchoring and the subsequent sentiment and tone labeling, making it a gating factor for downstream ABSA accuracy and explainability [110], [114]–[117] Transition cue to methodology The methods chapter will outline OCR + layout-aware extraction pipelines, multimodal ABSA integration, and evaluation plans for cross-language multimodal ESG documents.

6) *Reproducibility, explainability, and regulator-facing outputs*: What the literature shows There is a growing emphasis on explainability (XAI) and reproducibility in NLP/ESG analytics, but many studies do not publish complete experimental pipelines, prompts, schemas, or ground-truth corpora required for external replication. The need for auditable outputs, especially for regulatory review and investor due diligence, is repeatedly called out [3], [7], [10], [48], [50], [51], [63], [88] What is missing A reproducible, auditable ABSA pipeline with explicit JSON schemas for outputs, seven prompt templates, two model baselines, and a publicly available bilingual ESG corpus with tone annotations. A formal evaluation protocol that captures schema stability and cross-language consistency is also lacking. Why this matters for ESG ABSA Regulators and investors require transparent methodologies with traceable evidence chains. A reproducible framework enables independent verification, confidence in model outputs, and credible governance of automated ESG analysis, critical for trust in ESG disclosures and for meeting regulatory expectations [21]–[23], [61], [62]

7) *How this thesis fills the gaps*:

a) *Positioning the study*: This thesis targets the identified gaps with an integrated, tone-aware, bilingual ESG ABSA pipeline grounded in a formal ontology aligned to GRI/SASB/TCFD. It introduces a practical tone taxonomy (commitment/action/outcome) as a core methodological contribution for automated greenwashing analysis, and embeds agreement/disagreement diagnostics, cross-model comparisons, and explainability features designed for regulator usability.

b) *The methodological core*: End-to-end pipeline: PDF/ scanned-doc → OCR/layout-aware extraction → sentence- and region-grounded ABSA with tone tagging → ontology mapping to GRI/SASB/TCFD → knowledge-graph-grounded explanations → regulator-friendly JSON outputs. Tone-aware ABSA: explicit tagging of commitment, action, and outcome signals at the sentence level, with bilingual annotation guidelines and cross-language mappings. Cross-language ontology: Indonesian-English term mappings, cross-framework alignment, and a bilingual ontology path that supports explainable ABSA outputs. Multimodal and OCR-aware processing: integration of tables and charts extraction with text, tying numeric signals to ESG aspects.

Explainability and reproducibility: seven prompt templates, two model baselines, an open schema for outputs, and a reproducibility package that enables cross-language benchmarking and regulator-facing interpretability.

c) *Expected impact*: The integrated, explainable, tone-aware pipeline will provide regulators with auditable evidence chains for greenwashing risk, enable investors to discern credible progress from rhetoric, and support the Indonesian market's move toward standardized, trustworthy ESG disclosures. This aligns with established theoretical frameworks (stakeholder theory, legitimacy theory) and with practical needs for materiality and accountability in ESG reporting.

d) *Transition to the methodology chapter*: The Literature Gap Summary naturally leads into Chapter 3 (Methodology), where the detailed pipeline design, annotation schemas, model configurations, ontology mappings, evaluation plans, and reproducibility artifacts will be described, enabling rigorous implementation and replication.

8) *References to Ground the Gap Argument*: Foundational ABSA and cross-language work, including general ABSA surveys, cross-lingual ABSA studies, and climate-focused ABSA approaches, provide the methodological scaffolding for an integrated bilingual ABSA pipeline [7], [11]–[14], [44], [46], [47], [52]. Ontology and regulatory-aligned information extraction frameworks underpin the explanatory and auditable outputs needed for governance and compliance in ESG reporting [10], [81], [83], [84], [86]. XAI and disagreement diagnostics support the need for transparent model behavior, regulator-friendly explanations, and robust evaluation across languages and modalities [63], [86], [88], [91]–[93]. OCR, multimodal document understanding and ToC/table/chart extraction literature provide the practical prerequisites for pushing ABSA beyond text-only analyses to credible, real-world ESG document processing [109]–[113], [117], [118].

Note on references

The above synthesis references the core ideas and target claims in the literature you provided and the candidate references list. In your final manuscript, please substitute precise IEEE-formatted citations with DOIs and complete bibliographic details corresponding to the exact sources you choose to rely on. If you want, I can generate a fully formatted IEEE references section aligned with this summary. Literature Gap Summary for Research Design: ESG ABSA with Tone-Aware, Multilingual, Explainable Pipelines

Executive framing Across the reviewed literature on ESG ABSA, a coherent, end-to-end, bilingual (Indonesian/English) pipeline that delivers sentence- and region-grounded ESG evidence with explicit tone distinctions (commitment, action, outcome) and regulator-aligned outputs remains elusive. The common thread is a fragmentation of approaches: many studies address either sentence-level ABSA, or cross-lingual ABSA, or climate-domain ABSA, or ethics and explainability in isolation, but few integrate all these threads into a single auditable architecture suitable for OJK/IDX and international investors. Moreover, although numerous domain-adapted models (ClimateBERT, FinBERT, ESG-BERT) show promise, there is limited evidence on how to orchestrate multiple LLMs and prompt templates in a controlled, reproducible manner for ESG disclosures that include multimodal elements (tables, figures) and non-English language, while guaranteeing explainability and regulatory traceability. This synthesis identifies how these gaps cohere and why a dedicated research design is necessary to advance robust, trustworthy ESG ABSA in practice, and it motivates the subsequent methodology chapter.

9) *Gap: Lack of an integrated, end-to-end, bilingual ESG ABSA pipeline with tone awareness*: What the literature shows Numerous works demonstrate ABSA at the sentence level or across languages, or specialize in climate/disclosure domains, yet few deliver a single, unified pipeline that (i) processes bilingual Indonesian/English sustainability reports, (ii) grounds analysis in a formal tone taxonomy (commitment, action, outcome), (iii) aligns extractions to a joint ESG ontology anchored to GRI/SASB/TCFD, and (iv) produces regulator-ready, auditable outputs end-to-end. Multiple strands exist in isolation (monolingual ABSA, cross-lingual ABSA, climate-target ABSA, and ontology-based extraction), but there is sparse evidence of seamless integration with end-to-end evaluation on Indonesian disclosures (Seow, 2025; , (Šmíd Král, 2025; , Gorovaia Makrominas, 2024; , Hang, 2025; , Heever et al., 2024; , Garrido-Merchán et al., 2023; , Hassani et al., 2025; , Schimanski et al., 2023), Kim et al., 2024). What's missing A cohesive architecture that transforms PDFs or scans into sentence- and region-grounded ESG records with provenance, performs ABSA with a four-dimensional signal (aspectterm, sentimentlabel, tonetype, regulatoryanchor), and links outputs to GRI/SASB/TCFD concepts within a bilingual ontology. There is also a lack of end-to-end evaluation on Indonesian disclosures under OJK/IDX with bilingual performance metrics and regulatory traceability. Implications Without an integrated pipeline, comparisons across firms and jurisdictions are unreliable, auditability is weak, and the ability to satisfy investor due diligence and regulator inquiries is compromised. This gap motivates a design that combines OCR, multilingual ABSA, ontology grounding, and explainable outputs in a reproducible framework.

10) *Gap: Insufficient emphasis on a formal tone taxonomy (commitment/action/outcome) as a diagnostic signal for credibility and greenwashing*: What the literature shows Most ABSA studies treat sentiment as a single polarity per aspect, with limited attention to epistemic modality or the regulatory relevance of forward-

looking commitments versus realized outcomes. Although some climate-domain work investigates targets (net-zero) and commitments, there is no widely adopted, validated tone taxonomy that explicitly discriminates commitment, action, and outcome across ESG facets in a bilingual regulatory context (Seow, 2025; , Gorovaia Makrominas, 2024; , Heever et al., 2024; , Garrido-Merchán et al., 2023; , Hassani et al., 2025; , Schimanski et al., 2023), Kim et al., 2024). What's missing A validated tone taxonomy that (i) segments statements by tone type (commitment, action, outcome) and (ii) ties tone to regulatory anchors and measurable KPIs across GRI/SASB/TCFD concepts, with bilingual annotation guidelines and robust inter-annotator agreement metrics. There is also a need for corpus resources annotated with tone types in both Indonesian and English ESG disclosures. Implications The absence of a formal tone taxonomy handicaps the detection of greenwashing and weakens auditability. Embedding a tone-aware ABSA framework enables more precise credibility assessment and supports governance signaling research aligned with stakeholder theory and legitimacy theory.

11) Gap: Insufficient cross-language, regulator-aligned ontology integration for Indonesian/English ESG disclosures: What the literature shows Cross-lingual ABSA literature demonstrates the feasibility of multilingual pipelines, but Indonesian-specific regulatory lexicons and bridging Indonesian-English ESG vocabulary to GRI/SASB/TCFD nodes remain underdeveloped. Ontology-based extraction in ESG is gaining traction, yet many efforts rely on surface lexicons rather than formal, machine-interpretable ontologies with cross-language mappings and provenance. There is a paucity of bilingual corpora annotated to ontology nodes and lack of standardized mappings to multiple frameworks (GRI, SASB, TCFD) in Indonesian contexts (Šmíd Král, 2025; , Gorovaia Makrominas, 2024; , Hang, 2025; , Kansal et al., 2024; , JIA et al., 2025; , Brauwiers Fräsincar, 2022). What's missing A bilingual ESG ontology with Indonesian-English term mappings, cross-language synonym sets, and formal alignment to GRI/SASB/TCFD indicators. A pipeline that uses this ontology to drive ABSA, ensure cross-language consistency, and maintain regulatory traceability across Indonesian and international standards. Implications Without a robust bilingual ontology, multilingual ABSA will struggle with term drift, inconsistent aspect labeling, and misalignment with regulatory requirements. The ontology is essential for explainability, cross-document comparability, and regulator-friendly outputs.

12) Gap: Underdeveloped multimodal and document-structure aware ESG ABSA (OCR, tables, charts, layout): What the literature shows ESG reports are multimodal, containing narratives, tables with KPI data, and charts. While multimodal NLP and document understanding are advancing, there is still a lack of end-to-end ESG ABSA pipelines that integrate OCR quality, table/chart extraction, and cross-language grounding within an ontology-driven ABSA framework. The literature highlights the risk that OCR errors propagate into ABSA and compromise sentiment and regulation mappings, especially in long, complex sustainability documents (Gao, 2025; , Wang, 2025; , Gu et al., 2022; , Aggarwal et al., 2023; , Li et al., 2024; , Wang et al., 2023; , Yu, 2025). What's missing An integrated pipeline that preserves document structure via page-batching, performs layout-aware OCR, extracts and aligns tables/charts with textual sentiment, and jointly reasons across modalities to produce sentence-level ABSA with regulatory tags. Evaluation protocols should quantify how OCR and multimodal extraction quality affect ABSA outcomes and how cross-language content (Indonesian/English) is handled in tables and figures. Implications Without multimodal integration, ABSA will miss or misinterpret essential evidentiary signals, undermining regulatory alignment and stakeholder trust. A multimodal ABSA design is critical for credible ESG analysis in Indonesian markets.

13) Gap: Limited reproducibility, explainability, and regulator-facing outputs in ESG ABSA with ground-truth labeling scarcity: What the literature shows While XAI techniques (LIME, SHAP, attention visualization) are increasingly applied in ESG contexts, many studies do not publish complete reproducibility artifacts (prompts, schemas, ground-truth corpora) or provide regulator-oriented explanations that join textual signals with regulatory mapping. The absence of expert-labeled ground truth in many ESG ABSA settings constrains rigorous evaluation and replication, making it difficult to compare methods or demonstrate auditability to regulators and investors (Seow, 2025; , Moreno Caminero, 2020; , Srivastava Anand, 2023; , Lin et al., 2024; , Cardoni et al., 2019; , Yang, 2025). What's missing A comprehensive reproducibility package including seven prompt templates, two baseline models, an explicit JSON schema for outputs, bilingual annotated corpora with tone labels and ontology mappings, and publicly accessible evaluation protocols. A clear plan for expert-in-the-loop labeling and incremental ground-truth generation over time is needed. Implications In the absence of ground-truth labels, the design must rely on robust diagnostics, disagreement analyses, and cross-model validation to triangulate performance, while building a scalable plan to generate expert-labeled data that will eventually anchor evaluation.

14) Gap: Experimental design and evaluation paradigms without expert-labeled ground truth: What the literature shows A number of ESG ABSA studies employ supervised learning on domain-specific labels, yet many lack expert-labeled ground truth that would serve as a gold standard for sentence-level tone and ontology mappings. This makes it difficult to quantify absolute performance and to validate models for regulatory credibility. There is also a need for cross-language evaluation metrics and for domain-aware calibration of

model confidence in bilingual settings (Seow, 2025; , Gorovaia Makrominas, 2024; , Hang, 2025; , Heever et al., 2024; , Garrido-Merchán et al., 2023; , Hassani et al., 2025; , Schimanski et al., 2023). What’s missing An experimental design that (i) acknowledges the absence of a full ground-truth, (ii) uses robust proxy metrics (disagreement analyses, calibration curves, cross-language concordance), (iii) includes a plan for expert-driven ground-truth annotation in stages, and (iv) defines a reproducibility plan that ensures future analysts can rebuild the experiments. Implications The research must embrace a flexible, exploratory design that can generate interim, task-specific ground-truth data while providing rigorous diagnostics for reliability, stability, and interpretability.

15) How this thesis positions to fill the gaps: Integrated, explainable, tone-aware bilingual ABSA: This research proposes a holistic pipeline that ingests PDFs or scans, preserves document structure via page-batching, performs sentence- and region-level ABSA with a four-field signal (aspectterm, sentimentlabel, tone-type, regulatoryanchor), and grounds outputs to a bilingual ESG ontology aligned to GRI/SASB/TCFD. It introduces a formal tone taxonomy (commitment, action, outcome) as a core methodological contribution for automated greenwashing analysis, and utilizes a knowledge-graph grounding layer to support explainability and regulator-facing outputs. Multi-model and prompt-template comparative strategy: The design explicitly incorporates multiple LLMs (e.g., generic multilingual encoders and climate/domain-adapted models) and seven prompt templates to explore variance, enable ensemble strategies, and build a robust diagnostic framework. This comparative strategy is the primary mechanism to cope with the absence of expert-labeled ground truth by triangulating signals across models and prompts. Ground-truth development plan: A staged approach is proposed to generate expert-labeled data over time, starting with bilingual sentence- and clause-level annotations for a curated corpus of Indonesian/English ESG disclosures, focusing on tone labels and ontology mappings. This plan acknowledges the current ground-truth gap and provides a concrete trajectory toward expert-validated evaluation. Reproducibility and regulator alignment: The design emphasizes reproducibility artifacts (PROMPTS, JSON schemas, model configurations, prompts/templates, evaluation scripts) and regulator-friendly explainability (ontology paths, provenance tracking, finance/regulatory mappings), ensuring outputs are auditable and actionable for investors and regulators. Transition to methodology The following methodology chapter will detail data collection, annotation schemas, the end-to-end pipeline architecture, the seven prompt templates, model configurations, ontology mapping strategies, evaluation plans (including disagreement analyses and calibration metrics), and the reproducibility package. It will also outline the ground-truth generation plan and staged expert-in-the-loop labeling protocol to move from exploratory evaluation toward a formal gold standard. References and grounding

Ground the design in ABSA literature across domains (environmental, financial, climate), cross-language ABSA, and ontology-driven information extraction; incorporate XAI approaches (LIME, SHAP, attention visualization) for regulator-facing interpretability; and anchor to governance theories (stakeholder theory, legitimacy theory) to justify tone-focused analysis and greenwashing detection. The seven prompt templates and two model configurations will be grounded in the cited works on prompt variation, RAG-augmented extraction, and domain-adapted models (ClimateBERT, FinBERT, ESG-BERT). Where possible, explicit DOIs and venues should be inserted in the final manuscript. Note: If you provide the exact seven prompt templates and the two models you plan to use, I can tailor the justification to your concrete setup and generate a polished, IEEE-formatted References section with precise DOIs and bibliographic details.

III. METHODOLOGY

A. 3.1 Research Design

1) subsection Purpose: This subsection defines the methodological foundation for the ESG disclosure tone analysis pipeline. The design follows a mixed-method research logic because the project combines document engineering, NLP model experimentation, LLM-based information extraction, and interpretive evaluation of disclosure tone. Prior work on corporate disclosure, sustainability reporting, ESG analytics, climate disclosure mining, ABSA, and machine-learning-based ESG research supports this integrated design [R003], [R006], [R007], [R011], [R012].

2) Research Positioning: The pipeline is exploratory rather than purely confirmatory. The current workspace does not yet contain a complete expert-labeled ground-truth dataset, so the research cannot honestly claim final supervised classification validity. Instead, the methodology treats model outputs, ClimateBERT-style labels, ontology mappings, and manually inspectable records as triangulated evidence. This is appropriate for an early-stage ESG text-mining study where the first objective is to build a reproducible research instrument and identify where future annotation should focus.

The methodological unit of analysis is the ESG disclosure record. A record may come from a sentence, paragraph, table fragment, or page batch. Each record stores text, ESG pillar, aspect, tone, sentiment, labels, reasoning, source target, prompt, and model metadata. This structure allows the research to move beyond document-level ESG scoring toward evidence-level interpretation.

3) *Operational Design*: The design uses four linked operations:

1. Convert sustainability reports into markdown through OCR and page-level processing. 2. Extract structured ESG records using LLM prompts. 3. Compare record-level tone against ClimateBERT-style climate labels and ABSA outputs. 4. Visualize and document the resulting patterns for auditability.

The current empirical snapshot includes 332 extracted ESG records from 110 successful ESG extraction runs, using 7 prompt templates and 2 LLM backends/models. The tone distribution is `commitment=115`, `missing=61`, `action=58`, `outcome=50`, `none=48`. These figures are used as methodological evidence of feasibility, not as final population estimates.

4) *Rationale*: A single-method design would be too narrow for this topic. Manual ESG scoring is interpretable but slow; pure LLM extraction is flexible but schema-sensitive; ClimateBERT is strong for climate-related signals but does not directly classify commitment, action, and outcome; rule-based ABSA is transparent but brittle. Combining these methods creates a defensible research design: each layer compensates for a weakness in the others.

The use of multiple prompt templates is also methodological rather than incidental. Zero-shot prompts measure direct instruction-following, few-shot prompts test whether examples stabilize schema behavior, and chain-of-thought style prompts test whether reasoning-oriented instructions improve classification detail. English and Indonesian variants reflect the bilingual nature of the reports.

5) *Validity Controls*: The design uses transparency controls: raw outputs are preserved, parsed records are saved in JSON, visualizations can be regenerated, and the Streamlit dashboard allows filtering by tone, ESG pillar, source document, and prompt. The main validity limitation is that weak labels and model-derived labels cannot replace expert annotation. Therefore, this subsection recommends treating the current pipeline as a benchmark-construction apparatus.

6) *Expected Output*: The expected output of this methodological subsection is a clear explanation of how the pipeline transforms reports into structured evidence and how that evidence can later support formal evaluation. The subsection should be read alongside `results/esg_records.json`, `results/climatebert_results.json`, and the generated visualization tables.

7) *subsection-Specific Report*: This research uses a mixed-method design because the object of study has both computational and interpretive dimensions. The computational dimension concerns OCR, JSON extraction, prompt execution, model comparison, and visualization. The interpretive dimension concerns whether ESG text communicates a commitment, an action, an outcome, or no meaningful disclosure. A purely quantitative design would understate the interpretive uncertainty of ESG language, while a purely qualitative design would not provide enough scalability for large sustainability reports.

The design is exploratory because no complete expert-labeled ground truth exists yet. Instead of forcing an artificial supervised-evaluation claim, the study uses weak labels, ClimateBERT-style semantic checks, prompt comparisons, and dashboard-based review. The purpose is to build a reproducible benchmark-construction pipeline that can later be validated through manual annotation.

The comparative use of multiple LLMs and prompt templates is central to the design. The seven prompt templates operationalize zero-shot, few-shot, and chain-of-thought styles in English and Indonesian. This makes prompt sensitivity observable rather than hidden. The two model families provide a first comparison of model behavior, although the current results should be interpreted as exploratory because model distribution is unbalanced.

B. 3.2 Pipeline Architecture

1) *subsection Purpose*: This subsection defines the methodological foundation for the ESG disclosure tone analysis pipeline. The design follows a mixed-method research logic because the project combines document engineering, NLP model experimentation, LLM-based information extraction, and interpretive evaluation of disclosure tone. Prior work on corporate disclosure, sustainability reporting, ESG analytics, climate disclosure mining, ABSA, and machine-learning-based ESG research supports this integrated design [R004], [R010], [R019], [R038], [R039].

2) *Research Positioning*: The pipeline is exploratory rather than purely confirmatory. The current workspace does not yet contain a complete expert-labeled ground-truth dataset, so the research cannot honestly claim final supervised classification validity. Instead, the methodology treats model outputs, ClimateBERT-style labels, ontology mappings, and manually inspectable records as triangulated evidence. This is appropriate for an early-stage ESG text-mining study where the first objective is to build a reproducible research instrument and identify where future annotation should focus.

The methodological unit of analysis is the ESG disclosure record. A record may come from a sentence, paragraph, table fragment, or page batch. Each record stores text, ESG pillar, aspect, tone, sentiment, labels, reasoning, source target, prompt, and model metadata. This structure allows the research to move beyond document-level ESG scoring toward evidence-level interpretation.

3) *Operational Design*: The design uses four linked operations:

1. Convert sustainability reports into markdown through OCR and page-level processing. 2. Extract structured ESG records using LLM prompts. 3. Compare record-level tone against ClimateBERT-style climate labels and ABSA outputs. 4. Visualize and document the resulting patterns for auditability.

The current empirical snapshot includes 332 extracted ESG records from 110 successful ESG extraction runs, using 7 prompt templates and 2 LLM backends/models. The tone distribution is **commitment**=115, **missing**=61, **action**=58, **outcome**=50, **none**=48. These figures are used as methodological evidence of feasibility, not as final population estimates.

4) *Rationale*: A single-method design would be too narrow for this topic. Manual ESG scoring is interpretable but slow; pure LLM extraction is flexible but schema-sensitive; ClimateBERT is strong for climate-related signals but does not directly classify commitment, action, and outcome; rule-based ABSA is transparent but brittle. Combining these methods creates a defensible research design: each layer compensates for a weakness in the others.

The use of multiple prompt templates is also methodological rather than incidental. Zero-shot prompts measure direct instruction-following, few-shot prompts test whether examples stabilize schema behavior, and chain-of-thought style prompts test whether reasoning-oriented instructions improve classification detail. English and Indonesian variants reflect the bilingual nature of the reports.

5) *Validity Controls*: The design uses transparency controls: raw outputs are preserved, parsed records are saved in JSON, visualizations can be regenerated, and the Streamlit dashboard allows filtering by tone, ESG pillar, source document, and prompt. The main validity limitation is that weak labels and model-derived labels cannot replace expert annotation. Therefore, this subsection recommends treating the current pipeline as a benchmark-construction apparatus.

6) *Expected Output*: The expected output of this methodological subsection is a clear explanation of how the pipeline transforms reports into structured evidence and how that evidence can later support formal evaluation. The subsection should be read alongside **results/esg_records.json**, **results/climatebert_results.json**, and the generated visualization tables.

7) *subsection-Specific Report*: The pipeline architecture has four practical layers. First, the OCR layer ingests PDF sustainability reports and converts them into markdown pages with page-level traceability. This layer is necessary because ESG reports are often long, visually complex, bilingual, and table-heavy. The markdown output creates a stable intermediate representation for later processing.

Second, the T1 ClimateBERT layer provides external semantic validation. It is not treated as ground truth for tone, because ClimateBERT models are generally climate relevance, commitment, TCFD, sentiment, specificity, or environmental-claim classifiers. Its role is to answer whether extracted records that are labeled as commitments or environmental claims also look climate-related under an independent model family.

Third, the T2 ABSA layer includes rule-based lexicons, a classical TF-IDF/logistic-regression baseline, and a hybrid ontology-aware transformer. This layer allows the research to compare transparent lexical rules with statistical and contextual models. Fourth, the T3 LLM extraction layer produces structured JSON records across seven prompt templates and two LLM models.

The central artifact is **results/esg_records.json**. It links source target, prompt, model, extracted records, tone, sentiment, aspect, labels, and reasoning. Downstream visualizations and documentation are built from this artifact.

C. 3.3 Data Collection and Preprocessing

1) *subsection Purpose*: This subsection defines the methodological foundation for the ESG disclosure tone analysis pipeline. The design follows a mixed-method research logic because the project combines document engineering, NLP model experimentation, LLM-based information extraction, and interpretive evaluation of disclosure tone. Prior work on corporate disclosure, sustainability reporting, ESG analytics, climate disclosure mining, ABSA, and machine-learning-based ESG research supports this integrated design [R001], [R011], [R038], [R042], [R066].

2) *Research Positioning*: The pipeline is exploratory rather than purely confirmatory. The current workspace does not yet contain a complete expert-labeled ground-truth dataset, so the research cannot honestly claim final supervised classification validity. Instead, the methodology treats model outputs, ClimateBERT-style labels, ontology mappings, and manually inspectable records as triangulated evidence. This is appropriate for an early-stage ESG text-mining study where the first objective is to build a reproducible research instrument and identify where future annotation should focus.

The methodological unit of analysis is the ESG disclosure record. A record may come from a sentence, paragraph, table fragment, or page batch. Each record stores text, ESG pillar, aspect, tone, sentiment, labels, reasoning, source target, prompt, and model metadata. This structure allows the research to move beyond document-level ESG scoring toward evidence-level interpretation.

3) *Operational Design*: The design uses four linked operations:

1. Convert sustainability reports into markdown through OCR and page-level processing. 2. Extract structured ESG records using LLM prompts. 3. Compare record-level tone against ClimateBERT-style climate labels and ABSA outputs. 4. Visualize and document the resulting patterns for auditability.

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The use of multiple prompt templates is also methodological rather than incidental. Zero-shot prompts measure direct instruction-following, few-shot prompts test whether examples stabilize schema behavior, and chain-of-thought style prompts test whether reasoning-oriented instructions improve classification detail. English and Indonesian variants reflect the bilingual nature of the reports.

5) *Validity Controls*: The design uses transparency controls: raw outputs are preserved, parsed records are saved in JSON, visualizations can be regenerated, and the Streamlit dashboard allows filtering by tone, ESG pillar, source document, and prompt. The main validity limitation is that weak labels and model-derived labels cannot replace expert annotation. Therefore, this subsection recommends treating the current pipeline as a benchmark-construction apparatus.

6) *Expected Output*:

7) *subsection-Specific Report*: The data collection process focuses on sustainability reports from Indonesian listed companies and related corporate reports. The selection is appropriate because Indonesian sustainability reports often contain both Indonesian and English text, creating a realistic bilingual ESG analysis setting.

Preprocessing follows a page-aware strategy. Reports are converted into markdown pages, then selected pages or page batches are sent to the extraction pipeline. Batching is important because LLM context windows are limited and because ESG statements often depend on surrounding subsection headings, tables, or adjacent paragraphs.

The current comparison uses 34 source targets and 39 tone-comparison runs, while the broader extraction file includes 110 successful runs and 40 source targets. Text selection emphasizes ESG-relevant pages and batches rather than whole-document blind extraction. Quality issues include OCR artifacts, table formatting noise, repeated headers/footers, bilingual duplication, and occasional fragmented sentences. These issues explain why missing tone and schema drift are treated as expected diagnostic phenomena.

D. 3.4 Model Design

1) *subsection Purpose*: This subsection defines the methodological foundation for the ESG disclosure tone analysis pipeline. The design follows a mixed-method research logic because the project combines document engineering, NLP model experimentation, LLM-based information extraction, and interpretive evaluation of disclosure tone. Prior work on corporate disclosure, sustainability reporting, ESG analytics, climate disclosure mining, ABSA, and machine-learning-based ESG research supports this integrated design [R006], [R007], [R011], [R015], [R019].

2) *Research Positioning*: The pipeline is exploratory rather than purely confirmatory. The current workspace does not yet contain a complete expert-labeled ground-truth dataset, so the research cannot honestly claim final supervised classification validity. Instead, the methodology treats model outputs, ClimateBERT-style labels, ontology mappings, and manually inspectable records as triangulated evidence. This is appropriate for an early-stage ESG text-mining study where the first objective is to build a reproducible research instrument and identify where future annotation should focus.

The methodological unit of analysis is the ESG disclosure record. A record may come from a sentence, paragraph, table fragment, or page batch. Each record stores text, ESG pillar, aspect, tone, sentiment, labels, reasoning, source target, prompt, and model metadata. This structure allows the research to move beyond document-level ESG scoring toward evidence-level interpretation.

3) *Operational Design*: The design uses four linked operations:

1. Convert sustainability reports into markdown through OCR and page-level processing. 2. Extract structured ESG records using LLM prompts. 3. Compare record-level tone against ClimateBERT-style climate labels and ABSA outputs. 4. Visualize and document the resulting patterns for auditability.

The current empirical snapshot includes 332 extracted ESG records from 110 successful ESG extraction runs, using 7 prompt templates and 2 LLM backends/models. The tone distribution is `commitment=115`,

missing=61, action=58, outcome=50, none=48. These figures are used as methodological evidence of feasibility, not as final population estimates.

4) *Rationale*: A single-method design would be too narrow for this topic. Manual ESG scoring is interpretable but slow; pure LLM extraction is flexible but schema-sensitive; ClimateBERT is strong for climate-related signals but does not directly classify commitment, action, and outcome; rule-based ABSA is transparent but brittle. Combining these methods creates a defensible research design: each layer compensates for a weakness in the others.

The use of multiple prompt templates is also methodological rather than incidental. Zero-shot prompts measure direct instruction-following, few-shot prompts test whether examples stabilize schema behavior, and chain-of-thought style prompts test whether reasoning-oriented instructions improve classification detail. English and Indonesian variants reflect the bilingual nature of the reports.

5) *Validity Controls*: The design uses transparency controls: raw outputs are preserved, parsed records are saved in JSON, visualizations can be regenerated, and the Streamlit dashboard allows filtering by tone, ESG pillar, source document, and prompt. The main validity limitation is that weak labels and model-derived labels cannot replace expert annotation. Therefore, this subsection recommends treating the current pipeline as a benchmark-construction apparatus.

6) *Expected Output*:

7) *subsection-Specific Report*: The model design contains four components. The rule-based component uses ESG lexicons to identify aspects, polarity, and tone markers. It is transparent and useful as a baseline, but it can miss paraphrases and bilingual variation.

The classical ML component uses TF-IDF features and logistic regression. It offers interpretable coefficients and a stronger statistical baseline than rules alone. Its limitation is that it depends on weak labels and may reproduce lexical bias.

The hybrid ontology-aware transformer combines sentence-level representation with ontology paths and subsection context. This design is intended to capture semantic similarity between disclosure text and ESG categories while preserving a structured ESG taxonomy.

The LLM prompting component uses zero-shot, few-shot, and chain-of-thought prompt templates in English and Indonesian. These seven templates test whether examples, reasoning instructions, and language alignment improve structured JSON extraction. The LLM component is responsible for the main `esg_records.json` output.

E. 3.5 Metrics Definition

1) *subsection Purpose*: This subsection defines the methodological foundation for the ESG disclosure tone analysis pipeline. The design follows a mixed-method research logic because the project combines document engineering, NLP model experimentation, LLM-based information extraction, and interpretive evaluation of disclosure tone. Prior work on corporate disclosure, sustainability reporting, ESG analytics, climate disclosure mining, ABSA, and machine-learning-based ESG research supports this integrated design [R012], [R026], [R027], [R033], [R056].

2) *Research Positioning*: The pipeline is exploratory rather than purely confirmatory. The current workspace does not yet contain a complete expert-labeled ground-truth dataset, so the research cannot honestly claim final supervised classification validity. Instead, the methodology treats model outputs, ClimateBERT-style labels, ontology mappings, and manually inspectable records as triangulated evidence. This is appropriate for an early-stage ESG text-mining study where the first objective is to build a reproducible research instrument and identify where future annotation should focus.

The methodological unit of analysis is the ESG disclosure record. A record may come from a sentence, paragraph, table fragment, or page batch. Each record stores text, ESG pillar, aspect, tone, sentiment, labels, reasoning, source target, prompt, and model metadata. This structure allows the research to move beyond document-level ESG scoring toward evidence-level interpretation.

3) *Operational Design*: The design uses four linked operations:

1. Convert sustainability reports into markdown through OCR and page-level processing.
2. Extract structured ESG records using LLM prompts.
3. Compare record-level tone against ClimateBERT-style climate labels and ABSA outputs.
4. Visualize and document the resulting patterns for auditability.

The current empirical snapshot includes 332 extracted ESG records from 110 successful ESG extraction runs, using 7 prompt templates and 2 LLM backends/models. The tone distribution is commitment=115, missing=61, action=58, outcome=50, none=48. These figures are used as methodological evidence of feasibility, not as final population estimates.

4) *Rationale*: A single-method design would be too narrow for this topic. Manual ESG scoring is interpretable but slow; pure LLM extraction is flexible but schema-sensitive; ClimateBERT is strong for climate-related signals but does not directly classify commitment, action, and outcome; rule-based ABSA is transparent but brittle. Combining these methods creates a defensible research design: each layer compensates for a weakness in the others.

The use of multiple prompt templates is also methodological rather than incidental. Zero-shot prompts measure direct instruction-following, few-shot prompts test whether examples stabilize schema behavior, and chain-of-thought style prompts test whether reasoning-oriented instructions improve classification detail. English and Indonesian variants reflect the bilingual nature of the reports.

5) *Validity Controls*: The design uses transparency controls: raw outputs are preserved, parsed records are saved in JSON, visualizations can be regenerated, and the Streamlit dashboard allows filtering by tone, ESG pillar, source document, and prompt. The main validity limitation is that weak labels and model-derived labels cannot replace expert annotation. Therefore, this subsection recommends treating the current pipeline as a benchmark-construction apparatus.

6) *Expected Output*:

7) *subsection-Specific Report*: Metrics are defined at five levels. OCR metrics include page coverage, successful document conversion, extracted image count, and future character/table accuracy measures. Extraction metrics include JSON parse success, record count per run, field completion rate, missing tone rate, and schema drift rate.

ABSA metrics should eventually include aspect precision, recall, and F1 once expert labels are available. At the current stage, ABSA outputs are used mainly for comparison and diagnosis. Ontology alignment is interpreted as the semantic consistency between extracted text and ESG ontology paths.

Cross-model agreement measures whether rule-based, ML, hybrid, LLM, and ClimateBERT-style outputs converge. The greenwashing index is defined heuristically as a commitment-to-outcome imbalance per document or target. A higher commitment/outcome ratio may indicate that disclosure emphasizes promises more than demonstrated results, but this requires validation against expert labels or external ESG assessments.

F. 3.6 Benchmark and Ground Truth Design

1) *subsection Purpose*: This subsection defines the methodological foundation for the ESG disclosure tone analysis pipeline. The design follows a mixed-method research logic because the project combines document engineering, NLP model experimentation, LLM-based information extraction, and interpretive evaluation of disclosure tone. Prior work on corporate disclosure, sustainability reporting, ESG analytics, climate disclosure mining, ABSA, and machine-learning-based ESG research supports this integrated design [R011], [R017], [R039], [R051], [R056].

2) *Research Positioning*: The pipeline is exploratory rather than purely confirmatory. The current workspace does not yet contain a complete expert-labeled ground-truth dataset, so the research cannot honestly claim final supervised classification validity. Instead, the methodology treats model outputs, ClimateBERT-style labels, ontology mappings, and manually inspectable records as triangulated evidence. This is appropriate for an early-stage ESG text-mining study where the first objective is to build a reproducible research instrument and identify where future annotation should focus.

The methodological unit of analysis is the ESG disclosure record. A record may come from a sentence, paragraph, table fragment, or page batch. Each record stores text, ESG pillar, aspect, tone, sentiment, labels, reasoning, source target, prompt, and model metadata. This structure allows the research to move beyond document-level ESG scoring toward evidence-level interpretation.

3) *Operational Design*: The design uses four linked operations:

1. Convert sustainability reports into markdown through OCR and page-level processing.
2. Extract structured ESG records using LLM prompts.
3. Compare record-level tone against ClimateBERT-style climate labels and ABSA outputs.
4. Visualize and document the resulting patterns for auditability.

The current empirical snapshot includes 332 extracted ESG records from 110 successful ESG extraction runs, using 7 prompt templates and 2 LLM backends/models. The tone distribution is commitment=115, missing=61, action=58, outcome=50, none=48. These figures are used as methodological evidence of feasibility, not as final population estimates.

4) *Rationale*: A single-method design would be too narrow for this topic. Manual ESG scoring is interpretable but slow; pure LLM extraction is flexible but schema-sensitive; ClimateBERT is strong for climate-related signals but does not directly classify commitment, action, and outcome; rule-based ABSA is transparent but brittle. Combining these methods creates a defensible research design: each layer compensates for a weakness in the others.

The use of multiple prompt templates is also methodological rather than incidental. Zero-shot prompts measure direct instruction-following, few-shot prompts test whether examples stabilize schema behavior, and chain-of-thought style prompts test whether reasoning-oriented instructions improve classification detail. English and Indonesian variants reflect the bilingual nature of the reports.

5) *Validity Controls*: The design uses transparency controls: raw outputs are preserved, parsed records are saved in JSON, visualizations can be regenerated, and the Streamlit dashboard allows filtering by tone, ESG pillar, source document, and prompt. The main validity limitation is that weak labels and model-derived labels cannot replace expert annotation. Therefore, this subsection recommends treating the current pipeline as a benchmark-construction apparatus.

6) *Expected Output*:

7) *subsection-Specific Report*: The benchmark design uses JSON and JSONL-style persistence so experiments can be resumed, audited, and compared. Each record should eventually contain stable identifiers, source document, page or batch reference, text, predicted fields, ground-truth fields, ClimateBERT outputs, and review notes.

The current approach relies on weak labels because the dataset has not yet been fully annotated by experts. This is acceptable for exploratory pipeline design but not sufficient for final accuracy claims. Specific limitations are already visible: 61 records have missing tone, the ClimateBERT remote sample contains only three inputs, and schema drift appears when `commitment` is written into the sentiment field.

The proposed manual annotation process should use stratified sampling by tone, ESG pillar, source document, and disagreement type. At least two annotators should label a subset so inter-annotator agreement can be measured.

IV. IMPLEMENTATION AND RESULTS

A. 4.1 System Implementation Overview

1) *subsection Purpose*: This subsection reports the implementation and empirical outputs produced by the ESG disclosure analysis pipeline. It connects the research questions to concrete artifacts, including JSON result files, CSV summaries, static charts, and the Streamlit dashboard. The implementation reflects recent literature on ESG data analytics, sustainability report mining, climate disclosure NLP, LLM-based disclosure tools, and explainable AI [R004], [R017], [R019], [R039], [R042].

2) *Implementation Context*: The implemented system is a Python and Streamlit multi-page application. It organizes source documents under **data**, processing pages under **pages**, analytical modules under **code**, prompt templates under **prompt**, experiment outputs under **results**, and documentation under **docs** and **documentation**. The central empirical artifact for this report is **results/esg_records.json**.

The current extraction file contains 110 successful ESG extraction runs and 332 extracted ESG records. The records were produced by 7 prompt templates and 2 LLM backends/models. The results demonstrate that PDF-derived sustainability disclosure can be transformed into structured ESG evidence with fields for text, aspect, labels, ESG pillar, sentiment, tone, and reasoning.

3) *Results Summary*: The tone distribution is commitment=115, missing=61, action=58, outcome=50, none=48. The ESG pillar distribution is E=179, G=121, none=27, S=4, missing=1. The most important ClimateBERT-style validation pattern is commitment with climate-commitment=91, climate-d=57, and environmental-claims=48. This alignment is valuable because it suggests that many future-oriented ESG statements were also recognized by climate-related labels as commitment-like or environmental-claim-like.

At the same time, the results reveal quality issues. There are 61 missing tone records. The sentiment field also contains schema drift, including 18 records where commitment appears as sentiment. Social-pillar records are underrepresented with only four records. These weaknesses are not failures of the research design; they are diagnostic findings that show where annotation and prompt refinement should be targeted.

4) *Visualization Artifacts*: The implementation produces five key static charts:

`tone_distribution.png` `esg_by_tone.png` `climatebert_label_by_tone.png` `aspect_by_tone_heatmap.png` `climatebert_remote_top_scores.png`

The interactive dashboard in `pages/0_9_Tone_ClimateBERT_Visualization.py` provides five tabs: Overview, Tone Comparison, ClimateBERT Runs, Records, and Documentation. The dashboard supports filtering by tone, ESG pillar, prompt, and source document.

5) *Interpretation*: The implementation confirms the feasibility of automated tone-aware ESG disclosure analysis. It also shows why visualization is essential. Aggregate counts reveal the dominance of commitment-oriented disclosure, while heatmaps reveal whether those commitments correspond to climate-related labels or appear as ambiguous/missing records. The dashboard therefore functions as both an analytical instrument and a quality-control interface.

6) *Reproducibility*: The static artifacts can be regenerated with:

The Streamlit dashboard can be launched with:

This reproducibility design supports transparent reporting and future extension.

7) *subsection-Specific Report*: The implemented system uses Python and Streamlit as a multi-page research application. The main folders are data for source and OCR-derived data, pages for active Streamlit pages, hidden_pages for experimental pages, code for analytical modules, prompt for prompt templates, results for model outputs, logs for processing logs, and documentation or docs for written outputs.

OCR is handled through the report ingestion workflow and produces markdown pages. LLM access is designed around cloud and local options, including OpenRouter and LM Studio-style workflows. The visualization dashboard contains five tabs: Overview, Tone Comparison, ClimateBERT Runs, Records, and Documentation. Reproducibility is supported by code/visualize_tone_climatebert.py, which regenerates flat CSV files, static PNG charts, and the short comparison documentation.

B. 4.2 RQ1-RQ2 Results

1) *subsection Purpose*: This subsection reports the implementation and empirical outputs produced by the ESG disclosure analysis pipeline. It connects the research questions to concrete artifacts, including JSON result files, CSV summaries, static charts, and the Streamlit dashboard. The implementation reflects recent literature on ESG data analytics, sustainability report mining, climate disclosure NLP, LLM-based disclosure tools, and explainable AI [R003], [R008], [R010], [R019], [R038].

2) *Implementation Context*: The implemented system is a Python and Streamlit multi-page application. It organizes source documents under data, processing pages under pages, analytical modules under code, prompt templates under prompt, experiment outputs under results, and documentation under docs and documentation. The central empirical artifact for this report is results/esg_records.json.

The current extraction file contains 110 successful ESG extraction runs and 332 extracted ESG records. The records were produced by 7 prompt templates and 2 LLM backends/models. The results demonstrate that PDF-derived sustainability disclosure can be transformed into structured ESG evidence with fields for text, aspect, labels, ESG pillar, sentiment, tone, and reasoning.

3) *Results Summary*: The tone distribution is commitment=115, missing=61, action=58, outcome=50, none=48. The ESG pillar distribution is E=179, G=121, none=27, S=4, missing=1. The most important ClimateBERT-style validation pattern is commitment with climate-commitment=91, climate-d=57, and environmental-claims=48. This alignment is valuable because it suggests that many future-oriented ESG statements were also recognized by climate-related labels as commitment-like or environmental-claim-like.

At the same time, the results reveal quality issues. There are 61 missing tone records. The sentiment field also contains schema drift, including 18 records where commitment appears as sentiment. Social-pillar records are underrepresented with only four records. These weaknesses are not failures of the research design; they are diagnostic findings that show where annotation and prompt refinement should be targeted.

4) *Visualization Artifacts*: The implementation produces five key static charts:

tone_distribution.png esg_by_tone.png climatebert_label_by_tone.png aspect_by_tone_heatmap.png climatebert_remote_top_scores.png

The interactive dashboard in pages/0_9_Tone_ClimateBERT_Visualization.py provides five tabs: Overview, Tone Comparison, ClimateBERT Runs, Records, and Documentation. The dashboard supports filtering by tone, ESG pillar, prompt, and source document.

5) *Interpretation*: The implementation confirms the feasibility of automated tone-aware ESG disclosure analysis. It also shows why visualization is essential. Aggregate counts reveal the dominance of commitment-oriented disclosure, while heatmaps reveal whether those commitments correspond to climate-related labels or appear as ambiguous/missing records. The dashboard therefore functions as both an analytical instrument and a quality-control interface.

6) *Reproducibility*: This reproducibility design supports transparent reporting and future extension.

7) *subsection-Specific Report*: RQ1 asks whether sustainability reports can be transformed into structured ESG evidence. The implementation demonstrates this by converting document text into extractable page batches and then into JSON records. Although OCR quality has not yet been measured with formal character error rate, the saved markdown pages and source targets provide traceability from structured records back to report context.

RQ2 asks whether ESG statements can be categorized by aspect, ESG pillar, sentiment, and tone. The current output contains 332 extracted records from 110 successful runs across seven prompt templates and two models. Prompt run counts are: data.md=20, tone_zero_shot_indonesian.md=16,

tone_chain_of_thought_english.md=16, tone_chain_of_thought_indonesian.md=15, tone_few_shot_english.md=15, tone_zero_shot_english.md=14, and tone_few_shot_indonesian.md=14. The 61 missing tone records show that extraction works but schema control needs improvement.

C. 4.3 RQ3-RQ4 Results

1) *subsection Purpose*: This subsection reports the implementation and empirical outputs produced by the ESG disclosure analysis pipeline. It connects the research questions to concrete artifacts, including JSON result files, CSV summaries, static charts, and the Streamlit dashboard. The implementation reflects recent literature on ESG data analytics, sustainability report mining, climate disclosure NLP, LLM-based disclosure tools, and explainable AI [R010], [R012], [R051], [R052], [R056].

2) *Implementation Context*: The implemented system is a Python and Streamlit multi-page application. It organizes source documents under data/, processing pages under pages/, analytical modules under code/, prompt templates under prompt/, experiment outputs under results/, and documentation under docs/ and documentation/. The central empirical artifact for this report is results/esg_records.json.

The current extraction file contains 110 successful ESG extraction runs and 332 extracted ESG records. The records were produced by 7 prompt templates and 2 LLM backends/models. The results demonstrate that PDF-derived sustainability disclosure can be transformed into structured ESG evidence with fields for text, aspect, labels, ESG pillar, sentiment, tone, and reasoning.

3) *Results Summary*: The tone distribution is commitment=115, missing=61, action=58, outcome=50, none=48. The ESG pillar distribution is E=179, G=121, none=27, S=4, missing=1. The most important ClimateBERT-style validation pattern is commitment with climate-commitment=91, climate-d=57, and environmental-claims=48. This alignment is valuable because it suggests that many future-oriented ESG statements were also recognized by climate-related labels as commitment-like or environmental-claim-like.

At the same time, the results reveal quality issues. There are 61 missing tone records. The sentiment field also contains schema drift, including 18 records where commitment appears as sentiment. Social-pillar records are underrepresented with only four records. These weaknesses are not failures of the research design; they are diagnostic findings that show where annotation and prompt refinement should be targeted.

4) *Visualization Artifacts*: The implementation produces five key static charts:

tone_distribution.png esg_by_tone.png climatebert_label_by_tone.png aspect_by_tone_heatmap.png climatebert_remote_top_scores.png

The interactive dashboard in pages/0_9_Tone_ClimateBERT_Visualization.py provides five tabs: Overview, Tone Comparison, ClimateBERT Runs, Records, and Documentation. The dashboard supports filtering by tone, ESG pillar, prompt, and source document.

5) *Interpretation*: The implementation confirms the feasibility of automated tone-aware ESG disclosure analysis. It also shows why visualization is essential. Aggregate counts reveal the dominance of commitment-oriented disclosure, while heatmaps reveal whether those commitments correspond to climate-related labels or appear as ambiguous/missing records. The dashboard therefore functions as both an analytical instrument and a quality-control interface.

6) *Reproducibility*: The static artifacts can be regenerated with:

This reproducibility design supports transparent reporting and future extension.

7) *subsection-Specific Report*: RQ3 evaluates the tone-to-ClimateBERT comparison. The strongest alignment is commitment with climate-commitment=91, climate-d=57, and environmental-claims=48. This is a meaningful validation signal because future-oriented ESG statements are being recognized by climate-related labels as commitment-like or environmental-claim-like.

Action records align most often with governance=25 and climate-d=23. Outcome records align with other=36, governance=20, and climate-d=19. None records align with none=24 and governance=21. These distributions suggest that climate and governance language require different interpretation rules.

RQ4 focuses on weakness detection. Missing tone co-occurs with climate-commitment=22 and environmental-claims=21, which suggests false negatives or schema failures. The sentiment field includes commitment in 18 records, indicating schema drift. The social pillar has only four records, indicating underrepresentation.

D. 4.4 RQ5-RQ6 Results

1) *subsection Purpose*: This subsection reports the implementation and empirical outputs produced by the ESG disclosure analysis pipeline. It connects the research questions to concrete artifacts, including JSON result files, CSV summaries, static charts, and the Streamlit dashboard. The implementation reflects recent literature on ESG data analytics, sustainability report mining, climate disclosure NLP, LLM-based disclosure tools, and explainable AI [R007], [R014], [R020], [R055], [R057].

2) *Implementation Context:* The implemented system is a Python and Streamlit multi-page application. It organizes source documents under data/, processing pages under pages/, analytical modules under code/, prompt templates under prompt/, experiment outputs under results/, and documentation under docs/ and documentation/. The central empirical artifact for this report is results/esg_records.json.

The current extraction file contains 110 successful ESG extraction runs and 332 extracted ESG records. The records were produced by 7 prompt templates and 2 LLM backends/models. The results demonstrate that PDF-derived sustainability disclosure can be transformed into structured ESG evidence with fields for text, aspect, labels, ESG pillar, sentiment, tone, and reasoning.

3) *Results Summary:* The tone distribution is commitment=115, missing=61, action=58, outcome=50, none=48. The ESG pillar distribution is E=179, G=121, none=27, S=4, missing=1. The most important ClimateBERT-style validation pattern is commitment with climate-commitment=91, climate-d=57, and environmental-claims=48. This alignment is valuable because it suggests that many future-oriented ESG statements were also recognized by climate-related labels as commitment-like or environmental-claim-like.

At the same time, the results reveal quality issues. There are 61 missing tone records. The sentiment field also contains schema drift, including 18 records where commitment appears as sentiment. Social-pillar records are underrepresented with only four records. These weaknesses are not failures of the research design; they are diagnostic findings that show where annotation and prompt refinement should be targeted.

4) *Visualization Artifacts:* The implementation produces five key static charts: tone_distribution.png esg_by_tone.png climatebert_label_by_tone.png aspect_by_tone_heatmap.png climatebert_remote_top_scores.png

The interactive dashboard in pages/0_9_Tone_ClimateBERT_Visualization.py provides five tabs: Overview, Tone Comparison, ClimateBERT Runs, Records, and Documentation. The dashboard supports filtering by tone, ESG pillar, prompt, and source document.

5) *Interpretation:* The implementation confirms the feasibility of automated tone-aware ESG disclosure analysis. It also shows why visualization is essential. Aggregate counts reveal the dominance of commitment-oriented disclosure, while heatmaps reveal whether those commitments correspond to climate-related labels or appear as ambiguous/missing records. The dashboard therefore functions as both an analytical instrument and a quality-control interface.

6) *Reproducibility:* The static artifacts can be regenerated with:

```
““bash python3 code/visualize_tone_climatebert.py ““
```

The Streamlit dashboard can be launched with:

```
““bash streamlit run app.py --server.port 8590 ““
```

This reproducibility design supports transparent reporting and future extension.

7) *subsection-Specific Report:* RQ5 concerns documentation and visualization. The implementation produces five static visualizations: tone distribution, ESG by tone, ClimateBERT label by tone, aspect by tone, and remote ClimateBERT top scores. The interactive Streamlit dashboard allows users to filter records and inspect both aggregate charts and row-level evidence.

RQ6 concerns model and prompt stability. The current run distribution is unbalanced: arcee-ai/trinity-large-preview:free contributes 90 runs, while openai/gpt-oss-120b:free contributes 20 runs. Prompt templates vary across zero-shot, few-shot, and chain-of-thought styles in English and Indonesian. This design supports sensitivity analysis, but future work should balance model and prompt runs before making strong comparative claims.

E. 4.5 Explainability Outputs

1) *subsection Purpose:* This subsection reports the implementation and empirical outputs produced by the ESG disclosure analysis pipeline. It connects the research questions to concrete artifacts, including JSON result files, CSV summaries, static charts, and the Streamlit dashboard. The implementation reflects recent literature on ESG data analytics, sustainability report mining, climate disclosure NLP, LLM-based disclosure tools, and explainable AI [R014], [R019], [R038], [R054], [R061].

2) *Implementation Context:* The implemented system is a Python and Streamlit multi-page application. It organizes source documents under data/, processing pages under pages/, analytical modules under code/, prompt templates under prompt/, experiment outputs under results/, and documentation under docs/ and documentation/. The central empirical artifact for this report is results/esg_records.json.

The current extraction file contains 110 successful ESG extraction runs and 332 extracted ESG records. The records were produced by 7 prompt templates and 2 LLM backends/models. The results demonstrate that PDF-derived sustainability disclosure can be transformed into structured ESG evidence with fields for text, aspect, labels, ESG pillar, sentiment, tone, and reasoning.

3) *Results Summary*: The tone distribution is commitment=115, missing=61, action=58, outcome=50, none=48. The ESG pillar distribution is E=179, G=121, none=27, S=4, missing=1. The most important ClimateBERT-style validation pattern is commitment with climate-commitment=91, climate-d=57, and environmental-claims=48. This alignment is valuable because it suggests that many future-oriented ESG statements were also recognized by climate-related labels as commitment-like or environmental-claim-like.

At the same time, the results reveal quality issues. There are 61 missing tone records. The sentiment field also contains schema drift, including 18 records where commitment appears as sentiment. Social-pillar records are underrepresented with only four records. These weaknesses are not failures of the research design; they are diagnostic findings that show where annotation and prompt refinement should be targeted.

4) *Visualization Artifacts*: The implementation produces five key static charts:

tone_distribution.png esg_by_tone.png climatebert_label_by_tone.png aspect_by_tone_heatmap.png climatebert_remote_top_scores.png

The interactive dashboard in pages/0_9_Tone_ClimateBERT_Visualization.py provides five tabs: Overview, Tone Comparison, ClimateBERT Runs, Records, and Documentation. The dashboard supports filtering by tone, ESG pillar, prompt, and source document.

5) *Interpretation*: The implementation confirms the feasibility of automated tone-aware ESG disclosure analysis. It also shows why visualization is essential. Aggregate counts reveal the dominance of commitment-oriented disclosure, while heatmaps reveal whether those commitments correspond to climate-related labels or appear as ambiguous/missing records. The dashboard therefore functions as both an analytical instrument and a quality-control interface.

6) *Reproducibility*: The static artifacts can be regenerated with:

```
““bash python3 code/visualize_tone_climatebert.py ““
```

The Streamlit dashboard can be launched with:

```
““bash streamlit run app.py --server.port 8590 ““
```

This reproducibility design supports transparent reporting and future extension.

7) *subsection-Specific Report*: Explainability is implemented through several mechanisms. Rule-based outputs expose lexical triggers, such as commitment markers (will, target, committed), action markers (implemented, conducted, established), and outcome markers (achieved, reduced, completed). Classical ML outputs can expose TF-IDF coefficients that show which terms contribute most strongly to aspect, sentiment, or tone predictions.

Ontology-aware outputs map records to ESG paths, helping users understand whether a statement is environmental, social, governance, or generic. subsection influence can be inspected by comparing where disclosures appear in the report. Representative examples should include text, predicted tone, ESG pillar, ClimateBERT-style labels, and reasoning so reviewers can evaluate whether the model’s classification is plausible.

V. DISCUSSION

A. 5.1 Comparison with Existing Methods

1) *subsection Purpose*: This subsection discusses the meaning of the empirical findings and compares the proposed system with existing ESG analysis methods. It draws on literature concerning corporate disclosure, sustainability reporting, ESG data analytics, ABSA, greenwashing detection, materiality, and climate disclosure analysis [R002], [R006], [R007], [R012], [R026].

2) *Discussion*: The main empirical finding is that commitment-oriented ESG disclosure dominates the current extracted dataset. Out of 332 records, 115 are classified as commitment, while 50 are classified as outcome. This imbalance matters because commitment language often signals strategic intention, legitimacy, or aspiration, whereas outcome language provides stronger evidence of realized performance.

The ClimateBERT-style comparison provides a partial validation signal. Commitment tone co-occurs with climate-commitment 91 times, with climate-d 57 times, and with environmental-claims 48 times. This pattern suggests that the extraction layer is not randomly assigning commitment labels; many commitment records also carry climate-related semantic evidence.

However, the same evidence exposes limitations. Missing tones co-occur with climate-commitment 22 times and environmental-claims 21 times. These cases are likely false negatives or schema failures because climate-relevant claims should usually have an interpretable disclosure tone. Governance disclosures also spread across action, commitment, outcome, none, and missing categories, which indicates that governance language is less easily captured by environmental-centric tone cues.

3) *Comparison With Existing Methods:* Compared with manual ESG scoring, this pipeline is more scalable and produces traceable text-level evidence, but it lacks expert judgment unless annotation is added. Compared with pure LLM classification, the pipeline is more auditable because it stores raw outputs, parsed JSON, prompt metadata, visualizations, and ClimateBERT comparison labels. Compared with ClimateBERT-only approaches, the pipeline is broader because it includes social and governance records and explicitly models disclosure tone. Compared with rule-based taxonomies, the pipeline is more flexible for bilingual and context-dependent language, while still preserving interpretability through lexicons and ontology paths.

4) *Theoretical Implications:* The findings are consistent with legitimacy-oriented interpretations of sustainability reporting: firms may use ESG disclosure to signal responsibility and future alignment even when measured outcomes are less frequent. The low number of social records also raises a stakeholder-theory question about whether selected report pages, extraction prompts, or corporate disclosure practices emphasize environmental and governance audiences more strongly than social stakeholders.

For ABSA theory, the results show why sentiment alone is insufficient. A neutral statement can still be an important commitment or outcome. For greenwashing research, the commitment/outcome imbalance suggests a measurable warning signal, but it must be validated before being treated as a definitive greenwashing index.

5) *Limitations Discussed:* The most important limitation is the absence of full expert-labeled ground truth. The ClimateBERT remote sample has only three inputs, and ClimateBERT is not a direct tone classifier. OCR quality has not been measured with character error rate or table accuracy. Prompt sensitivity remains a concern because seven prompt templates are used. Finally, the ontology may not fully cover Indonesian-specific sustainability disclosure language.

6) *Implication for the Thesis:* The discussion should position the project as an exploratory but reproducible research framework. Its main contribution is not that it has already solved ESG classification, but that it creates the infrastructure for formal benchmark construction and richer ESG disclosure interpretation.

7) *subsection-Specific Report:* Compared with manual ESG scoring, this system is more scalable and produces traceable text-level evidence, but it still requires expert validation. Compared with pure LLM classification, it is more auditable because prompts, raw outputs, parsed records, labels, and visualizations are preserved. Compared with ClimateBERT-only approaches, it covers broader ESG categories and introduces tone as a separate disclosure dimension. Compared with rule-based taxonomies, it is more flexible for bilingual and context-dependent report language.

The system adds value by integrating tone, ontology alignment, ClimateBERT-style validation, and explainability. This combination supports both exploratory research and practical review workflows.

B. 5.2 Interpretation of Key Findings

1) *subsection Purpose:* This subsection discusses the meaning of the empirical findings and compares the proposed system with existing ESG analysis methods. It draws on literature concerning corporate disclosure, sustainability reporting, ESG data analytics, ABSA, greenwashing detection, materiality, and climate disclosure analysis [R003], [R012], [R020], [R029], [R065].

2) *Discussion:* The main empirical finding is that commitment-oriented ESG disclosure dominates the current extracted dataset. Out of 332 records, 115 are classified as `commitment`, while 50 are classified as `outcome`. This imbalance matters because commitment language often signals strategic intention, legitimacy, or aspiration, whereas outcome language provides stronger evidence of realized performance.

The ClimateBERT-style comparison provides a partial validation signal. Commitment tone co-occurs with `climate-commitment` 91 times, with `climate-d` 57 times, and with `environmental-claims` 48 times. This pattern suggests that the extraction layer is not randomly assigning commitment labels; many commitment records also carry climate-related semantic evidence.

However, the same evidence exposes limitations. Missing tones co-occur with `climate-commitment` 22 times and `environmental-claims` 21 times. These cases are likely false negatives or schema failures because climate-relevant claims should usually have an interpretable disclosure tone. Governance disclosures also spread across action, commitment, outcome, none, and missing categories, which indicates that governance language is less easily captured by environmental-centric tone cues.

3) *Comparison With Existing Methods:* Compared with manual ESG scoring, this pipeline is more scalable and produces traceable text-level evidence, but it lacks expert judgment unless annotation is added. Compared with pure LLM classification, the pipeline is more auditable because it stores raw outputs, parsed JSON, prompt metadata, visualizations, and ClimateBERT comparison labels. Compared with ClimateBERT-only approaches, the pipeline is broader because it includes social and governance

records and explicitly models disclosure tone. Compared with rule-based taxonomies, the pipeline is more flexible for bilingual and context-dependent language, while still preserving interpretability through lexicons and ontology paths.

4) *Theoretical Implications:* The findings are consistent with legitimacy-oriented interpretations of sustainability reporting: firms may use ESG disclosure to signal responsibility and future alignment even when measured outcomes are less frequent. The low number of social records also raises a stakeholder-theory question about whether selected report pages, extraction prompts, or corporate disclosure practices emphasize environmental and governance audiences more strongly than social stakeholders.

For ABSA theory, the results show why sentiment alone is insufficient. A neutral statement can still be an important commitment or outcome. For greenwashing research, the commitment/outcome imbalance suggests a measurable warning signal, but it must be validated before being treated as a definitive greenwashing index.

5) *Limitations Discussed:* The most important limitation is the absence of full expert-labeled ground truth. The ClimateBERT remote sample has only three inputs, and ClimateBERT is not a direct tone classifier. OCR quality has not been measured with character error rate or table accuracy. Prompt sensitivity remains a concern because seven prompt templates are used. Finally, the ontology may not fully cover Indonesian-specific sustainability disclosure language.

6) *Implication for the Thesis:* The discussion should position the project as an exploratory but reproducible research framework. Its main contribution is not that it has already solved ESG classification, but that it creates the infrastructure for formal benchmark construction and richer ESG disclosure interpretation.

7) *subsection-Specific Report:* The dominance of commitment tone, with 115 out of 332 records, suggests that Indonesian corporate ESG disclosure in the sampled data often emphasizes goals, intentions, and strategic direction. This does not mean the reports are weak, but it shows why commitment and outcome must be separated analytically.

The alignment between commitment tone and `climate-commitment` labels, with 91 co-occurrences, is a positive validation signal. It means the extracted tone is often semantically consistent with independent climate-related labels. The 61 missing tone records, especially in governance-related disclosures, show that governance language is harder to classify with tone categories originally motivated by environmental performance narratives.

C. 5.3 Theoretical Implications

1) *subsection Purpose:* This subsection discusses the meaning of the empirical findings and compares the proposed system with existing ESG analysis methods. It draws on literature concerning corporate disclosure, sustainability reporting, ESG data analytics, ABSA, greenwashing detection, materiality, and climate disclosure analysis [R003], [R028], [R031], [R032], [R033].

2) *Discussion:* The main empirical finding is that commitment-oriented ESG disclosure dominates the current extracted dataset. Out of 332 records, 115 are classified as `commitment`, while 50 are classified as `outcome`. This imbalance matters because commitment language often signals strategic intention, legitimacy, or aspiration, whereas outcome language provides stronger evidence of realized performance.

The ClimateBERT-style comparison provides a partial validation signal. Commitment tone co-occurs with `climate-commitment` 91 times, with `climate-d` 57 times, and with `environmental-claims` 48 times. This pattern suggests that the extraction layer is not randomly assigning commitment labels; many commitment records also carry climate-related semantic evidence.

However, the same evidence exposes limitations. Missing tones co-occur with `climate-commitment` 22 times and `environmental-claims` 21 times. These cases are likely false negatives or schema failures because climate-relevant claims should usually have an interpretable disclosure tone. Governance disclosures also spread across action, commitment, outcome, none, and missing categories, which indicates that governance language is less easily captured by environmental-centric tone cues.

3) *Comparison With Existing Methods:* Compared with manual ESG scoring, this pipeline is more scalable and produces traceable text-level evidence, but it lacks expert judgment unless annotation is added. Compared with pure LLM classification, the pipeline is more auditable because it stores raw outputs, parsed JSON, prompt metadata, visualizations, and ClimateBERT comparison labels. Compared with ClimateBERT-only approaches, the pipeline is broader because it includes social and governance records and explicitly models disclosure tone. Compared with rule-based taxonomies, the pipeline is more flexible for bilingual and context-dependent language, while still preserving interpretability through lexicons and ontology paths.

4) *Theoretical Implications:* The findings are consistent with legitimacy-oriented interpretations of sustainability reporting: firms may use ESG disclosure to signal responsibility and future alignment even when measured outcomes are less frequent. The low number of social records also raises a stakeholder-theory question about whether selected report pages, extraction prompts, or corporate disclosure practices emphasize environmental and governance audiences more strongly than social stakeholders.

For ABSA theory, the results show why sentiment alone is insufficient. A neutral statement can still be an important commitment or outcome. For greenwashing research, the commitment/outcome imbalance suggests a measurable warning signal, but it must be validated before being treated as a definitive greenwashing index.

5) *Limitations Discussed:* The most important limitation is the absence of full expert-labeled ground truth. The ClimateBERT remote sample has only three inputs, and ClimateBERT is not a direct tone classifier. OCR quality has not been measured with character error rate or table accuracy. Prompt sensitivity remains a concern because seven prompt templates are used. Finally, the ontology may not fully cover Indonesian-specific sustainability disclosure language.

6) *Implication for the Thesis:* The discussion should position the project as an exploratory but reproducible research framework. Its main contribution is not that it has already solved ESG classification, but that it creates the infrastructure for formal benchmark construction and richer ESG disclosure interpretation.

7) *subsection-Specific Report:* From a legitimacy-theory perspective, the dominance of commitment language is consistent with companies using ESG reports to signal responsibility and future alignment. From a stakeholder-theory perspective, the pillar distribution ($E=179$, $G=121$, $S=4$) suggests that environmental and governance audiences receive more visible attention in the extracted records than social stakeholders.

For ABSA theory, the results show that sentiment polarity is not enough. A neutral sentence may still be an important commitment or outcome. For greenwashing literature, the commitment/outcome imbalance provides a measurable risk signal, but it should remain a heuristic until validated against expert labels or known cases.

D. 5.4 Limitations

1) *subsection Purpose:* This subsection discusses the meaning of the empirical findings and compares the proposed system with existing ESG analysis methods. It draws on literature concerning corporate disclosure, sustainability reporting, ESG data analytics, ABSA, greenwashing detection, materiality, and climate disclosure analysis [R011], [R017], [R042], [R057], [R060].

2) *Discussion:* The main empirical finding is that commitment-oriented ESG disclosure dominates the current extracted dataset. Out of 332 records, 115 are classified as `commitment`, while 50 are classified as `outcome`. This imbalance matters because commitment language often signals strategic intention, legitimacy, or aspiration, whereas outcome language provides stronger evidence of realized performance.

The ClimateBERT-style comparison provides a partial validation signal. Commitment tone co-occurs with `climate-commitment` 91 times, with `climate-d` 57 times, and with `environmental-claims` 48 times. This pattern suggests that the extraction layer is not randomly assigning commitment labels; many commitment records also carry climate-related semantic evidence.

However, the same evidence exposes limitations. Missing tones co-occur with `climate-commitment` 22 times and `environmental-claims` 21 times. These cases are likely false negatives or schema failures because climate-relevant claims should usually have an interpretable disclosure tone. Governance disclosures also spread across action, commitment, outcome, none, and missing categories, which indicates that governance language is less easily captured by environmental-centric tone cues.

3) *Comparison With Existing Methods:* Compared with manual ESG scoring, this pipeline is more scalable and produces traceable text-level evidence, but it lacks expert judgment unless annotation is added. Compared with pure LLM classification, the pipeline is more auditable because it stores raw outputs, parsed JSON, prompt metadata, visualizations, and ClimateBERT comparison labels. Compared with ClimateBERT-only approaches, the pipeline is broader because it includes social and governance records and explicitly models disclosure tone. Compared with rule-based taxonomies, the pipeline is more flexible for bilingual and context-dependent language, while still preserving interpretability through lexicons and ontology paths.

4) *Theoretical Implications:* The findings are consistent with legitimacy-oriented interpretations of sustainability reporting: firms may use ESG disclosure to signal responsibility and future alignment even when measured outcomes are less frequent. The low number of social records also raises a stakeholder-theory question about whether selected report pages, extraction prompts, or corporate disclosure practices emphasize environmental and governance audiences more strongly than social stakeholders.

For ABSA theory, the results show why sentiment alone is insufficient. A neutral statement can still be an important commitment or outcome. For greenwashing research, the commitment/outcome imbalance suggests a measurable warning signal, but it must be validated before being treated as a definitive greenwashing index.

5) *Limitations Discussed:* The most important limitation is the absence of full expert-labeled ground truth. The ClimateBERT remote sample has only three inputs, and ClimateBERT is not a direct tone classifier. OCR quality has not been measured with character error rate or table accuracy. Prompt sensitivity remains a concern because seven prompt templates are used. Finally, the ontology may not fully cover Indonesian-specific sustainability disclosure language.

6) *Implication for the Thesis:* The discussion should position the project as an exploratory but reproducible research framework. Its main contribution is not that it has already solved ESG classification, but that it creates the infrastructure for formal benchmark construction and richer ESG disclosure interpretation.

7) *subsection-Specific Report:* The first limitation is the absence of complete expert-labeled ground truth. The 332 records currently rely on model-predicted fields. The second limitation is weak ClimateBERT coverage: only three remote inputs are available, so ClimateBERT cannot yet function as a benchmark.

The third limitation is unmeasured OCR quality. Character error rate, word error rate, and table extraction accuracy are not yet computed. The fourth limitation is LLM prompt sensitivity across seven templates. The fifth limitation is that the greenwashing index is unvalidated. The sixth limitation is ontology scope: Indonesian-specific disclosure topics may not be fully represented.

VI. CONCLUSION

A. 6.1 Contribution Summary

1) *subsection Purpose:* This conclusion subsection synthesizes the research contributions, practical implications, and future work arising from the ESG disclosure tone analysis pipeline. The conclusion is grounded in literature on ESG analytics, sustainability disclosure, climate NLP, machine learning for ESG, and greenwashing detection [R003], [R006], [R007], [R009], [R018].

2) *Contribution Summary:* The research contributes an end-to-end workflow that converts sustainability reports into structured ESG records. It introduces a tone-aware disclosure taxonomy, compares extracted tone with ClimateBERT-style labels, provides visual dashboards, preserves reproducible artifacts, and defines a path toward formal ground-truth benchmarking.

The empirical evidence supports feasibility. The pipeline produced 332 extracted ESG records from 110 successful ESG extraction runs. The dominant tone was commitment with 115 records. The strongest validation pattern was the 91 co-occurrences between commitment and climate-commitment. These numbers show that automated tone-aware ESG analysis can surface meaningful disclosure patterns.

3) *Answers to Research Questions:* RQ1 is answered by the OCR and extraction workflow: PDF-derived sustainability reports can be transformed into markdown and then into structured ESG records.

RQ2 is answered by the extracted fields: records include aspect, ESG pillar, sentiment, tone, labels, and reasoning.

RQ3 is answered by the tone/ClimateBERT comparison: commitment labels align strongly with climate-commitment labels, while missing tones reveal review candidates.

RQ4 is answered by the diagnostic findings: 61 missing tone records, schema drift in sentiment fields, and social underrepresentation expose weaknesses in the current pipeline.

RQ5 is answered by the documentation and dashboard outputs: static charts, CSV tables, markdown reports, and the Streamlit app make the work inspectable and reproducible.

4) *Practical Implications:* For ESG analysts, the pipeline accelerates evidence extraction and helps locate commitment/outcome imbalances. For sustainability report authors, it provides feedback on whether disclosures read as promises, actions, or results. For regulators, especially in contexts such as OJK-aligned sustainability reporting, automated tone classification could support preliminary review and risk screening. For investors, tone distributions could supplement traditional ESG ratings by identifying whether ESG narratives are backed by reported outcomes.

5) *Future Work:* The next stage should build an expert-labeled ground truth dataset. ClimateBERT should be run over all 332 extracted record texts to create one-to-one comparison rows. OCR quality should be measured using character error rate and table extraction accuracy. The greenwashing index should be validated against external ratings or known cases. The ESG ontology should be expanded for Indonesian disclosure topics. Finally, multilingual transformer fine-tuning should be tested using Indonesian and English ESG text.

6) *Closing Statement*: The combined findings demonstrate that automated tone-aware ESG disclosure analysis is feasible, but formal validity depends on stronger ground truth. The current pipeline is therefore best understood as a complete research prototype and benchmark-construction platform.

7) *subsection-Specific Report*: The research contributes six concrete outputs: an end-to-end document-to-record pipeline, a tone-aware ESG taxonomy, ClimateBERT-style comparison, interactive visualization, reproducible artifacts, and a benchmark design for future annotation.

RQ1 is answered by the OCR-to-record workflow. RQ2 is answered by the 332 structured ESG records. RQ3 is answered by the 91 co-occurrences between `commitment` and `climate-commitment`. RQ4 is answered by diagnostic weaknesses such as 61 missing tones. RQ5 is answered through static charts, CSV files, markdown documentation, and the Streamlit dashboard. Together, these findings demonstrate the feasibility of automated tone-aware ESG disclosure analysis.

B. 6.2 Practical Implications

1) *subsection Purpose*: This conclusion subsection synthesizes the research contributions, practical implications, and future work arising from the ESG disclosure tone analysis pipeline. The conclusion is grounded in literature on ESG analytics, sustainability disclosure, climate NLP, machine learning for ESG, and greenwashing detection [R001], [R002], [R009], [R018], [R025].

2) *Contribution Summary*: The research contributes an end-to-end workflow that converts sustainability reports into structured ESG records. It introduces a tone-aware disclosure taxonomy, compares extracted tone with ClimateBERT-style labels, provides visual dashboards, preserves reproducible artifacts, and defines a path toward formal ground-truth benchmarking.

The empirical evidence supports feasibility. The pipeline produced 332 extracted ESG records from 110 successful ESG extraction runs. The dominant tone was `commitment` with 115 records. The strongest validation pattern was the 91 co-occurrences between `commitment` and `climate-commitment`. These numbers show that automated tone-aware ESG analysis can surface meaningful disclosure patterns.

3) *Answers to Research Questions*: RQ1 is answered by the OCR and extraction workflow: PDF-derived sustainability reports can be transformed into markdown and then into structured ESG records.

RQ2 is answered by the extracted fields: records include aspect, ESG pillar, sentiment, tone, labels, and reasoning.

RQ3 is answered by the tone/ClimateBERT comparison: `commitment` labels align strongly with `climate-commitment` labels, while missing tones reveal review candidates.

RQ4 is answered by the diagnostic findings: 61 missing tone records, schema drift in sentiment fields, and social underrepresentation expose weaknesses in the current pipeline.

RQ5 is answered by the documentation and dashboard outputs: static charts, CSV tables, markdown reports, and the Streamlit app make the work inspectable and reproducible.

4) *Practical Implications*: For ESG analysts, the pipeline accelerates evidence extraction and helps locate `commitment/outcome` imbalances. For sustainability report authors, it provides feedback on whether disclosures read as promises, actions, or results. For regulators, especially in contexts such as OJK-aligned sustainability reporting, automated tone classification could support preliminary review and risk screening. For investors, tone distributions could supplement traditional ESG ratings by identifying whether ESG narratives are backed by reported outcomes.

5) *Future Work*: The next stage should build an expert-labeled ground truth dataset. ClimateBERT should be run over all 332 extracted record texts to create one-to-one comparison rows. OCR quality should be measured using character error rate and table extraction accuracy. The greenwashing index should be validated against external ratings or known cases. The ESG ontology should be expanded for Indonesian disclosure topics. Finally, multilingual transformer fine-tuning should be tested using Indonesian and English ESG text.

6) *Closing Statement*: The combined findings demonstrate that automated tone-aware ESG disclosure analysis is feasible, but formal validity depends on stronger ground truth. The current pipeline is therefore best understood as a complete research prototype and benchmark-construction platform.

7) *subsection-Specific Report*: For ESG analysts, the pipeline accelerates evidence extraction and surfaces `commitment/outcome` imbalances. For sustainability report authors, tone-aware feedback can show whether disclosures are dominated by promises rather than measurable outcomes. For regulators, automated tone classification could support preliminary review of sustainability reporting quality under OJK-aligned disclosure expectations. For investors, a tone-derived greenwashing risk signal could supplement traditional ESG ratings by distinguishing narrative ambition from reported performance.

C. 6.3 Future Work

1) *subsection Purpose*: This conclusion subsection synthesizes the research contributions, practical implications, and future work arising from the ESG disclosure tone analysis pipeline. The conclusion is grounded in literature on ESG analytics, sustainability disclosure, climate NLP, machine learning for ESG, and greenwashing detection [R007], [R011], [R019], [R042], [R051].

2) *Contribution Summary*: The research contributes an end-to-end workflow that converts sustainability reports into structured ESG records. It introduces a tone-aware disclosure taxonomy, compares extracted tone with ClimateBERT-style labels, provides visual dashboards, preserves reproducible artifacts, and defines a path toward formal ground-truth benchmarking.

The empirical evidence supports feasibility. The pipeline produced 332 extracted ESG records from 110 successful ESG extraction runs. The dominant tone was `commitment` with 115 records. The strongest validation pattern was the 91 co-occurrences between `commitment` and `climate-commitment`. These numbers show that automated tone-aware ESG analysis can surface meaningful disclosure patterns.

3) *Answers to Research Questions*: RQ1 is answered by the OCR and extraction workflow: PDF-derived sustainability reports can be transformed into markdown and then into structured ESG records.

RQ2 is answered by the extracted fields: records include aspect, ESG pillar, sentiment, tone, labels, and reasoning.

RQ3 is answered by the tone/ClimateBERT comparison: `commitment` labels align strongly with climate-commitment labels, while missing tones reveal review candidates.

RQ4 is answered by the diagnostic findings: 61 missing tone records, schema drift in sentiment fields, and social underrepresentation expose weaknesses in the current pipeline.

RQ5 is answered by the documentation and dashboard outputs: static charts, CSV tables, markdown reports, and the Streamlit app make the work inspectable and reproducible.

4) *Practical Implications*: For ESG analysts, the pipeline accelerates evidence extraction and helps locate commitment/outcome imbalances. For sustainability report authors, it provides feedback on whether disclosures read as promises, actions, or results. For regulators, especially in contexts such as OJK-aligned sustainability reporting, automated tone classification could support preliminary review and risk screening. For investors, tone distributions could supplement traditional ESG ratings by identifying whether ESG narratives are backed by reported outcomes.

5) *Future Work*: The next stage should build an expert-labeled ground truth dataset. ClimateBERT should be run over all 332 extracted record texts to create one-to-one comparison rows. OCR quality should be measured using character error rate and table extraction accuracy. The greenwashing index should be validated against external ratings or known cases. The ESG ontology should be expanded for Indonesian disclosure topics. Finally, multilingual transformer fine-tuning should be tested using Indonesian and English ESG text.

6) *Closing Statement*: The combined findings demonstrate that automated tone-aware ESG disclosure analysis is feasible, but formal validity depends on stronger ground truth. The current pipeline is therefore best understood as a complete research prototype and benchmark-construction platform.

7) *subsection-Specific Report*: Future work should first build a larger expert-labeled ground truth dataset with inter-annotator agreement. Second, OCR quality should be measured using character error rate, word error rate, and table extraction accuracy. Third, the greenwashing index should be validated against known cases or external ESG ratings. Fourth, the ontology should be expanded for Indonesian-specific disclosure topics. Fifth, multilingual transformer models should be fine-tuned on Indonesian and English ESG text. Sixth, ClimateBERT should be run over all 332 extracted records to create a true one-to-one benchmark.

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TABLE I: Gap Analysis Summary

Gap	What existing methods do	What is missing	Implications for practice and theory	Supporting references
Gap 1: Absence of sentence-level ESG evidence extraction	Contemporary ESG NLP work often emphasizes document-level sentiment or coarse-grained topic signals (e.g., general tone of a sustainability report, or high-level topic modeling across entire documents) [8], [9], [12], [28]. These approaches capture broad impressions but do not systematically attach sentiment to discrete ESG evidence sentences or clause-level textual cues. Some ABSA implementations target fine-grained aspect extraction within domains such as general text or non-financial domains, yet there is limited evidence of rigorously validated sentence-level ABSA pipelines specifically tuned to ESG disclosures, where regulatory-relevant evidence often resides in particular sentences, sentences within tables, or footnotes [7], [10], [12], [14] Where sentence-level processing exists, it is frequently confined to generic sentiment at the sentence level or to shallow extraction of opinion-bearing phrases, without a formal linkage to precise ESG aspects, indicators, or regulatory lexicons.	A formal sentence-level ABSA framework for ESG filings that (i) maps each sentence to a concrete ESG aspect (e.g., emissions, governance oversight, supply-chain ethics), (ii) assigns sentence-level sentiment or evidentiary polarity with respect to that aspect, and (iii) propagates confidence metrics and explainability to regulators and investors. A standardized annotation schema and corpus for Indonesian and English ESG disclosures that anchors aspect terms to GRI/SASB/TCFD-aligned concepts at the sentence level, enabling reproducible model development and cross-study comparability. Evaluation protocols that quantify the precision, recall, and calibration of sentence-level aspect-sentiment associations, including robustness to multi-aspect sentences and to the presence of measured numerical indicators within the same sentence.	Without sentence-level evidence extraction, ABSA outputs remain insufficiently actionable for due diligence, risk assessment, and regulatory verification, because decision-makers require precise textual evidence linking sentiment signals to specific ESG topics and regulatory criteria [3], [7], [12] A sentence-level approach would enable more transparent risk signaling, facilitate traceability for greenwashing detection, and improve the interpretability of ABSA outputs through direct citation of textual evidence.	ABSA literature highlights the need for fine-grained, aspect-level extraction to reveal granular signals in ESG narratives [7], [10], [12], [14] Cross-lingual ABSA and multilingual ESG contexts further motivate sentence-level analysis to capture language-specific expressions of risk or compliance indicators in Indonesian and English text [11]–[13]
Gap 2: Conflation of tone and sentiment in existing approaches	Many studies treat ESG "tone" (the overall sentiment polarity of a document or section) as synonymous with "sentiment" toward ESG topics, effectively collapsing contextual cues into a single polarity score [3], [7], [9] Some traditional sentiment analyses rely on bag-of-words or lexicon-based approaches that assign a global sentiment to documents, or to broad sections, without distinguishing whether sentiment reflects a positive environmental action, neutral governance reporting, or negative social impact within a specific topic [12], [28] ABSA literature emphasizes the distinction between sentiment polarity and opinion-bearing aspects, but practice in ESG contexts often lags in consistently separating topic-level sentiment from the broader document tone, particularly when sentences mention multiple ESG topics with conflicting cues.	A clear theoretical and methodological separation between tone (global document or section-level polarity) and targeted sentiment toward discrete ESG aspects, ensuring that ABSA outputs reflect aspect-specific sentiment rather than aggregate tone. Evaluation frameworks that can detect and quantify conflation errors, such as misattributing an English-positive sentence to an environmental aspect when the sentiment is actually about governance or labor practices. Multimodal and numerical signal integration to disambiguate tone from explicit sentiment about an ESG aspect, for example when numerical performance indicators accompany otherwise neutral text.	Conflation reduces the reliability of ABSA outputs for decision-makers who rely on precise signal attribution, such as due diligence teams and regulators assessing material ESG risks and governance reforms. Clear separation improves model interpretability and supports explainable AI (XAI) approaches, enabling stakeholders to trace the rationale behind an aspect-level sentiment verdict.	Systematic ESG analytics reviews emphasize challenges around explainability, interpretability, and the need to map sentiments to concrete ESG dimensions [3], [7], [10] Cross-lingual ABSA research underscores the importance of maintaining topic-sentiment separation across languages to preserve semantic fidelity [11]–[13]

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TABLE I – Continued

Gap	What existing methods do	What is missing	Implications for practice and theory	Supporting references
Gap 3: LLM output instability across prompt templates	Large language models (LLMs) are increasingly applied to extract, summarize, and reason about ESG content via prompt-based or retrieval-augmented generation (RAG) paradigms. However, reported behavior often varies with prompt wording, template selection, and in-context examples, leading to instability in outputs across prompts or sessions [29], [30] Some deployments adopt fixed prompts or single-shot prompts, which can yield inconsistent results when applied to diverse ESG documents, especially given domain-specific vocabulary and regulatory lexicons [29], [30] There is growing recognition of the need for prompt engineering, prompt ensembles, and verification strategies to stabilize outputs and improve reproducibility [29], [30]	Systematic studies that quantify prompt-induced variability for ESG ABSA tasks in Indonesian/English bilingual corpora, including metrics of stability (e.g., variance in aspect extraction, sentiment labeling, and regulatory mappings across templates). Robust prompt design guidelines and ensemble strategies (e.g., multiple prompts, self-consistency checks, majority voting) tailored to ESG ABSA with multilingual and legal/regulatory constraints. Integration of verification layers, such as post-hoc fact-checking against regulatory lexicons (GRI/SASB/TCFD) and cross-language alignment checks, to improve reliability of LLM outputs.	Without addressing prompt instability, LLM-based ABSA systems risk inconsistent outputs, undermining trust, regulatory audibility, and investor confidence. Stable, hybrid architectures combining LLMs with deterministic components (rule-based lexicons, supervised ABSA models) can reduce risk while preserving the benefits of generative capabilities.	The literature on LLMs in finance and ESG emphasizes model behavior, prompts, and the need for reliability and explainability, including considerations of bias, privacy, and regulatory compliance [?], [31], [33] Recent ESG document QA and RAG approaches illustrate how retrieval-augmented generation can improve accuracy and traceability, but also highlight instability risks in prompt-driven systems [29].
Gap 4: Neglect of OCR quality in document-level pipelines	Many ESG NLP studies assume textual input is available as clean machine-readable text, effectively bypassing the OCR bottleneck typical of scanned or PDF-disclosed documents. In practice, several document-analysis pipelines acknowledge OCR as a precursor step but do not systematically evaluate how OCR quality (character error rate, layout extraction accuracy) propagates into downstream ABSA performance, especially in multilingual settings where scripts and diacritics differ. Some studies of scanned or semi-structured documents from other domains (healthcare, historical archives) explicitly show that OCR quality, layout understanding, and document structure critically affect information extraction performance, suggesting similar implications for ESG reports that are often distributed as PDFs with embedded images or tables [34]–[37]	An end-to-end evaluation of how OCR quality, layout analysis, and document structure influence sentence-level ABSA and aspect-term extraction in Indonesian/English ESG disclosures. Techniques that couple OCR refinement (e.g., layout-aware OCR, region-based extraction, post-OCR text normalization) with ABSA pipelines, including strategies to handle multilingual content, tables, footnotes, and image-based figures that contain text. Standards for reporting OCR-related uncertainty in ESG analytics, including error-aware sentiment and evidence mapping to specific document sections.	Ignoring OCR quality leads to noise and bias in ABSA outputs, particularly in Indonesian contexts with mixed Indonesian/English text and domain-specific vocabulary. This undermines reliability for regulatory and investor use. A tightly integrated OCR-ABSA workflow with layout-aware processing improves signal fidelity and supports reproducibility, auditability, and scalability.	Document- and OCR-centric studies illustrate the impact of OCR and layout on downstream NLP tasks, including historical document digitization and healthcare document processing where layout information significantly affects extraction quality [34]–[36], [38]–[41] Multilingual and cross-language ABSA literature emphasizes the need for robust tokenization and alignment across languages, which are sensitive to OCR output quality in bilingual documents (Šmíd & Král, 2025), (Gorovaia & Makrominas, 2024), (Hang, 2025).

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TABLE I – Continued

Gap	What existing methods do	What is missing	Implications for practice and theory	Supporting references
Gap 5: Underdeveloped greenwashing detection methods	A subset of ESG NLP research focuses on identifying greenwashing signals by analyzing the volume, framing, and accessibility of environmental claims, sometimes using length, readability, copiousness of environmental content, or discrepancy between stated commitments and disclosed performance [12], [42], [43] Some studies explicitly link linguistic features to greenwashing risk, finding that longer or more favorable environmental narratives do not always correlate with substantive environmental performance, implying the need for evidence-based detection pipelines that tie textual claims to measurable indicators [12] Cross-domain literature shows that detection of misleading sustainability content benefits from ABSA to locate sentiment around environmental topics and to cross-check with governance disclosures and quantified targets [9], [12]	A rigorous, formally defined greenwashing detection framework that integrates sentence- or aspect-level ABSA with verifiable evidence from numerical performance data, regulatory disclosures, and third-party verification, specifically adapted to Indonesian and English ESG reporting. Methodologies that quantify inconsistency between qualitative statements and quantitative signals (e.g., emissions reductions, energy use intensity, board oversight metrics) and that produce explainable justifications for greenwashing flags. Benchmark datasets and evaluation protocols for greenwashing detection in bilingual ESG disclosures, including annotation guidelines that capture deceptive framing, selectivity in disclosures, or omissions.	Without robust greenwashing detection, investors and regulators face elevated risk of misinterpreting ESG commitments, potentially leading to misallocations of capital or regulatory blind spots. A multidisciplinary approach combining ABSA, factual verification, and explainable anomaly detection can strengthen trust in ESG communications and support more reliable sustainability governance.	Studies on greenwashing detection via NLP show that longer, more positive but less readable ESG content may indicate greenwashing risk, highlighting the need for deeper, evidence-aligned analyses beyond superficial sentiment [12] ESG analytics literature emphasizes the importance of explainability and regulatory-aligned signals to counter greenwashing and enhance the credibility of ESG disclosures [3], [7], [10]

TABLE II: Research Questions

RQ	Aims and scope	Methodological components	Expected outputs	Grounding references
RQ1. How can a PDF-to-structured-ESG transformation pipeline convert Indonesian/English sustainability reports and filings into a governance- and regulation-aligned, sentence- and paragraph-level ECS (ESG Content Structure) representation that supports ABSA at the sentence level?	design and evaluate an end-to-end pipeline that (i) performs robust PDF parsing, (ii) extracts and normalizes textual content, tables, and figures, (iii) aligns content to a joint ESG taxonomy anchored in GRI/SASB/TCFD concepts, and (iv) outputs a structured JSON/NER-ready representation with sentence-level section identifiers and regulatory tag mappings.	document layout analysis, OCR quality handling, cross-language alignment, topic-taxonomy mapping, and sentence-level evidence tagging.	a reproducible, auditable data representation that enables sentence-level ABSA and regulator-oriented tracing of evidence.	Climate-focused ABSA pipelines and document-to-structure transformations emphasize the need to anchor text to regulatory concepts and to manage multi-language inputs in structured formats [44], [48]–[51], [119], [120]. Foundational work on ClimateBERT and domain-adapted transformers informs the feasibility of accurate sentence-level extraction within climate and ESG contexts [44]–[47]
RQ2. How should ESG be categorized by aspect/pillar, sentiment, and tone in bilingual Indonesian/English corporate disclosures to enable fine-grained ABSA, while preserving cross-language comparability and regulatory alignment?	define and validate a multi-dimensional annotation schema that (i) maps sentences to ESG aspects (environment, social, governance), (ii) assigns aspect-level sentiment polarity and evaluative tone, and (iii) accounts for document-level tone as a contextual backdrop. The schema should support cross-language consistency and interoperability with GRI/SASB/TCFD mappings.	ontology design, cross-language term mapping (Indonesian-English), annotation guidelines, inter-annotator agreement assessment, and a pilot corpus annotated in both languages.	a reference corpus and a formal taxonomy enabling comparability across Indonesian and English disclosures and enabling downstream model training for ABSA.	ABSA literature stresses sentence- and aspect-level extraction and cross-lingual considerations, with cross-lingual ABSA surveys highlighting the need for consistent aspect taxonomies across languages [7], [11]–[14]. FinBERT/ClimateBERT-style domain models support sentence-level interpretation in ESG domains [44], [47], [49].
RQ3. Do tone-based ABSA outputs differ meaningfully from ClimateBERT-style label classifications for climate-related disclosures, and what is the relationship between detected tone and climate-specific targets or commitments?	compare traditional tone classification (document/section-level polarity) with climate-focused label outputs produced by ClimateBERT-like models, including assessments of alignment with explicit climate targets (e.g., net-zero, emissions reductions). Investigate whether tone signals correlate with or diverge from target-specific classifications.	parallel labeling experiments using (i) a tone-focused ABSA pipe (global polarity) and (ii) a climate-target-oriented classifier (ClimateBERT-NetZero-like or ClimateBERT variants), followed by statistical analyses of concordance, discordance, and predictive validity for reported targets.	empirical criteria and guidelines for choosing or combining tone and climate-focused labels to maximize decision-usefulness and regulatory alignment.	ClimateBERT and zero-shot climate classification literature demonstrate how domain-specific transformers capture climate-related signals beyond generic sentiment, and several studies compare climate-target detection with broader tone signals [47], [48], [50]–[52]. Foundational climate-disclosure detection work shows that target specificity requires specialized models for robust performance [44]–[46].
RQ4. What weaknesses arise in ABSA extraction outputs, and how can a diagnostics framework detect and quantify extraction errors (e.g., misidentified aspects, incorrect sentiment polarity, or missed regulatory mappings) to inform model improvement?	develop a sentence-level ABSA diagnostics framework that (i) identifies common error modes (noise, misclassification, lexical drift, cross-language misalignment), (ii) quantifies error types with interpretable metrics, and (iii) provides actionable feedback loops to improve model components (lexicon updates, annotation guidelines, cross-language alignment).	error taxonomy development, curated test sets with known ground truth, per-sentence calibration analyses, ablation studies, and model-retraining protocols.	a robust quality assurance regime for ESG ABSA systems, including dashboards for developers and regulators to monitor performance and track improvements over time.	ABSA and ESG ML literature repeatedly highlight challenges in explainability, calibration, and domain adaptation; cross-lingual ABSA studies stress the need for rigorous evaluation across languages to ensure reliable outputs [3], [7], [10]–[14]. Explainability-focused integration with ABSA is recommended to support regulator-facing outputs [3], [7], [10].

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TABLE II – Continued

RQ	Aims and scope	Methodological components	Expected outputs	Grounding references
RQ5. How can documentation and visualization practices be designed to maximize reproducibility and auditability of ESG ABSA experiments, including annotation schemas, model configurations, prompts/templates, and evaluation results?	propose comprehensive documentation standards and visualization schemes that (i) capture data provenance (source documents, language, version), (ii) describe annotation schemas and inter-annotator agreement metrics, (iii) enumerate model architectures, hyperparameters, datasets, and prompts/templates, (iv) present evaluation metrics and error analyses, and (v) provide reproducible notebooks and data-sharing guidelines.	a reproducibility blueprint, standardized metadata schemas, version-controlled annotation guidelines, and visualization dashboards for performance, bias, and calibration across languages and frameworks.	a reproducibility package including annotated corpora, code, prompts/templates, and evaluation reports, enabling independent replication and regulatory audit.	across climate and ESG NLP, systematic documentation and reproducibility considerations are emphasized; works on climate claim detection and zero-shot analyses illustrate the need for transparent methodology and datasets [48], [50], [51]. The broader ESG literature also stresses standardized reporting of framework mappings and cognitive traceability [21]–[23]
RQ6. What is the stability of ABSA outputs across cross-model and cross-prompt configurations, and what ensemble or verification strategies yield the most reliable and regulator-friendly results in Indonesian/English bilingual ESG analysis?	quantify output stability across multiple models (e.g., generic ABSA, FinBERT-based ESG classifiers, ClimateBERT variants, multilingual transformers) and across prompts/templates, and evaluate ensemble strategies (e.g., majority voting, self-consistency checks, retrieval-augmented approaches) for reducing variance and increasing reliability.	prompt engineering experiments, cross-model comparisons under controlled data splits, stability metrics (e.g., per-sentence label agreement, variance in aspect extraction, consistency of regulatory mappings), and verifier modules that cross-check outputs against regulatory lexicons and numerical indicators.	practical guidance for practitioners on selecting robust configurations, with quantified trade-offs between performance, stability, and interpretability.	

TABLE III: Research Objectives

Objective	Explanation	Deliverables	Supporting basis
Objective 1 – Build the OCR-to-record transformation pipeline (PDF-to-structured-ESG representation)	To design and implement an end-to-end OCR-enabled pipeline that converts Indonesian and English sustainability reports and regulatory filings (PDF, scanned or native) into a sentence- and paragraph-level, regulator-aligned structured representation (JSON/NER-ready schema) with explicit mappings to GRI/SASB/TCFD concepts. To evaluate the pipeline on a diverse set of 60–100 document samples (across Indonesian and cross-language variants) in terms of (a) OCR accuracy, (b) layout-aware extraction quality, (c) fidelity of section and sentence segmentation, and (d) correctness of ESG-topic alignment, reporting regression testing, and cross-language alignment.	a reproducible OCR-to-record toolchain, annotated preprocessing logs, and a documented data schema linking document regions to ESG aspects, with verifiable performance metrics (character error rate, sentence boundary precision/recall, and taxonomy-mapping accuracy).	prior work emphasizes the necessity of robust document structure extraction, OCR/layout-aware processing, and cross-language alignment to support downstream ABSA tasks in multilingual ESG contexts (Lamiroy et al., 2011; , Hsu et al., 2022; , Nandurbarkar et al., 2025; , Ali et al., 2022; , Lafia et al., 2023; , Clausner et al., 2017; , Chiatti et al., 2018; , Rehm et al., 2019). Climate- and ESG-specific ABSA literature demonstrates the value of anchoring textual content to regulatory concept mappings for interpretability and auditability (Garrido-Merchán et al., 2023; , (Bingler et al., 2021; , Huang et al., 2023; , Li et al., 2024; , Auzepy et al., 2023; , Trajanov et al., 2023; , Kim et al., 2024).
Objective 2 – Implement tone-aware ABSA for bilingual ESG disclosures	To design and implement a sentence- and aspect-level ABSA pipeline that yields: (i) aspect-term extraction and mapping to ESG pillars (environment, social, governance), (ii) sentence-level sentiment polarity per aspect, and (iii) document-level tonal context to situate per-sentence signals. To evaluate the ABSA system using a bilingual Indonesian/English corpus with gold-standard sentence-level annotations, computing precision, recall, F1 for aspect-term extraction, sentiment accuracy per aspect, and calibration metrics (confidence intervals, reliability diagrams) across languages.	a fully documented ABSA model with domain-adapted fine-tuning (e.g., Indonesian-English multilingual transformer with domain tokens), evaluation report including language-wise performance, and an open-schema for aspect-to-regulatory mapping.	ABSA literature consistently argues for sentence- and aspect-level granularity to achieve actionable ESG signals, and cross-lingual ABSA work underscores the need for robust multilingual mappings and domain adaptation when handling Indonesian/English content (Seow, 2025; , Šmíd & Král, 2025; , Gorovaia & Makrominas, 2024; , Hang, 2025; , Heever et al., 2024). Tone-aware design advances beyond generic sentiment by ensuring the polarity is tied to concrete ESG aspects, thereby improving interpretability for investors and regulators Srivastava & Anand, 2023), Moreno & Caminero, 2020; .
Objective 3 – Compare tone outputs with ClimateBERT-label classifications for climate-related disclosures	To conduct a controlled comparison between (a) tone-based ABSA outputs (document/section-level polarity and aspect-level sentiment) and (b) dedicated ClimateBERT-style climate-target labels (net-zero/reduction targets and climate-specific classifications) on identical climate-related passages. To quantify concordance/discordance between the two labeling regimes, assess predictive validity for reported climate targets and commitments, and identify systematic divergences that inform model selection or ensemble strategies.	a comparative study report with statistical analyses (e.g., Cohen’s kappa for agreement, confusion matrices for target labels vs sentiment, and regression analyses linking labels to disclosed targets), plus recommendations for when to fuse or separate climate-target signals from tone signals.	emerging climate-disclosure NLP work demonstrates that domain-specific models (e.g., ClimateBERT variants) capture climate signals beyond generic tone, and comparing these with tone-centric ABSA outputs reveals when specialized targets improve decision-useful signal extraction (Garrido-Merchán et al., 2023; , Hassani et al., 2025; , Schimanski et al., 2023; , Kim et al., 2024). The objective directly enables evidence-based taxonomy and model selection for ESG analytics in climate-relevant domains.
Objective 4 – Evaluate extraction quality and identify failure patterns (diagnostics for ABSA)	To develop and deploy a diagnostics framework that automatically identifies extraction weaknesses, including (i) incorrect/omitted ESG-aspect mappings, (ii) misattribution of sentiment polarity to wrong aspects, (iii) language-specific lexical drift and regulatory lexicon gaps, and (iv) failures due to near-duplicate or multi-aspect sentences. To quantify failure modes with interpretable metrics (per-sentence error taxonomy, per-document error rates, calibration of confidence scores) and implement a closed-loop process for model improvement (lexicon updates, annotation guideline revisions, re-training or fine-tuning schedules).	an ABSA failure-analytics dashboard, a formal error taxonomy document, and a protocol for iterative model refinement with traceability to ground-truth annotations and regulatory mappings.	Section 1.3 highlighted the critical need for extraction diagnostics and failure analysis to drive robust ABSA in ESG contexts (Seow, 2025; , Moreno & Caminero, 2020; , Srivastava & Anand, 2023). A formal diagnostics framework operationalizes this need, enabling continuous improvement and regulator-friendly transparency.

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TABLE III – Continued

RQ	Aims and scope	Methodological components	Expected outputs
Grounding references			
Objective 5 – Create reproducible visualization and documentation artifacts for auditability	To establish standardized visualization and documentation artifacts that support reproducibility and regulatory auditability, including: (a) an annotated corpus with language tags, aspect taxonomies, and ABSA labels, (b) model configuration records (architecture, hyperparameters, seeds), (c) prompts/templates and in-context example sets for prompts-based LLM components, (d) evaluation results with language- and framework-specific breakdowns, and (e) executable notebooks and data-access guidelines that enable independent replication. To evaluate the usefulness of artifacts through a replication study with independent researchers attempting to reproduce key results within predefined time constraints, reporting any deviations and remediation steps.	a reproducibility package comprising data, code, prompts/templates, schemas, and visualization dashboards, accompanied by a formal reproducibility report validating the ability to reproduce core findings.	Literature on climate- and ESG-NLP emphasizes the importance of transparent methodology, standardized documentation, and reproducible pipelines to ensure trust, accountability, and regulatory auditability in ESG analytics (Bingler et al., 2021; , Auzepy et al., 2023; , Trajanov et al., 2023; , Abhayawansa & Adams, 2021; , Rowbottom, 2022; , O’Dwyer & Unerman, 2020). This objective ensures that technical contributions are accessible, auditable, and citable.
Objective 6 – Assess cross-model and cross-prompt stability; develop robust ensemble/verification strategies	To quantify ABSA output stability across multiple models (monolingual and multilingual transformers, climate-focused variants, general ABSA baselines) and across diverse prompt templates (including zero-shot and few-shot configurations), measuring per-sentence agreement, label stability, and variability in aspect extraction. To design and evaluate ensemble strategies (e.g., majority voting, self-consistency prompts, retrieval-augmented generation with external regulatory lexicons) and verification layers that improve reliability and regulator-friendly interpretability of outputs.	a stability benchmark suite with defined datasets, a set of recommended ensemble/verification configurations, and guidance for practitioners on achieving stable, auditable ESG ABSA outputs in bilingual Indonesian/English reporting.	The need to address prompt-induced instability and model variability in LLM-augmented ABSA is well documented in recent work on ESG NLP and ABSA robustness (Khak et al., 2025; , Xu, 2022; , Adeoye et al., 2024; , Mishra et al., 2024; , Hassani et al., 2025; , Schimanski et al., 2023; , Viñán-Ludeña & Campos, 2024; , Ming-shan et al., 2024). This objective provides formal measurement and practical remedies to ensure reliable deployment in regulatory and investment contexts.

TABLE IV: Research Questions

Contribution	What this contribution delivers	Novelty	Evaluation and metrics	Deliverables	Rationale and grounding
Contribution 1: An end-to-end OCR-to-record ESG processing pipeline from PDF to structured records	A fully integrated pipeline that ingests PDF sustainability reports and regulatory filings (including scanned and native PDFs), applies page- and layout-aware OCR, performs multi-language text extraction, normalizes and aligns content to a joint ESG taxonomy (GRI/SASB/TCFD), and outputs a structured, machine-readable record format (JSON/NER-ready) with sentence- and paragraph-level annotations and explicit ESG aspect mappings.	Combines layout-aware OCR, multilingual text normalization, and cross-framework topic alignment into a single, reproducible data product specifically designed for ABSA in ESG disclosures. Unlike prior work that treats PDFs as plain text or bypasses layout, this pipeline preserves document structure to enable sentence-level ABSA and regulatory traceability. It integrates a sentence- and region-level provenance map that ties extracted content to exact document regions (sections, tables, footnotes) and to regulatory concepts, enabling auditable downstream analysis [34]–[41]. The pipeline is designed for bilingual Indonesian/English reports, with integrated cross-language alignment modules and a joint ontology linking Indonesian terms to English counterparts and to GRI/SASB/TCFD anchors, facilitating cross-language comparability and regulatory consistency [11]–[14]	OCR quality metrics (character error rate, word error rate) stratified by language; layout accuracy metrics (section/column segmentation accuracy); sentence boundary detection precision/recall; ESG-topic alignment accuracy against a gold standard; and end-to-end regeneration fidelity (regression tests across 60–100 document samples).	A reusable, documented OCR-to-record toolkit, an explicit JSON schema for ESG aspects, language tags, section identifiers, and regulatory mappings; and a benchmark dataset with ground-truth region-level annotations for Indonesia-related disclosures.	This contribution operationalizes the PDF-to-structure transformation essential for ABSA in ESG contexts. It builds on the demonstrated need for layout-aware processing and cross-language alignment in multilingual ESG texts, and leverages prior work on document structure extraction and regulatory mapping to deliver a robust, auditable data product [34]–[41], [44], [48]. The Indonesian regulatory context (OJK/IDX) motivates an implementation that supports regulator-facing traceability and standardization [1], [3], [4]
Contribution 2: A page-aware, prompt-driven extraction framework with seven templates across bilingual reports	A page-aware extraction engine that uses seven deliberate prompt templates to drive extraction of ESG evidence, aspect terms, and sentiment signals from Indonesian and English pages. Templates are designed to respect page-level cues (headers, tables, footnotes, figures) and to capture evidence across multiple document regions (text blocks, tables, captions).	Introduces a disciplined, multi-template prompt strategy for ABSA in ESG disclosures, explicitly engineered for bilingual Indonesian/English corpora. It extends beyond generic document-level prompts by injecting page-context signals, region identifiers, and regulatory anchors into prompts to improve alignment of extracted content with ESG aspects and regulatory concepts. This mirrors found work on prompt-template diversity and prompt ensembles but applies them to the ESG ABSA domain with cross-language considerations [29], [46], [47], [52]. The seven templates are procedurally defined to elicit: (a) sentence-level aspect extraction, (b) sentence-level polarity per aspect, (c) cross-language term normalization, (d) regulatory mapping hints, (e) target/commitment vs. tone cues, (f) evidence tagging with document-region provenance, and (g) explanations of the rationale behind each extraction decision.	Template-wise ablation studies measuring extraction precision/recall for aspects and sentiments per template; inter-template agreement; and cross-language consistency of outputs. Stability metrics across prompts (variance in extractions for the same text region) will be reported.	The seven-template prompt suite, a registry of prompt configurations, and an evaluation report comparing template performance across Indonesian/English inputs, including guidance for practitioners on template selection and combination.	This contribution directly addresses the practical need for robust extraction in ESG ABSA pipelines, especially in complex, bilingual reports where content spans paragraphs, tables, and captions. It aligns with literature on prompt-based NLP in finance/ESG contexts and the importance of domain-specific prompts for reliable outputs, while explicitly addressing cross-language inputs and regulatory alignment [29], [31], [46], [47], [52]

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TABLE IV – Continued

Contribution	What this contribution delivers	Novelty	Evaluation and metrics	Deliverables	Rationale and grounding
Contribution 3: A tone-aware ESG taxonomy distinguishing commitment, action, and outcome	A multi-dimensional taxonomy and annotation framework that separates tone from explicit commitments, actions, and outcomes. It operationalizes a three-tiered tone taxonomy (commitment, action, outcome) layered atop the ESG pillars (environment, social, governance) to enable fine-grained aspect-level ABSA that discriminates between intent, activity, and measurable results.	Expands beyond conventional global sentiment or document tone to introduce a structured, action-oriented tonal ontology that maps to real-world governance and performance signals. This taxonomy enables researchers and practitioners to disentangle motivational language (commitment) from operational changes (action) and verifiable results (outcome), thereby reducing conflation and increasing interpretability for regulators and investors. This addresses a gap noted in ABSA literature regarding distinguishing tone from concrete sentiment toward ESG aspects [3], [7], [10] The taxonomy is designed for bilingual application, with cross-language alignment rules to ensure consistency of commitment/action/outcome labels across Indonesian and English expressions and across GRI/SASB/TCFD mappings.	Inter-annotator agreement on the taxonomy; cross-language agreement statistics; calibration of tone labels with actual disclosed outcomes (where targets or KPIs exist); precision/recall for each subcategory per aspect.	A formal, ontology-backed tone taxonomy, annotation guidelines, and an openly accessible reference corpus with tone-annotated sentences in both languages.	This contribution directly addresses the need to distinguish tone from substantive ESG signals (commitment, action, outcome) to improve the fidelity and auditability of ABSA outputs. It resonates with cross-lingual ABSA and domain-adapted sentiment literatures calling for nuanced, topic- and context-sensitive sentiment representations and emphasizes interpretability for governance contexts [7], [10]–[14]
Contribution 4: An ontology-based ABSA layer for ESG aspect classification	An ontology-driven ABSA layer that provides explicit mappings from extracted phrases to ESG aspects (environmental, social, governance) and sub-aspects (e.g., emissions, energy efficiency, governance oversight, supply chain, diversity) with cross-reference to GRI/SASB/TCFD concepts. The layer includes an explicit hierarchical structure, relations among aspects, and an evidence-linking mechanism to regulatory sections.	Introduces an explicit, machine-readable ESG ontology tailored to Indonesian/English disclosures, enabling consistent aspect classification across languages and frameworks. The ontology supports explainable ABSA by providing a pathway from textual cues to regulatory topics, metrics, and potential action items. While ABSA often relies on flat taxonomy or domain-adapted lexicons, this work elevates taxonomy to a formal ontology with hierarchical relationships and regulatory tag propagation. The ontology is designed to be extensible to additional frameworks (TCFD, ISSB) and to accommodate multilingual alignment through cross-lingual term mappings and synonym sets.	ontology coverage metrics (recall/precision of aspect mappings), hierarchical consistency checks, and cross-language mapping accuracy. End-user evaluation with regulator-facing scenarios (mapping to required disclosures) to validate interpretability.	The ESG ontology with a formal schema (OWL/RDF-compatible), mapping tables to GRI/SASB/TCFD indicators, and a bilingual synonym/translation file set for Indonesian-English terms.	This contribution provides a principled backbone for ABSA in ESG analytics, enabling consistent, auditable aspect classification and regulatory traceability. It aligns with literature on ESG standardization and ontology-driven information extraction, addressing the need for structured, cross-framework alignment in a multilingual Indonesian context [19], [21]–[24]

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TABLE IV – Continued

Contribution	What this contribution delivers	Novelty	Evaluation and metrics	Deliverables	Rationale and grounding
Contribution 5: A ClimateBERT comparison framework for external semantic validation	A formal framework to compare ABSA-derived climate signals with external semantic validators (e.g., ClimateBERT-derived climate-target labels, ClimateBERT-NetZero outputs, and other domain-specific validators) to assess alignment and provide external validation of climate-related signals.	Introduces a validation architecture that couples internal ABSA outputs with external climate-domain models to detect divergence, calibrate climate-specific signals, and identify when ClimateBERT-style labels provide additional predictive or regulatory value. This cross-model validation addresses concerns about relying solely on internal ABSA outputs for climate signaling and supports robust external verification for regulators and investors. The framework supports bilingual Indonesian/English data by enabling cross-language validation pathways, using multilingual or domain-adapted validators where appropriate.	Concordance measures (e.g., Cohen’s kappa) between ABSA-derived climate signals and ClimateBERT-derived classifications; predictive validity analyses linking approvals of climate targets to disclosed commitments; calibration analyses across languages.	A validation protocol, a computational framework for cross-model validation, and a set of guidance metrics for when to trust internal ABSA labels versus external validator signals.	This contribution directly addresses the need for external semantic validation of climate-related ESG signals, leveraging established ClimateBERT frameworks and related domain models. It aligns with the literature on domain-specific climate NLP models and cross-model validation practices in ESG analytics [44], [46], [47], [52], [121]–[123]
Contribution 6: Explainability outputs including lexical triggers, TF-IDF coefficients, and ontology paths	A comprehensive explainability module that produces (i) lexical triggers (key terms and phrases that drive aspect-level sentiment), (ii) TF-IDF-based feature importance for per-aspect explanations, and (iii) ontology-path visualizations showing the reasoning chain from textual cues to ESG aspects and regulatory mappings.	Combines rule-based lexical cues with data-driven feature importance (TF-IDF) and an ontology-path traversal to provide multi-faceted explanations for ABSA decisions. This triad supports regulator-friendly transparency by explicitly identifying why an extraction was made, what features influenced sentiment, and which ontology nodes guided the classification. This holistic explainability approach extends beyond standard post-hoc XAI methods (e.g., SHAP/LIME) by embedding ontology-informed reasoning into the explanation, improving traceability to ESG frameworks. The explainability outputs are designed to be language-agnostic at the architectural level, with bilingual support through lexicon and ontology mappings, facilitating cross-language interpretability and auditability.	Coverage of lexical triggers per aspect; correlation of TF-IDF weights with human judgments of importance; user studies with regulators to assess the clarity and usefulness of ontology-path visualizations; fidelity of explanations to the actual content in the PDF-to-record pipeline.	An explainability toolkit integrated into the ABSA pipeline, including: (a) a lexical trigger repository, (b) TF-IDF explanation dashboards, and (c) ontology-path visualization components.	Explainability is central to ESG analytics for stakeholder trust and regulatory acceptance. This contribution synthesizes established XAI approaches with ontology-driven reasoning to deliver regulator-friendly outputs and actionable insights, addressing the imperative for interpretability highlighted across ABSA and ESG literature [3], [7], [10], [14], [48], [50], [51]